

One-Step Regularized Spatial Weight Matrix and Fixed Effects Estimation with Instrumental Variables

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Abstract

Analysis in spatial econometrics is often conditioned on the knowledge of the spatial weight matrix, choice that often makes use of previous expert information or proxies of related variables. We first argue that its misspecification leads to serious estimation bias of spatial parameters. We then propose estimating the spatial weight matrices directly, along with spatial parameters, with recourse to a regularization based on the adaptive LASSO with instrumental variables (IVLASSO) under a model with fixed effects. In the special case in which the covariates are exogenous, they themselves can be used as the instruments. We assume the matrix is sparse or approximately sparse, condition likely to be met in a number of scenarios, for example when interaction pools are limited. In this way, we are able to both conduct spatial estimation without pre-specification of the spatial weight matrix and uncover empirical interaction patterns supported exclusively by data. Consistency of the estimators are thoroughly studied when the number of individuals N can be larger than the sample size T , including the partial sign consistency for the IVLASSO estimators. The rate of estimation error is spelt out for the estimation of componentwise spatial fixed effects. We propose and prove a one-step estimator which enjoys the same theoretical properties but can be computed much quicker than the fully iterated estimator. The estimator works well in the simulated experiment. We finally apply the method to study the passover of international oil prices into disaggregated consumer price index taking into account unobserved sectoral linkages.

Key words and phrases. Spatial econometrics; instrumental variables; adaptive LASSO; spatial fixed effects; spatial weight matrices; partial sign consistency.

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1 Introduction

The study of social interactions are of increasing importance in economics. Examples range from learning in classrooms (Angrist and Lang, 2004; Ammermuller and Pischke, 2009), technology adoption (Foster and Rosenzweig, 1995; Conley and Udry, 2010), crime and delinquent behavior (Glaeser et al., 1996) to contagion between financial markets (Longstaff, 2010). However, existing empirical methods assume the knowledge of the interaction matrix prior to estimation. This has been regarded as a challenge for applied research because of limited availability of data regarding such interactions.

In this paper, we propose a method to alleviate this restriction by estimating the elements of the interaction matrices, also called the spatial weight matrices, of a spatial autoregressive model from data. The aims in estimating the spatial weight matrices are twofold. Firstly, we remove the need to specify the spatial weight matrices prior to estimation. It is well known that the quality of the estimates depend heavily on the specification of the spatial weight matrix. See for example, Anselin (2002), Fingleton (2003) or Lee and Yu (2012), among others. Even in nonparametric models, when misspecification of the spatial weight matrices is not leading to bias theoretically, it can lead to reduced efficiency of the estimators, and severely affect the estimates when the sample size is small (Kostov, 2013). Furthermore, Lam and Souza (2014) has provided an error upper bound for the spatial regression parameter estimators in a spatial autoregressive model, showing that misspecification can indeed introduce large bias in the final estimates. We quote a part of the criticism in Arbia and Fingleton (2008), “(...) arbitrary nature of weight matrix (...) are not the results obtained conditional on somewhat arbitrary decisions taken about its structure?”. In this paper, with instruments available, we also provide an upper bound for the bias of the estimators for the parameters of the model in the presence of misspecification of the spatial weight matrices.

Secondly, in many cases the spatial weight matrices are themselves the object of interest. For example, Lam and Souza (2015) introduces sparse estimation of unknown block structures of such a matrix, which is of interest in visualizing voting polarization pattern. Estimating these matrices entirely from data reveals the dependence structures that our assumptions can miss out. In Section 6, we provide an example in which we estimate the contagion of international oil prices on the disaggregated consumer price index in the United States. By using this methodology, we can both identify the most influential sectors and build impulse-response functions to oil price shocks taking into account complex sectoral inter-linkages that are often unobserved.

Few researchers attempted to estimate the spatial weight matrices, including a well known paper by Pinkse et al. (2002). They estimate a nonparametric smooth function $\hat{g}(\cdot)$ assuming normality of data, and the (i, j) -th element of the matrix $\mathbf{W}_1^*$ is estimated as $\hat{g}(d_{ij})$, where d_{ij} is a distance measure specified by the user. Beenstock and Felsenstein (2012) suggested using a moment estimator for the spatial weight matrix. Bhattacharjee and Jensen-Butler (2013) proposes to estimate the spatial weight matrix by first estimating the error covariance matrix. However,

estimating a large error covariance matrix can be inaccurate if the number of individuals is high and can be close to the sample size. Other attempts in estimating spatial weight matrices or providing specification procedures include Stakhovych and Bijmolt (2009), Benjanuvatra (2013) or Chupp (2014), to quote but a few.

Recently, Ahrens and Bhattacharjee (2014) proposes to estimate the spatial weight matrix in a spatial autoregressive model with exogenous instruments by using a two-step LASSO estimation, imitating a two-stage least square estimation but with penalization. Their model assumes $\mathbf{W}_2^* = \mathbf{I}_N$ and $\boldsymbol{\mu}^* = \mathbf{0}$ in our model (2.1) to be proposed in section 2, so that their model does not have a spatial fixed effects to be estimated. While consistency and rate of convergence of the spatial weight matrix and the regression parameters are spelt out and proved also, they do not have a sign consistency result like what we have proved for our one-step estimator in this paper; see Theorem 4 for details. Moreover, their paper assumes that the spatial weight matrix is exactly sparse, which is relaxed in our paper to approximate sparseness; see section 3.2 for the assumption of approximate sparseness and the details therein. Finally, the rate of convergence for the estimator of $\boldsymbol{\beta}^*$ is better in our paper; see the discussions after Theorem 3.

We assume in this paper that the matrices themselves are sparse or approximately sparse – i.e., with limited number of large elements but many other small elements – a notion which will be discussed in details in Section 3.2. Arguably, in real-life scenarios those matrices are only approximately sparse, in the sense that there are few dominating elements in addition to a large number of small elements, representing a more realistic spatial interactions structure. Assuming that the data regarding outcomes and covariates are available for one group over many periods of time, we apply the Adaptive LASSO of Zou (2006) to estimate the elements of the spatial weight matrices. We also allow for endogeneity of the covariates when instrumental variables are available. At the same time, with the availability of instruments, spatial fixed effects are incorporated into our model and can be estimated accurately elementwise. We also define and prove the partial sign-consistency for our spatial weight matrices estimators, where the small and zero spatial interactions are all estimated at zero, with the signs of all large spatial interactions estimated correctly with probability converging to one. Error upper bounds are also provided for all of our estimators. Compared to Lam and Souza (2014), on top of allowing for spatial fixed effects, we lifted the assumption of decaying disturbance variance when instruments there are not available. We also introduce a modified block coordinate descent (BCD) algorithm and a modified BIC for tuning parameter selection. We only need to iterate the modified BCD algorithm once to arrive at our final estimator, which gives the name “once-step” estimator for our spatial weight matrices.

The rest of the paper is organized as follows. In Section 2, we introduce the spatial autoregressive model considered and presents the model in a compact form. We then introduce the minimization problems for obtaining the estimators of the sparse spatial weight matrices. These estimators are analyzed in Section 3 using a relatively new concept of time dependence in time

series data, with rates of convergence spelt out for the errors of our estimators, as well as partial-sign consistency presented. Section 4 discusses the computational issue of our estimators, and presented a modified block coordinate descent algorithm as a solution. Section 5 presents our extensive simulation results and real data analysis. The paper concludes with Section 7, outlining our main contributions and some future research directions.

2 Model and Methodology

We consider the regularization of the spatial lag model with spatial fixed effect,

$$\mathbf{y}_t = \mathbf{W}_1^* \mathbf{y}_t + \mathbf{W}_2^* \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\mu}^* + \boldsymbol{\epsilon}_t, \quad t = 1, \dots, T, \quad (2.1)$$

where $\mathbf{y}_t = (y_{t1}, \dots, y_{tN})^\top$ is an $N \times 1$ vector of dependent time series variables, $\mathbf{W}_j^*$ for $j = 1, 2$ are the $N \times N$ spatial weight matrices to be estimated. The elements in $\mathbf{W}_1^*$ and $\mathbf{W}_2^*$ can be negative. The covariates matrix $\mathbf{X}_t$ has size $N \times K$. The true regression parameter is $\boldsymbol{\beta}^*$. The innovation process $\{\boldsymbol{\epsilon}_t\}$ has mean $\mathbf{0}$ and covariance matrix $\boldsymbol{\Sigma}_\epsilon$. The spatial weight matrix $\mathbf{W}_1^*$ has 0 on the main diagonal while $\mathbf{W}_2^*$ has 1.

The above model generalizes that in Lam and Souza (2014) by incorporating the spatial fixed effects as characterized by $\boldsymbol{\mu}^*$. The introduction of the spatial weight matrix $\mathbf{W}_2^*$, which can be different from $\mathbf{W}_1^*$, allows for flexibility in modeling local spatial spillover effects (Anselin, 2002).

The parameter $\boldsymbol{\mu}^*$ is a constant vector. It is not difficult to see that the spatial fixed effects of the model is then $(\mathbf{I}_N - \mathbf{W}_1^*)^{-1} \boldsymbol{\mu}^*$. For identifiability of such, we assume without loss of generality that $E(\mathbf{X}_t) = \mathbf{0}$. If not, we can write

$$\mathbf{W}_2^* \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\mu}^* = \mathbf{W}_2^* (\mathbf{X}_t - E(\mathbf{X}_t)) \boldsymbol{\beta}^* + (\boldsymbol{\mu}^* + \mathbf{W}_2^* E(\mathbf{X}_t) \boldsymbol{\beta}^*),$$

so that the spatial fixed effects is now captured by $\boldsymbol{\mu}^* + \mathbf{W}_2^* E(\mathbf{X}_t) \boldsymbol{\beta}^*$, and the covariates are of mean $\mathbf{0}$.

Similar to Lam and Souza (2014), we assume $\|\mathbf{W}_2^*\|_\infty < 1$, $\|\mathbf{W}_1^*\|_\infty < \eta < 1$ uniformly as $N \rightarrow \infty$, where the L_∞ -norm is defined as $\|M\|_\infty = \max_i \sum_j |M_{ij}|$ for a matrix M . See Section 3.2 for more detailed assumptions.

Unlike Lam and Souza (2014), for the above model, we assume that $\mathbf{X}_t$ and $\boldsymbol{\epsilon}_t$ are correlated in general. However, we assume that there are instrumental variables $\mathbf{B}_t$ of size $N \times K$ for each t , i.e., for each component of $\mathbf{X}_t$ there is a corresponding instrument in $\mathbf{B}_t$. Furthermore, the instrument $\mathbf{B}_t$ is assumed to be independent of $\boldsymbol{\epsilon}_t$ for each t , but is correlated with $\mathbf{X}_t$. Hence $\mathbf{B}_t$ is a proper instrument for model (2.1), since it only correlates with $\mathbf{y}_t$ and $\mathbf{X}_t$ but not $\boldsymbol{\epsilon}_t$. As a special case, if $\mathbf{X}_t$ is independent of $\boldsymbol{\epsilon}_t$ for each t , i.e., $\mathbf{X}_t$ is exogenous, then clearly $\mathbf{X}_t$ is an instrument to model (2.1), so that we can take $\mathbf{B}_t = \mathbf{X}_t$. See Remark 1 at the end of this section when there are more instruments than covariates.

Rewrite model (2.1) in regression form for the weight matrices:

$$\begin{aligned}\mathbf{y} &= \mathbf{Z}\boldsymbol{\xi}_1^* + \mathbf{X}_{\beta^*}\boldsymbol{\xi}_2^* + \boldsymbol{\mu}^* \otimes \mathbf{1}_T + \boldsymbol{\epsilon} \\ &= \mathbf{M}_{\beta^*}\boldsymbol{\xi}^* + \boldsymbol{\mu}^* \otimes \mathbf{1}_T + \boldsymbol{\epsilon},\end{aligned}\tag{2.2}$$

where $\mathbf{y} = \text{vec}\{(\mathbf{y}_1, \dots, \mathbf{y}_T)^\top\}$, $\mathbf{Z} = \mathbf{I}_N \otimes (\mathbf{y}_1, \dots, \mathbf{y}_T)^\top$, $\mathbf{X}_{\beta^*} = \mathbf{I}_N \otimes \{(\mathbf{I}_T \otimes \boldsymbol{\beta}^{*\top})(\mathbf{X}_1, \dots, \mathbf{X}_T)^\top\}$, $\boldsymbol{\xi}_j^* = \text{vec}(\mathbf{W}_j^{*\top})$ for $j = 1, 2$, and $\boldsymbol{\epsilon} = \text{vec}\{(\boldsymbol{\epsilon}_1, \dots, \boldsymbol{\epsilon}_T)^\top\}$. The notation $\otimes$ represents the Kronecker product, and $\mathbf{1}_m$ is a vector of ones of size m . We also have $\boldsymbol{\xi}^* = (\boldsymbol{\xi}_1^{*\top}, \boldsymbol{\xi}_2^{*\top})^\top$, $\mathbf{M}_{\beta^*} = (\mathbf{Z}, \mathbf{X}_{\beta^*})$.

With respect to model (2.2), define $\bar{\mathbf{B}} = T^{-1} \sum_{t=1}^T \mathbf{B}_t$, and the instrument

$$\mathbf{B} = T^{-1/2} N^{-a/2} (\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma) = T^{-1/2} N^{-a/2} \mathbf{I}_N \otimes \{(\mathbf{I}_T \otimes \boldsymbol{\gamma}^\top)(\mathbf{B}_1 - \bar{\mathbf{B}}, \dots, \mathbf{B}_T - \bar{\mathbf{B}})^\top\}, \tag{2.3}$$

where $\boldsymbol{\gamma} = K^{-1} \mathbf{1}_K$, so that $\|\boldsymbol{\gamma}\|_1 = 1$. For $\mathbf{B}$ to be non-zero it suffices that $\mathbf{B}_t$ varies over time, such that $\mathbf{B}_t \neq \bar{\mathbf{B}}$, for at least one $t \in \{1, \dots, T\}$. The constant $a \in [0, 1]$ is defined in Assumptions R1 and R2 in Section 3.2, and in general a larger a means the instruments are correlated with more covariates. In practice, since we do not know a , we compute $\mathbf{B}$ by dividing by $N^{1/2}$ and adjust the value of the tuning parameter γ_T in (2.5) and γ'_T in (2.6). See Section 4 for practical details. The augmented model for our problem is then

$$\mathbf{B}^\top \mathbf{y} = \mathbf{B}^\top \mathbf{M}_{\beta^*} \boldsymbol{\xi}^* + \mathbf{B}^\top \boldsymbol{\epsilon}. \tag{2.4}$$

The spatial fixed effects are gone since it is easy to show that $\mathbf{B}^\top (\boldsymbol{\mu}^* \otimes \mathbf{1}_T) = \mathbf{0}$. This model corresponds to modeling the transformed data

$$\tilde{\mathbf{y}}_j = T^{-1/2} N^{-a/2} \sum_{t=1}^T \boldsymbol{\gamma}^\top (\mathbf{b}_{t,j} - \bar{\mathbf{b}}_j) \mathbf{y}_t, \quad j = 1, \dots, N,$$

where $\mathbf{b}_{t,j}^\top$ is the j -th row of $\mathbf{B}_t$, and $\bar{\mathbf{b}}_j$ is the sample mean of $\{\mathbf{b}_{t,j}\}_{1 \leq t \leq T}$.

We are now in position to introduce the regularized estimation for $\boldsymbol{\xi}^*$, $\boldsymbol{\beta}^*$ and $\boldsymbol{\mu}^*$. Since $\mathbf{W}_1^*$ and $\mathbf{W}_2^*$ are assumed to be sparse or approximately sparse (see Section 3 for the definition of approximate sparseness), and model (2.4) is in a regression form with response $\mathbf{B}^\top \mathbf{y}$ and covariates $\mathbf{B}^\top \mathbf{M}_{\beta^*}$, we suggest the LASSO penalization

$$\begin{aligned}\tilde{\boldsymbol{\xi}} &= \arg \min_{\boldsymbol{\xi}} \frac{1}{2T} \|\mathbf{B}^\top \mathbf{y} - \mathbf{B}^\top \mathbf{M}_{\beta(\boldsymbol{\xi})} \boldsymbol{\xi}\|^2 + \gamma_T \|\boldsymbol{\xi}\|_1, \quad \text{subj. to } \sum_{j \neq i} |w_{1,ij}|, \sum_{j \neq i} |w_{2,ij}| < 1, \text{ with} \\ \beta(\boldsymbol{\xi}) &= \left\{ \mathbf{X}^\top \mathbf{W}_2^{\otimes \top} \mathbf{W}_2^{\otimes} \mathbf{B}^v \mathbf{B}^{v\top} \mathbf{W}_2^{\otimes \top} \mathbf{W}_2^{\otimes} \mathbf{X} \right\}^{-1} \mathbf{X}^\top \mathbf{W}_2^{\otimes \top} \mathbf{W}_2^{\otimes} \mathbf{B}^v \mathbf{B}^{v\top} \mathbf{W}_2^{\otimes \top} (\mathbf{I}_{TN} - \mathbf{W}_1^{\otimes}) \mathbf{y}^v, \end{aligned} \tag{2.5}$$

where $\mathbf{B}^v = ((\mathbf{B}_1 - \bar{\mathbf{B}})^\top, \dots, (\mathbf{B}_T - \bar{\mathbf{B}})^\top)^\top$, $\mathbf{X} = (\mathbf{X}_1^\top, \dots, \mathbf{X}_T^\top)^\top$, $\mathbf{y}^v = (\mathbf{y}_1^\top, \dots, \mathbf{y}_T^\top)^\top$ and for any matrix M , we have $M^{\otimes} = \mathbf{I}_T \otimes M$. The notation $\|\cdot\|_1$ represents the L_1 -norm, and the definition of $\mathbf{M}_{\beta}$, $\boldsymbol{\xi}$ and $\boldsymbol{\beta}$ are parallel to those in model (2.2). The estimator for $\boldsymbol{\beta}^*$ is then $\tilde{\boldsymbol{\beta}} = \beta(\tilde{\boldsymbol{\xi}})$. See

Remark 3 at the end of this section for the reason we choose to optimize (2.5) rather than to use the least square estimator for estimating β^* .

The tuning parameter γ_T will be specified in Section 3.2, when we introduce various important rates in our main assumptions in this paper. Since the L_1 penalty has implicit bias issues (Fan and Li, 2001) with restrictive requirements¹ for asymptotic sign consistency of the estimators, we use the Adaptive LASSO proposed by Zou (2006):

$$\hat{\xi} = \arg \min_{\xi} \frac{1}{2T} \|\mathbf{B}^T \mathbf{y} - \mathbf{B}^T \mathbf{M}_{\beta(\xi)} \xi\|^2 + \gamma'_T \mathbf{v}^T |\xi|, \quad \text{subj. to } \sum_{j \neq i} |w_{1,ij}|, \sum_{j \neq i} |w_{2,ij}| < 1, \quad (2.6)$$

where $|\xi| = (|\xi_1|, \dots, |\xi_{2N^2}|)^T$ and $\mathbf{v} = (|\tilde{\xi}_1|^{-1}, \dots, |\tilde{\xi}_{2N^2}|^{-1})^T$. The function $\beta(\xi)$ is as defined in (2.5). Note that $\mathbf{M}_{\beta(\xi)}$ depends on the unknown ξ , and hence this is not the standard Adaptive LASSO problem. The tuning parameter γ'_T can be the same as γ_T when the true spatial weight matrices $\mathbf{W}_1^*$ and $\mathbf{W}_2^*$ are sparse. However, if they are approximately sparse as defined in Section 3.2, we may need $\gamma_T = o(\gamma'_T)$, depending on how large and how many non-zero elements are there in the matrices. Exactly how γ'_T is different from γ_T is spelt out in Assumption R6 in Section 3.2.

The above non-standard adaptive LASSO estimators is not convex in the parameters since β is a nonlinear function of ξ . Instead, in this paper we are investigate the properties of a one-step estimator $\hat{\xi}_{\text{OS}}$ for ξ^* , obtained by solving the following adaptive LASSO problem

$$\hat{\xi}_{\text{OS}} = \arg \min_{\xi} \frac{1}{2T} \|\mathbf{B}^T \mathbf{y} - \mathbf{B}^T \mathbf{M}_{\beta(\tilde{\xi})} \xi\|^2 + \gamma'_T \mathbf{v}^T |\xi|, \quad \text{subj. to } \sum_{j \neq i} |w_{1,ij}|, \sum_{j \neq i} |w_{2,ij}| < 1. \quad (2.7)$$

This is indeed an appropriate adaptive LASSO problem since $\mathbf{M}_{\beta(\tilde{\xi})}$ is known. This problem is much simpler to solve, with only one step of the modified block coordinate descent algorithm required. In section 3, we present some desirable properties of this estimator. To find the LASSO solution, we propose an iterative block coordinate descent algorithm, to be introduced in section 4. Simulation results for this estimator will also be presented in section 6.

The estimator of μ^* is $\hat{\mu} = (\mathbf{I}_N - \widehat{\mathbf{W}}_1) \bar{\mathbf{y}}$, where $\widehat{\mathbf{W}}$ is obtained by reconstruction from the one-step estimator $\hat{\xi}_{\text{OS}}$. However, if $E(\mathbf{X}_t) \neq \mathbf{0}$ initially, this estimator does not reflect the true spatial fixed effects. Hence, we use instead the estimator

$$\hat{\mu} = (\mathbf{I}_N - \widehat{\mathbf{W}}_1) \bar{\mathbf{y}} - \widehat{\mathbf{W}}_2 \bar{\mathbf{X}} \tilde{\beta}, \quad (2.8)$$

so that the estimated spatial fixed effects is given by

$$(\mathbf{I}_N - \widehat{\mathbf{W}}_1)^{-1} \hat{\mu} = \bar{\mathbf{y}} - (\mathbf{I}_N - \widehat{\mathbf{W}}_1)^{-1} \widehat{\mathbf{W}}_2 \bar{\mathbf{X}} \tilde{\beta}. \quad (2.9)$$

¹See e.g. Zhao and Yu (2006) in the case of classical linear model. It is difficult to show that the Irrepresentable Condition therein holds when spatial dependence is considered, as in the current paper.

Remark 1. If there are $L > K$ columns of instruments for each component of $\mathbf{y}_t$, i.e. we have instruments $\mathbf{B}_t$ of size $N \times L$, we can still define $\mathbf{B}$ as in (2.3), but now with $\boldsymbol{\gamma}$ a constant vector of size L . Practical experience shows that defining $\boldsymbol{\gamma} = L^{-1}\mathbf{1}_L$ works very well.

Remark 2. If $\mathbf{W}_2^* = \mathbf{I} - \mathbf{W}_1^*$, then model (2.1) is reduced to

$$\mathbf{y}_t = \mathbf{X}_t\boldsymbol{\beta}^* + (\mathbf{I} - \mathbf{W}_1^*)^{-1}\boldsymbol{\mu}^* + (\mathbf{I} - \mathbf{W}_1^*)^{-1}\boldsymbol{\epsilon}_t,$$

which is a spatial error model with spatial fixed effect. Hence model (2.1) is more general than a spatial error model and the traditional spatial lag model.

Remark 3. The estimator $\tilde{\boldsymbol{\beta}} = \boldsymbol{\beta}(\tilde{\boldsymbol{\xi}})$ from (2.5) is not a least square-type estimator. In fact, for least square-type estimator, the problem can be posed as

$$(\tilde{\boldsymbol{\xi}}, \tilde{\boldsymbol{\beta}}) = \arg \min_{\boldsymbol{\xi}, \boldsymbol{\beta}} \frac{1}{2T} \|\mathbf{B}^T \mathbf{y} - \mathbf{B}^T \mathbf{M}_{\boldsymbol{\beta}} \boldsymbol{\xi}\|^2 + \gamma_T \|\boldsymbol{\xi}\|_1, \quad \text{subj. to } \sum_{j \neq i} |w_{1,ij}|, \sum_{j \neq i} |w_{2,ij}| < 1. \quad (2.10)$$

Reconstructing $\widetilde{\mathbf{W}}_1$ and $\widetilde{\mathbf{W}}_2$ using $\tilde{\boldsymbol{\xi}}_1$ and $\tilde{\boldsymbol{\xi}}_2$ respectively, the least square-type estimator $\tilde{\boldsymbol{\beta}}$ is

$$\tilde{\boldsymbol{\beta}} = \left[\mathbf{X}^T \left\{ \mathbf{C}_{\boldsymbol{\gamma}}^T \mathbf{C}_{\boldsymbol{\gamma}} \otimes \widetilde{\mathbf{W}}_2^T \widetilde{\mathbf{W}}_2 \right\} \mathbf{X} \right]^{-1} \mathbf{X}^T \left\{ \mathbf{C}_{\boldsymbol{\gamma}}^T \mathbf{C}_{\boldsymbol{\gamma}} \otimes \widetilde{\mathbf{W}}_2^T \right\} (\mathbf{I}_{TN} - \widetilde{\mathbf{W}}_1^{\otimes}) \mathbf{y}^v, \quad (2.11)$$

where $\mathbf{C}_{\boldsymbol{\gamma}} = ((\mathbf{B}_1 - \bar{\mathbf{B}})\boldsymbol{\gamma}, \dots, (\mathbf{B}_T - \bar{\mathbf{B}})\boldsymbol{\gamma})$. Not shown in this paper, the empirical performance of the estimators from (2.5) and (2.10) are extensively compared, and the results are very similar in all cases. We choose (2.5) over (2.10) in this paper due to easier analysis for the theoretical properties of (2.5).

3 Properties of the Estimators

We first present the notations and assumptions for the time dependence of the time series variables in our model in Section 3.1. We introduce approximate sparse spatial weight matrices used in this paper, together with other technical assumptions in Section 3.2. Main theoretical results are presented in Section 3.3. All the proofs of theorems are relegated to the Appendix.

3.1 Time dependence of time series variables

Denote $\{\mathbf{b}_t\} = \{\text{vec}(\mathbf{B}_t)\}$ and $\{\mathbf{x}_t\} = \{\text{vec}(\mathbf{X}_t)\}$ the vectorized processes for the instrumental variables and the covariates respectively. For each N and $t = 1, \dots, T$, the $NK \times 1$ vectors $\mathbf{b}_t$, $\mathbf{x}_t$ and the noise vector $\boldsymbol{\epsilon}_t$ are defined by

$$\mathbf{x}_t = [f_j(\mathcal{F}_t)]_{1 \leq j \leq NK}, \quad \mathbf{b}_t = [g_j(\mathcal{G}_t)]_{1 \leq j \leq NK}, \quad \boldsymbol{\epsilon}_t = [h_j(\mathcal{H}_t)]_{1 \leq j \leq NK}, \quad (3.1)$$

where the $f_j(\cdot)$'s, $g_j(\cdot)$'s and $h_j(\cdot)$'s are measurable functions defined on the real line. The shift processes $\mathcal{F}_t = (\dots, \mathbf{e}_{x,t-1}, \mathbf{e}_{x,t})$, $\mathcal{G}_t = (\dots, \mathbf{e}_{b,t-1}, \mathbf{e}_{b,t})$ and $\mathcal{H}_t = (\dots, \mathbf{e}_{\epsilon,t-1}, \mathbf{e}_{\epsilon,t})$ are defined by

independent and identically distributed processes $\{\mathbf{e}_{x,t}\}$, $\{\mathbf{e}_{b,t}\}$ and $\{\mathbf{e}_{\epsilon,t}\}$ respectively, with $\{\mathbf{e}_{b,t}\}$ independent of $\{\mathbf{e}_{\epsilon,t}\}$ but correlated with $\{\mathbf{e}_{x,t}\}$. This makes $\{\mathbf{B}_t\}$ proper instruments since it is independent of $\{\mathbf{\epsilon}_t\}$ but correlated with the covariates $\{\mathbf{X}_t\}$.

For measuring time dependence, instead of using traditional measures like mixing conditions for time series, we use the functional dependence measure introduced in Wu (2005). Similar to Lam and Souza (2014), this measure allows us to use a Nagaev-type inequality for obtaining the probability lower bound for our estimators to enjoy “nice” properties to be presented in Section 3.3. For the time series $\{\mathbf{x}_t\}$, $\{\mathbf{b}_t\}$ and $\{\mathbf{\epsilon}_t\}$ in (3.1), define for $d > 0$,

$$\begin{aligned}\theta_{t,d,j}^x &= \|x_{tj} - x'_{tj}\|_d = (E|x_{tj} - x'_{tj}|^d)^{1/d}, \\ \theta_{t,d,j}^b &= \|b_{tj} - b'_{tj}\|_d = (E|b_{tj} - b'_{tj}|^d)^{1/d}, \\ \theta_{t,d,\ell}^\epsilon &= \|\epsilon_{t\ell} - \epsilon'_{t\ell}\|_d = (E|\epsilon_{t\ell} - \epsilon'_{t\ell}|^d)^{1/d},\end{aligned}\tag{3.2}$$

where $j = 1, \dots, NK$, $\ell = 1, \dots, N$, and $x'_{tj} = f_j(\mathcal{F}'_t)$, $\mathcal{F}'_t = (\dots, \mathbf{e}_{x,-1}, \mathbf{e}'_{x,0}, \mathbf{e}_{x,1}, \dots, \mathbf{e}_{x,t})$, with $\mathbf{e}'_{x,0}$ independent of all other $\mathbf{e}_{x,j}$'s. Hence x'_{tj} is a coupled version of x_{tj} with $\mathbf{e}_{x,0}$ replaced by an i.i.d. copy $\mathbf{e}'_{x,0}$. Finally, we have similar definitions for b'_{tj} and $\epsilon'_{t\ell}$. Such a definition of “physical” or functional dependence of time series on past “inputs” is used in various papers, for example in Shao (2010) and Zhou (2010).

There are no direct relationships between the usual mixing conditions and this “physical” functional dependence measure. But this measure is easier to handle mathematically and leads to simpler and stronger proofs in our paper, through the Nagaev-type inequality in Lemma 1. Moreover, many well-known processes are not strong mixing, yet can be handled by using the dependence measure (3.2), like the Bernoulli shift process in Andrews (1984).

3.2 Main assumptions, notations, and partial sign consistency

There are four main categories of assumptions. They concern the spatial weight matrices, the distributional behavior of the time series processes involved, the time dependence of these processes, and the explicit rates of convergence for various quantities defined in the assumptions. We also present the concept of partial sign consistency of an estimator after Assumption T1.

Assumptions on the spatial weight matrices:

- M1. There exists a constant $\eta > 0$ such that $\|\mathbf{W}_1^*\|_\infty < \eta < 1$ and $\|\mathbf{W}_2^*\|_\infty - 1 < \eta < 1$ uniformly as $N \rightarrow \infty$. The elements in the spatial weight matrices can be negative.
- M2. (Approximate sparseness) There exists a constant $\tau > 0$ such that the elements $w_{1,ij}^*$ and $w_{2,ij}^*$ of the spatial weight matrices are constants as $N \rightarrow \infty$ whenever they are larger than or equal to τ in magnitude. For those elements smaller than τ , we have $w_{1,ij}^*, w_{2,ij}^* \rightarrow 0$ as $N \rightarrow \infty$. The rates in which the individual or the sum of these “small” elements satisfy will be explicitly spelt out in Assumptions R5 - R6 below. ■

Assumptions on the distributional behavior of time series variables:

- D1. The processes $\{\mathbf{B}_t\}$, $\{\mathbf{X}_t\}$ and $\{\epsilon_t\}$ are as defined in Section 3.1 and are second-order stationary, with $\{\mathbf{X}_t\}$ and $\{\epsilon_t\}$ having zero means. The instruments $\{\mathbf{B}_t\}$ are independent of the noise $\{\epsilon_t\}$.
- D2. Write $\epsilon_t = \Sigma_\epsilon^{1/2} \epsilon_t^*$, where Σ_ϵ is the covariance matrix for ϵ_t . Then the elements in Σ_ϵ are all less than $\sigma_{\max}^2$ uniformly as $N \rightarrow \infty$.
We also assume $\|\Sigma_\epsilon^{1/2}\|_\infty \leq S_\epsilon < \infty$ uniformly as $N \rightarrow \infty$, with $\{\epsilon_{t,j}^*\}_{1 \leq j \leq N}$ being a martingale difference with respect to the filtration generated by $\sigma(\epsilon_{t,1}^*, \dots, \epsilon_{t,j}^*)$.
- D3. The tail condition $P(|Z| > v) \leq D_1 \exp(-D_2 v^q)$ is satisfied for $B_{t,jk}, X_{t,jk}, \epsilon_{t,j}$ and $\epsilon_{t,j}^*$ by the same constants D_1, D_2 and q .
- D4. All the eigenvalues of $E(\mathbf{b}_t \mathbf{x}_t^T)$ are uniformly bounded away from 0 and infinity, where $\mathbf{b}_t$ and $\mathbf{x}_t$ are defined in (3.1). ■

Assumptions on the time dependence of time series variables:

T1. Define

$$\Theta_{m,a}^x = \sum_{t=m}^{\infty} \max_{1 \leq j \leq NK} \theta_{t,a,j}^x, \quad \Theta_{m,a}^b = \sum_{t=m}^{\infty} \max_{1 \leq j \leq NK} \theta_{t,a,j}^b, \quad \Theta_{m,a}^\epsilon = \sum_{t=m}^{\infty} \max_{1 \leq j \leq NK} \theta_{t,a,j}^\epsilon,$$

where $\theta_{t,a,j}^x, \theta_{t,a,j}^b$ and $\theta_{t,a,j}^\epsilon$ are defined in (3.2). The we assume that for some $w > 2$, $\Theta_{m,2w}^x, \Theta_{m,2w}^b, \Theta_{m,2w}^\epsilon \leq C m^{-\alpha}$ with $\alpha, C > 0$ being constants that can depend on w . The assumption also holds for $\{\epsilon_t^*\}_{1 \leq t \leq T}$. ■

To present the explicit rates assumed in this paper and the concept of partial sign consistency, we need some definitions. With respect to assumption M2, using the notations in (2.4), define

$$\begin{aligned} J_0 &= \{j : \xi_j^* = 0 \text{ and not corresponds to the diagonal of } \mathbf{W}_1^*\}, \\ J_1 &= \{j : |\xi_j^*| \geq \tau \text{ and not corresponds to the diagonal of } \mathbf{W}_2^*\}, \\ J_2 &= \{j : 0 < |\xi_j^*| < \tau \text{ and not corresponds to the diagonal of } \mathbf{W}_2^*\}. \end{aligned} \tag{3.3}$$

Essentially, J_0 is the index set for all the off-diagonal zeros in the matrices $\mathbf{W}_1^*$ and $\mathbf{W}_2^*$, J_2 is the index set for all the off-diagonal elements in $\mathbf{W}_1^*$ and $\mathbf{W}_2^*$ which are greater than the constant τ , and finally J_2 is the index set for all the non-zero off-diagonal elements of $\mathbf{W}_1^*$ and $\mathbf{W}_2^*$ which are smaller than τ . The idea is that if the spatial weight matrices $\mathbf{W}_1^*$ and $\mathbf{W}_2^*$ are approximately sparse, in the sense that there is a constant τ such that the absolute sum of elements indexed by J_2 is somehow “small”, then a partially sign consistent estimator should get the sign right for all the “large” elements in J_1 , as well as estimating all other elements in J_0 and J_2 to be 0. If J_2 is empty, such an estimator is exactly sign-consistent. A partially sign consistent estimator estimate

the “small” and zero elements to be exactly zero. See Theorem 4 for presenting the partial sign consistency of our one-step IVLASSO estimator.

We assume in M2 that τ and those non-zero elements larger than τ are constants. However, this assumption can be relaxed, so that τ and these “large” spatial weights decay slowly as $N \rightarrow \infty$, at the expense of longer proofs and more involved rates of convergence. We choose to assume this simply for the more presentable rates of convergence and more intuitive proofs.

We use the notation $\mathbf{A}_{IJ}$ to denote a matrix $\mathbf{A}$ with rows restricted to the index set I and columns to the index set J . Similarly we use $\mathbf{v}_J$ to denote a vector $\mathbf{v}$ with elements restricted to those in index J . We use $\|\mathbf{A}\|_{\max} = \max_{i,j} |A_{ij}|$ for a matrix $\mathbf{A}$. Finally, we use $\lambda_{\min}(\mathbf{A})$ and $\lambda_{\max}(\mathbf{A})$ to denote the minimum and maximum eigenvalues of matrix $\mathbf{A}$ respectively.

Assumptions on the explicit rates for various quantities:

R1. Denote $\mathbf{M}^* = \mathbf{M}_{\beta^*} = (\mathbf{Z}, \mathbf{X}_{\beta^*})$. Define the matrix, for $a \in [0, 1]$,

$$\mathbf{G} = N^{-a} E(T^{-1} \mathbf{M}^{*\top} (\mathbf{B}_{\gamma} - \bar{\mathbf{B}}_{\gamma})) E(T^{-1} (\mathbf{B}_{\gamma} - \bar{\mathbf{B}}_{\gamma})^{\top} \mathbf{M}^*).$$

Then there exists a constant $u > 0$ such that $\lambda_{\min}(\mathbf{G}_{J_1 J_1}) \geq u > 0$ uniformly as $N \rightarrow \infty$.

Furthermore, the number of elements belonging to J_1 in each row of $\mathbf{W}_1^*$ and $\mathbf{W}_2^*$ is bounded uniformly away from infinity.

R2. For the same constant a in Assumption R1, we have for each N ,

$$\max_{1 \leq i \leq N} \sum_{j=1}^N \|E(\mathbf{b}_{t,i} \mathbf{x}_{t,j}^{\top})\|_{\max}, \quad \max_{1 \leq j \leq N} \sum_{i=1}^N \|E(\mathbf{b}_{t,i} \mathbf{x}_{t,j}^{\top})\|_{\max} \leq C_{bx} N^a,$$

where $C_{bx} > 0$ is a constant and $\mathbf{b}_{t,i}$, $\mathbf{x}_{t,j}$ are the column vectors for the i th row of $\mathbf{B}_t$ and j th row of $\mathbf{X}_t$ respectively.

R3. (Restricted eigenvalue condition) Define

$$\kappa_{N,T}(r) = \min \left\{ \frac{\|\mathbf{G}^{1/2} \boldsymbol{\alpha}\|}{\|\boldsymbol{\alpha}_R\|}, \frac{\|\mathbf{G}^{1/2} \boldsymbol{\alpha}\|}{\|\boldsymbol{\alpha}_{R^c}\|} : |R| \leq r, \boldsymbol{\alpha} \in \mathbb{R}^{2N^2} / \{0\}, \|\boldsymbol{\alpha}_{R^c}\|_1 \leq 3\|\boldsymbol{\alpha}_R\|_1 + h_{N,T} \right\},$$

where $h_{N,T} = 4\|\boldsymbol{\xi}_{J_2}^*\|_1 + \lambda_T(s_2 + N^{\frac{1}{2} + \frac{1}{2w}} + N^{1/2} \log^{1/2}(T \vee N))$, with $s_2 = \|\boldsymbol{\xi}_{2,J_2}^*\|_1$ and $\lambda_T = cT^{-1/2} \log^{1/2}(T \vee N)$ for some constant $c > 0$. Then we assume

$$\kappa_{N,T}(n)/\kappa(n) \rightarrow 1,$$

with $\kappa(n)$ uniformly bounded away from 0 as $N, T \rightarrow \infty$, where $n = |J_1|$.

R4. The tuning parameter γ_T for (2.5) is such that $\gamma_T = C\lambda_T$ for some constant $C > 0$.

R5. The tuning parameter γ'_T for (2.6) is $\gamma'_T = \gamma_T$ if $\|\boldsymbol{\xi}_{J_2}^*\|_1 = O(\lambda_T)$. If $\lambda_T = o(\|\boldsymbol{\xi}_{J_2}^*\|_1)$, then $\gamma'_T = o(1)$ and satisfies

$$\|\boldsymbol{\xi}_{J_2}^*\|_1 = o\left(\frac{\gamma'_T}{\|\boldsymbol{\xi}_{J_2}^*\|_{\max}}, \frac{\gamma'_T}{\lambda_T n^{1/2}}, \frac{\gamma_T^{1/2}}{\lambda_T^{1/4} + N^{-1/4}}, \frac{\gamma'_T}{(\lambda_T^{1/2} + N^{-1/2})q_{N,T}}, \frac{\gamma_T^{4/5}}{\lambda_T^{3/5} n^{1/5}}, \frac{\gamma'_T}{\lambda_T^{3/4} n^{1/4} q_{N,T}^{1/4}}\right),$$

where $q_{N,T} = \lambda_T(s_2 + N^{\frac{1}{2} + \frac{1}{2w}} + N^{1/2} \log^{1/2}(T \vee N))$.

R6. In all the assumptions above, we assume that as $N, T \rightarrow \infty$,

$$\lambda_T n, \quad n N^{-1}, \quad \lambda_T h_{N,T}, \quad (\lambda_T + N^{-1}) h_{N,T}^2, \quad \|\boldsymbol{\xi}_{J_2}^*\|_{\max} = o(1),$$

while we have

$$\lambda_T(N^{1-a} + N^{1/2w} + s_2 N^{-1/2}) = O(1). \quad \blacksquare$$

Assumption M1 ensures that model (2.1) has a reduced form

$$\mathbf{y}_t = \boldsymbol{\Pi}_1^* \boldsymbol{\mu}^* + \boldsymbol{\Pi}_1^* \mathbf{W}_2^* \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\Pi}_1^* \boldsymbol{\epsilon}_t, \quad \boldsymbol{\Pi}_1^* = (\mathbf{I}_N - \mathbf{W}_1^*)^{-1}, \quad (3.4)$$

where the innovations $\boldsymbol{\Pi}_1^* \boldsymbol{\epsilon}_t$ have finite variances. See for example Corrado and Fingleton (2011) for a similar row sum condition for the spatial weight matrices in a slightly different spatial model specification. Assumption M2 is a relaxation to the need for straight sparseness of the matrices $\mathbf{W}_1^*$ and $\mathbf{W}_2^*$. Apart from several relatively large elements (defined in J_1), there can also be a lot of small non-zero elements (defined in J_2) in these matrices. How large these elements are allowed is spelt out in Assumptions R5 and R6. Assumption D1 defines $\{\mathbf{B}_t\}$ to be the proper instruments of model (2.1). The zero-mean assumption for $\{\mathbf{X}_t\}$ is for convenience of proofs only. All the theoretical results remain the same for $\{\mathbf{X}_t\}$ having non-zero mean, with only simple adjustments needed to the proofs. The uniform boundedness of the elements in $\boldsymbol{\Sigma}_\epsilon$ in Assumption D2 is implied by the tail Assumption D3, and is assumed for convenience only. The settings in (3.1) allows us to assume that $\{\boldsymbol{\epsilon}_{t,j}^*\}_{1 \leq j \leq N}$ to be a martingale difference, which is weaker than assuming that as an independent sequence. Assumption D3 is a relaxation to the normality assumption, while D4 is needed for the convergence of $\tilde{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\beta}}$ to $\boldsymbol{\beta}^*$.

Assumption T1 is the major assumption of time dependence of the processes $\{\mathbf{B}_t\}$, $\{\mathbf{X}_t\}$ and $\{\boldsymbol{\epsilon}_t\}$. Together with D3, they allow for the application of a Nagaev-type inequality in Lemma 1 for our results. Stationary Markov Chains and stationary linear processes are examples of time series that satisfy T1. See Chen et al. (2013).

Assumptions R1 and R2 indicate how strong are the correlations between the instruments in $\{\mathbf{B}_t\}$ and the covariates in $\{\mathbf{X}_t\}$. If $\mathbf{b}_{t,i}$ and $\mathbf{x}_{t,j}$ are correlated even when $|i - j|$ is large, then the constant a in Assumptions R1 and R2 is large, and vice versa. Assumption R3 is needed for establishing the upper bounds on the L_2 estimation error for the spatial weight matrices. Similar

restricted eigenvalue condition is used in Ahrens and Bhattacharjee (2014), for example. Assumptions R4 and R5 explicitly spelt out the rates required for the tuning parameters in problems (2.5) and (2.6). If the spatial weight matrices are exactly sparse, then $\|\boldsymbol{\xi}_{J_2}^*\|_1 = 0$, and hence we can use $\gamma_T = \gamma'_T$. Otherwise, we need γ'_T to satisfy Assumption R5 in general for partial sign consistency of the one-step estimator $\widehat{\boldsymbol{\xi}}_{\text{OS}}$. See Section 4 for the BIC criteria we use in practice for finding γ_T and γ'_T . Assumption R6 states the explicit rates for all theorems to hold in Section 3.3.

The assumptions above allow $\mathbf{X}_t$ to contain some past values of $\mathbf{y}_t$. For instance, if $\mathbf{X}_t = (\mathbf{y}_{t-1}, \dots, \mathbf{y}_{t-d}, \mathbf{z}_t)$ where $\mathbf{z}_t$ contains covariates other than $\{\mathbf{y}_t\}$, then

$$\mathbf{W}_2^* \mathbf{X}_t \boldsymbol{\beta}^* = \sum_{j=1}^d \beta_j^* \mathbf{W}_2^* \mathbf{y}_{t-j} + \mathbf{W}_2^* \mathbf{z}_t \boldsymbol{\beta}_2^*,$$

where $\boldsymbol{\beta}^* = (\beta_1^*, \dots, \beta_d^*, \boldsymbol{\beta}_2^{*\text{T}})^\text{T}$. Hence model (2.1) becomes

$$\mathbf{y}_t = \boldsymbol{\mu}^* + \mathbf{W}_1^* \mathbf{y}_t + \sum_{j=1}^d \beta_j^* \mathbf{W}_2^* \mathbf{y}_{t-j} + \mathbf{W}_2^* \mathbf{z}_t \boldsymbol{\beta}_2^* + \boldsymbol{\epsilon}_t.$$

Denote $\boldsymbol{\Pi}^* = (\mathbf{I}_N - \mathbf{W}_1^*)^{-1}$, the reduced form model is then

$$\mathbf{y}_t = \boldsymbol{\Pi}^* \boldsymbol{\mu}^* + \sum_{j=1}^d \beta_j^* \boldsymbol{\Pi}^* \mathbf{W}_2^* \mathbf{y}_{t-j} + \boldsymbol{\Pi}^* (\mathbf{W}_2^* \mathbf{z}_t \boldsymbol{\beta}_2^* + \boldsymbol{\epsilon}_t),$$

which has a vector autoregressive part with coefficient matrices $\beta_j^* \boldsymbol{\Pi}^* \mathbf{W}_2^*$. In fact, our real data example in Section 6 employs such a model to study the heterogeneity of price response in different sectors of the economy to unexpected macroeconomic shocks.

Remark 4. We can use another constant $b \in [0, 1]$ to replace a in Assumption R2 for more general results, since the constant a that makes R1 hold does not necessarily make R2 hold in general. All proofs can adapt to this change at the expense of longer proofs and more complicated rates.

3.3 Main results

Denote $\mu_{b,ij} = E(B_{t,ij})$ and $\mu_{b,\max} = \max_{i,j} |\mu_{b,ij}|$, where $B_{t,ij}$ is the (i,j) entry of $\mathbf{B}_t$, and similar definition for $X_{t,ij}$. Define the sets

$$\begin{aligned}
\mathcal{A}_1 &= \left\{ \max_{1 \leq i, k \leq N} \max_{1 \leq j, \ell \leq K} \left| \frac{1}{T} \sum_{t=1}^T [B_{t,ij} X_{t,k\ell} - E(B_{t,ij} X_{t,k\ell})] \right| < \lambda_T \right\}, \\
\mathcal{A}_2 &= \left\{ \max_{1 \leq i, k \leq N} \max_{1 \leq j \leq K} \left| \frac{1}{T} \sum_{t=1}^T B_{t,ij} \epsilon_{t,k} \right| < \lambda_T \right\}, \\
\mathcal{A}_3 &= \left\{ \max_{1 \leq k \leq K} \left| \frac{1}{T} \sum_{t=1}^T \sum_{s=1}^N B_{t,sk} \epsilon_{t,s} \right| < \lambda_T N^{\frac{1}{2} + \frac{1}{2w}} \right\}, \\
\mathcal{A}_4 &= \left\{ \max_{1 \leq i \leq N} \max_{1 \leq j \leq K} |\bar{B}_{\cdot,ij} - \mu_{b,ij}| < \lambda_T \right\}, \\
\mathcal{A}_5 &= \left\{ \max_{1 \leq j \leq N} |\bar{\epsilon}_{\cdot,j}| < \lambda_T \right\}, \\
\mathcal{A}_6 &= \left\{ \max_{1 \leq i \leq N} \max_{1 \leq j \leq K} |\bar{X}_{\cdot,ij}| < \lambda_T \right\}, \\
\mathcal{A}_7 &= \left\{ \max_{1 \leq k \leq K} \left| \sum_{s=1}^N \bar{B}_{\cdot,sk} \bar{\epsilon}_{\cdot,s} \right| < 2^{1/2} \lambda_T N^{1/2} \log^{1/2}(T \vee N) S_\epsilon(\mu_{b,\max} + \lambda_T) \right\},
\end{aligned} \tag{3.5}$$

where $\bar{B}_{\cdot,sk} = T^{-1} \sum_{t=1}^T B_{t,sk}$, $\bar{X}_{\cdot,ij} = T^{-1} \sum_{t=1}^T X_{t,ij}$ and $\bar{\epsilon}_{\cdot,s} = T^{-1} \sum_{t=1}^T \epsilon_{t,s}$. Our theorems presented below will be describing properties of estimators on the set $\mathcal{M} = \bigcap_{i=1}^7 \mathcal{A}_i$. It turns out that $P(\mathcal{M}) \rightarrow 1$ as $T, N \rightarrow \infty$, as shown in the following theorem.

Theorem 1 *Let all the assumptions in Section 3.2 hold. Suppose $\alpha > 1/2 - 1/w$ in assumption T1, and suppose for the applications of the Nagaev-type inequality in Lemma 1 for the processes in $\mathcal{A}_1$ to $\mathcal{A}_6$, the constants C_1 , C_2 and C_3 are the same. Then with $c \geq \sqrt{3/C_3}$ where c is the constant defined in $\lambda_T = cT^{-1/2} \log^{1/2}(T \vee N)$, we have*

$$P(\mathcal{M}) \geq 1 - 8C_1 K^2 \left(\frac{C_3}{3} \right)^{w/2} \frac{N^2}{T^{w/2-1} \log^{w/2}(T \vee N)} - \frac{8C_2 K^2 N^2}{T^3 \vee N^3} - \frac{2K}{T \vee N}.$$

It approaches 1 if we assume further that $N = o(T^{w/4-1/2} \log^{w/4}(T))$.

With Assumption D3, we can show that for any fixed $w > 0$, $\|B_{t,jk}\|_{2w}$, $\|X_{t,jk}\|_{2w}$ and $\|\epsilon_{tj}\|_{2w} < \infty$, so that Lemma 1 in the Appendix can be applied, allowing the probability bound in the above theorem to hold. And, there are many examples with $\Theta_{m,2w} \leq Cm^{-\alpha}$ where only the constant C is dependent on w . See for example the stationary linear process Example 2.2 in Chen et al. (2013). Therefore, we can set w to be large enough from the beginning so that $N = o(T^{w/4-1/2} \log^{w/4}(T))$ is satisfied, ensuring $P(\mathcal{M}) \rightarrow 1$. Note also that we can actually allow $\alpha \leq 1/2 - 1/w$ at the expense of more complicated rates and longer proofs of the theorems to be presented below.

All the error bounds in the remaining theorems below are assumed on the set $\mathcal{M}$, so that through Theorem 1, those are valid error upper bounds with probability approaching 1.

Theorem 2 *Let all the assumptions in Section 3.2 and in Theorem 1 hold. Let $\check{\mathbf{W}}_1$ and $\check{\mathbf{W}}_2$ be ANY estimators for $\mathbf{W}_1^*$ and $\mathbf{W}_2^*$ respectively (not necessarily the LASSO or the one-step LASSO estimators). Then $\check{\boldsymbol{\beta}} = \boldsymbol{\beta}(\check{\boldsymbol{\xi}})$ defined in (2.5) is well-defined, where $\check{\boldsymbol{\xi}} = (\text{vec}(\check{\mathbf{W}}_1^T), \text{vec}(\check{\mathbf{W}}_2^T))^T$. Furthermore, on the set $\mathcal{M}$,*

$$\|\check{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\|_1 \leq \frac{a_1 \lambda_T (s_2 + N^{\frac{1}{2} + \frac{1}{2w}} + N^{1/2} \log^{1/2}(T \vee N))}{N} + \frac{a_2}{N} \|\check{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\|_1,$$

for some large enough constants a_1 and a_2 .

This theorem presents the error upper bound for estimating $\boldsymbol{\beta}^*$ on the set $\mathcal{M}$, which occurs with high probability by Theorem 1. Since $s_2 \leq N$ and $\lambda_T \rightarrow 0$ as $T, N \rightarrow \infty$, the first term on the bound is going to 0. On the other hand, the second term is going to 0 only when $\|\check{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\|_1 = o(N)$. For instance, if $\check{\mathbf{W}}_1$ and $\check{\mathbf{W}}_2$ are pre-specified by expert knowledge, which is what practitioners usually do when fitting a spatial lag/error model to their data, the estimator of $\boldsymbol{\beta}^*$ can be inaccurate if it turns out that $\check{\mathbf{W}}_1$ and $\check{\mathbf{W}}_2$ are not accurate enough for $\mathbf{W}_1^*$ and $\mathbf{W}_2^*$ respectively. We provide some simulation results for the estimation of $\boldsymbol{\beta}^*$ in Section 6 using the LASSO solution $\tilde{\boldsymbol{\beta}} = \boldsymbol{\beta}(\tilde{\boldsymbol{\xi}})$ as well as the Adaptive LASSO solution $\hat{\boldsymbol{\beta}} = \boldsymbol{\beta}(\hat{\boldsymbol{\xi}}_{\text{OS}})$. Theoretical error upper bounds for $\boldsymbol{\beta}(\tilde{\boldsymbol{\xi}})$ and $\tilde{\boldsymbol{\xi}}$ are given in the following theorem.

Theorem 3 *Let the assumptions in Section 3.2 and in Theorem 1 hold. Then on the set $\mathcal{M}$, for large enough T, N and some large enough constant C , the LASSO solution $\tilde{\boldsymbol{\xi}}$ satisfies*

$$\begin{aligned} \|\tilde{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\|_1 &\leq 4n^{1/2} \|\tilde{\boldsymbol{\xi}}_{J_1} - \boldsymbol{\xi}_{J_1}^*\| + h_{N,T}, \text{ with} \\ \|\tilde{\boldsymbol{\xi}}_{J_1} - \boldsymbol{\xi}_{J_1}^*\| &\leq \frac{C \lambda_T n^{1/2}}{\kappa^2(n)} + \frac{C h_{N,T}^{1/2} (\lambda_T^{1/2} + \lambda_T^{1/2} h_{N,T}^{1/2} + N^{-1/2} h_{N,T}^{1/2})}{\kappa(n)}. \end{aligned}$$

Combining with the bound in Theorem 2, we have on the set $\mathcal{M}$, the LASSO solution $\tilde{\boldsymbol{\beta}} = \boldsymbol{\beta}(\tilde{\boldsymbol{\xi}})$ in (2.5) satisfies

$$\|\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\|_1 = O\left(\frac{\lambda_T n + h_{N,T}}{N}\right) = o(1).$$

Note that $h_{N,T}$ is not necessarily going to 0 as $T, N \rightarrow \infty$ since $\|\boldsymbol{\xi}_{J_2}^*\|_1$ is not necessarily small. Hence the error in estimating $\boldsymbol{\xi}^*$ by the LASSO estimator $\tilde{\boldsymbol{\xi}}$ is not necessarily going to 0. However, restricted to the set J_1 , the L_2 error $\|\tilde{\boldsymbol{\xi}}_{J_1} - \boldsymbol{\xi}_{J_1}^*\|$ is indeed going to 0 by Assumption R6. Moreover, $\tilde{\boldsymbol{\beta}} = \boldsymbol{\beta}(\tilde{\boldsymbol{\xi}})$ is an accurate estimator for $\boldsymbol{\beta}^*$ with high probability as $T, N \rightarrow \infty$ as seen from the corresponding upper bound on the set $\mathcal{M}$.

In Theorem 1 of Ahrens and Bhattacharjee (2014), using our notations, they proved that

$$\|\hat{\beta} - \beta^*\|_1 = O\left(\sqrt{\frac{\log(\log(L \vee T))}{T}}\right),$$

where L is the number of instruments for each variable y_{tj} . Compared with our rate for $\tilde{\beta}$ in Theorem 3 above, which can be expressed as

$$\|\tilde{\beta} - \beta^*\|_1 = O\left(\frac{\|\xi_{J_2}^*\|_1}{N} + \sqrt{\frac{\log(T \vee N)}{T}} \left(\frac{n + s_2 + N^{\frac{1}{2} + \frac{1}{2w}} + N^{1/2} \log^{1/2}(T \vee N)}{N}\right)\right),$$

it is easy to see that if $\|\xi_{J_2}^*\|_1 = o(NT^{-1/2})$ (which is satisfied automatically if the spatial weight matrices are exactly sparse), and $n, s_2 = o(N)$, then the rate of convergence for $\tilde{\beta}$ is better than $\hat{\beta}$ in Theorem 1 of Ahrens and Bhattacharjee (2014) since N is allowed to diverge to infinity with T . The better rate of convergence as $N \rightarrow \infty$ makes more sense intuitively as well, since the regression parameter β^* in model (2.1) (and in model (2.1) in Ahrens and Bhattacharjee (2014) as well) is the same across all variables y_{tj} , so that as $N \rightarrow \infty$, information across different y_{tj} are used towards estimating the same β^* .

The following theorem gives error upper bounds and shows the partial sign-consistency of the one-step IVLASSO estimator $\hat{\xi}_{OS}$.

Theorem 4 *Let the assumptions in Section 3.2 and in Theorem 1 hold. Then on the set $\mathcal{M}$, we have*

$$\|\hat{\xi}_{OS, J_1} - \xi_{J_1}^*\|_{\max} = O(\lambda_T + \gamma'_T + \|\xi_{J_2}^*\|_{\max} + \|\xi_{J_2}^*\|_1 N^{-1}).$$

Furthermore, on the set $\mathcal{M}$, with large enough T, N , we have

$$\text{sign}(\hat{\xi}_{OS, J_1}) = \text{sign}(\xi_{J_1}^*), \quad \hat{\xi}_{OS, J_0 \cup J_2} = \mathbf{0},$$

so that combining the above, we have on the set $\mathcal{M}$ with large enough T, N ,

$$\|\hat{\xi}_{OS} - \xi^*\| = O(\lambda_T n^{1/2} + \gamma'_T n^{1/2} + \|\xi_{J_2}^*\|_{\max} n^{1/2} + \|\xi_{J_2}^*\|_1 n^{1/2} N^{-1} + \|\xi_{J_2}^*\|).$$

For approximately sparse spatial weight matrices so that there are a lot of “small” entries, if $\|\xi_{J_2}^*\|$ is not large and $\gamma'_T n^{1/2} = o(1)$, then the one-step IVLASSO estimator $\hat{\xi}_{OS}$ is still accurate for estimating $\mathbf{W}_1^*$ and $\mathbf{W}_2^*$ in the L_2 error norm. If in particular, $\xi_{J_2}^* = \mathbf{0}$, i.e., the spatial weight matrices are exactly sparse, then the L_2 error norm $\|\hat{\xi}_{OS} - \xi^*\| = O(\lambda_T n^{1/2}) = o(1)$ by Assumption R6 that $\lambda_T n = o(1)$ and that γ'_T has the same order as λ_T when the spatial weight matrices are sparse enough (see Assumption R5). Moreover, $\hat{\xi}_{OS}$ enjoys partial sign consistency - large elements in the spatial weight matrices are estimated with the correct sign, while the zero or “small” elements in the spatial weight matrices are estimated as zeros.

We give the error for estimating the spatial fixed effects in the following theorem.

Theorem 5 *Let the assumptions in Section 3.2 and in Theorem 1 hold. Furthermore, let $\mathbf{w}_{1,j,J_2}^*$ be the column vector for the j th row of $\mathbf{W}_1^*$ restricted to the elements in J_2 . We assume that $\max_j \|\mathbf{w}_{1,j,J_2}^*\|_1 = o(1)$. Then on the set $\mathcal{M}$, with large enough T, N ,*

$$\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}^*\|_{\max} = O\left(\lambda_T + \gamma'_T + \max_j \|\mathbf{w}_{1,j,J_2}^*\|_1\right) = \|\hat{\boldsymbol{\Pi}}_1 \hat{\boldsymbol{\mu}} - \boldsymbol{\Pi}_1^* \boldsymbol{\mu}^*\|_{\max},$$

where $\hat{\boldsymbol{\Pi}}_1 \hat{\boldsymbol{\mu}} = (\mathbf{I} - \widehat{\mathbf{W}}_1)^{-1} \hat{\boldsymbol{\mu}}$ is the estimated spatial fixed effects given in (2.9).

It is clear from the theorem above that with high probability, the elementwise estimation of the spatial fixed effects is accurate, as long as the absolute sum of the “small” elements in each row of $\mathbf{W}_1^*$ is small.

3.4 Asymptotic Identification of Model Parameters

The identification of $\boldsymbol{\beta}^*$, $\boldsymbol{\xi}^*$ and $\boldsymbol{\mu}^*$ are done asymptotically assuming the same conditions as those listed in Section 3.2. In particular, assume that $\boldsymbol{\beta}^*, \boldsymbol{\xi}^*, \boldsymbol{\mu}^*$ and $\boldsymbol{\beta}^o, \boldsymbol{\xi}^o, \boldsymbol{\mu}^o$ are two sets of parameters for model (2.2). That is,

$$\mathbf{y} = \mathbf{M} \boldsymbol{\beta}^* \boldsymbol{\xi}^* + \boldsymbol{\mu}^* \otimes \mathbf{1}_T + \boldsymbol{\epsilon} = \mathbf{M} \boldsymbol{\beta}^o \boldsymbol{\xi}^o + \boldsymbol{\mu}^o \otimes \mathbf{1}_T + \boldsymbol{\epsilon}.$$

Then we have the following.

Corollary 6 *Let the assumptions in Section 3.2 and in Theorem 1 hold for both sets of parameters. Then on the set $\mathcal{M}$, for large enough T, N , we have for $n^* = |J_1^*|$ and $n^o = |J_1^o|$,*

$$\|\boldsymbol{\beta}^* - \boldsymbol{\beta}^o\| = O\left(\frac{\lambda_T(n^* + n^o) + h_{T,N}^* + h_{T,N}^o}{N}\right) = o(1),$$

where $h_{T,N}^* = 4\|\boldsymbol{\xi}_{J_2}^*\|_1 + C\lambda_T(\|\boldsymbol{\xi}_{2,J_2}^*\|_1 + N^{\frac{1}{2} + \frac{1}{2w}} + N^{1/2} \log^{1/2}(T \vee N))$, and $h_{T,N}^o$ is defined the same but with $\boldsymbol{\xi}^*$ replaced by $\boldsymbol{\xi}^o$ and J_2^* by J_2^o . The sets J_1^*, J_2^* are defined similar to those in (3.3) and are corresponding to $\boldsymbol{\xi}^*$, while J_1^o, J_2^o are corresponding to $\boldsymbol{\xi}^o$.

Also, on the set $\mathcal{M}$, with large enough T, N , we have $J_1^* = J_1^o$, so $n^* = |J_1^*| = |J_1^o| = n^o$, and

$$\begin{aligned} \text{sign}(\boldsymbol{\xi}_{J_1^*}^*) &= \text{sign}(\boldsymbol{\xi}_{J_1^o}^o), \quad \boldsymbol{\xi}_{J_1^{*c}}^* = \boldsymbol{\xi}_{J_1^{oc}}^o = \mathbf{0}, \\ \|\boldsymbol{\xi}^* - \boldsymbol{\xi}^o\| &= O(\|\boldsymbol{\xi}_{J_2^*}^*\| + \|\boldsymbol{\xi}_{J_2^o}^o\| + n^{*1/2}(\lambda_T + \gamma'_T + \|\boldsymbol{\xi}_{J_2^*}^*\|_{\max} + \|\boldsymbol{\xi}_{J_2^*}^*\|_1 N^{-1}) \\ &\quad + n^{o1/2}(\lambda_T + \gamma'_T + \|\boldsymbol{\xi}_{J_2^o}^o\|_{\max} + \|\boldsymbol{\xi}_{J_2^o}^o\|_1 N^{-1})). \end{aligned}$$

Finally, assuming further $\max_j \|\mathbf{w}_{1,j,J_2}^*\|_1, \max_j \|\mathbf{w}_{1,j,J_2}^o\|_1 = o(1)$, and the number of elements in each row of $\mathbf{W}_1^*$ and $\mathbf{W}_1^o$ with magnitude greater than τ is finite. Then on the set $\mathcal{M}$ with large

enough T, N ,

$$\|\boldsymbol{\mu}^* - \boldsymbol{\mu}^o\|_{\max} = O(\lambda_T + \gamma'_T + \max_j \|\mathbf{w}_{1,j,J_2^*}^*\|_1 + \max_j \|\mathbf{w}_{1,j,J_2^o}^o\|) = o(1).$$

This Corollary is an immediate consequence of Theorem 3, 4 and 5, and hence the proof is omitted. Identification of $\boldsymbol{\beta}$ and $\boldsymbol{\mu}$ is achieved with high probability by the result in Theorem 1 under the assumptions in Section 3.2, and that both spatial weight matrices are appropriately sparse or approximately sparse. Hypothesis M2 can then be thought of as a crucial identification hypothesis, since it also bounds the degree of simultaneity in the model. At the same time, if $\|\boldsymbol{\xi}_{J_2}\|, n^{1/2}(\lambda_T + \gamma'_T + \|\boldsymbol{\xi}_{J_2}\|_{\max} + \|\boldsymbol{\xi}_{J_2}\|_1 N^{-1}) = o(1)$ for both set of parameters, then identification of the two spatial weight matrices in L_2 norm is achieved with high probability. Note that if both set of spatial weight matrices are exactly sparse, then the rate requirement is reduced to $\lambda_T n^{1/2} = o(1)$, which is already assumed in Assumption R6 already. The zeros and the sign of larger elements in both spatial weight matrices are identified also with high probability.

4 Practical Implementation

In this section we provide details of the block coordinate descent algorithm (BCD) for carrying the LASSO minimization. Given the LASSO solution has been found, a one-step iteration the BCD algorithm adjusted for the Adaptive LASSO minimizations provides the final estimator. The algorithm makes use of the fact that, although the functions in these problems are not convex in $(\boldsymbol{\xi}, \boldsymbol{\beta})$, they are convex in $\boldsymbol{\xi}$ given $\boldsymbol{\beta}$ and vice-versa.

The BCD algorithm is closely related to the Iterative Coordinate Descent of Fan and Lv (2011), and is also discussed in Friedman et al. (2010) and Dicker et al. (2013). While it is difficult to establish global convergence of the BCD algorithm without convexity, it is easy to see that each iteration delivers an improvement of the objective functions since given one parameter, the objective functions are convex in the other. From our experience, starting from an appropriate initial value, a minimum will be achieved with good performance in practice. Indeed in the simulation experiments in Section 5, it is found that the algorithm is robust to a variety of initial values chosen.

We define $N+1$ blocks to make advantage of the intra-block convexity. The parameter $\boldsymbol{\beta}$ forms one block, and for $j = 1, \dots, N$, $\boldsymbol{\eta}_j^T = (\boldsymbol{\eta}_{1j}^T, \boldsymbol{\eta}_{2j}^T)^T$ which is equal to the j -th row of $(\mathbf{W}_1, \mathbf{W}_2)$ form the other N blocks. Given the value of $\boldsymbol{\beta}$ and $\boldsymbol{\eta}_{-j} = (\boldsymbol{\eta}_1^T, \dots, \boldsymbol{\eta}_{j-1}^T, \boldsymbol{\eta}_{j+1}^T, \dots, \boldsymbol{\eta}_N^T)^T$, $\boldsymbol{\eta}_j$ is solved by the Least Angle Regression algorithm (LARS) of Efron et al. (2004). Given $\boldsymbol{\eta}_j$ for $j = 1, \dots, N$, $\boldsymbol{\beta}$ is solved by the ordinary least squares estimator (OLS).

The One Step Block Coordinate Descent (one step BCD) Algorithm

0. Start with an initial value $\boldsymbol{\xi} = \boldsymbol{\xi}_0$. This can be obtained by using

$$\boldsymbol{\beta}_0 = (\mathbf{X}^T \mathbf{B}^v \mathbf{B}^{vT} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{B}^v \mathbf{B}^{vT} \mathbf{y}^v$$

and solves (2.5) given $\boldsymbol{\beta}_0$ using LARS. This gives $\boldsymbol{\xi}_0$.

1. At step $r > 0$, set

$$\boldsymbol{\beta}(\boldsymbol{\xi}_r) = \left\{ \mathbf{X}^T \mathbf{W}_{2,r}^{\otimes T} \mathbf{W}_{2,r}^{\otimes} \mathbf{B}^v \mathbf{B}^{vT} \mathbf{W}_{2,r}^{\otimes T} \mathbf{W}_{2,r}^{\otimes} \mathbf{X} \right\}^{-1} \mathbf{X}^T \mathbf{W}_{2,r}^{\otimes T} \mathbf{W}_{2,r}^{\otimes} \mathbf{B}^v \mathbf{B}^{vT} \mathbf{W}_{2,r}^{\otimes T} (\mathbf{I}_{TN} - \mathbf{W}_{1,r}^{\otimes}) \mathbf{y}^v$$

where $\mathbf{W}_{j,r}^{\otimes} = \mathbf{I}_N \otimes \mathbf{W}_{j,r}$, $\mathbf{W}_{1,r}$, $\mathbf{W}_{2,r}$ the spatial weight matrices recovered from $\boldsymbol{\xi}_r$.

2. Using LARS, solve sequentially for $j = 1, \dots, N$,

$$\boldsymbol{\eta}_{j,r} = \arg \min_{\boldsymbol{\eta}_j} \frac{1}{2T} \left\| \mathbf{B}^T \mathbf{y} - \mathbf{B}^T \mathbf{M}_{\boldsymbol{\beta}_r} \boldsymbol{\eta} \right\|^2 + \gamma_T \|\boldsymbol{\eta}_j\|_1, \quad \text{subj. to } \|\boldsymbol{\eta}_{1j}\|_1 < 1, \|\boldsymbol{\eta}_{2j}\|_1 < 2,$$

where $\boldsymbol{\eta} = (\tilde{\boldsymbol{\eta}}_1^T, \tilde{\boldsymbol{\eta}}_2^T)^T$ with $\tilde{\boldsymbol{\eta}}_i = (\boldsymbol{\eta}_{i,1,r-1}^T, \dots, \boldsymbol{\eta}_{i,j-1,r-1}^T, \boldsymbol{\eta}_{i,j}^T, \boldsymbol{\eta}_{i,j+1,r-1}^T, \dots, \boldsymbol{\eta}_{i,N,r-1}^T)^T$. Then

$$\boldsymbol{\xi}_r = (\boldsymbol{\eta}_{1,1,r}^T, \dots, \boldsymbol{\eta}_{1,N,r}^T, \boldsymbol{\eta}_{2,1,r}^T, \dots, \boldsymbol{\eta}_{2,N,r}^T)^T.$$

3. Iterate steps 1-2 until $\|\boldsymbol{\xi}_r - \boldsymbol{\xi}_{r-1}\|_1$ is smaller than some pre-set number. The LASSO solution is then $(\tilde{\boldsymbol{\beta}}, \tilde{\boldsymbol{\xi}}) = (\boldsymbol{\beta}_r, \boldsymbol{\xi}_r)$.

4. Repeat only once the steps 1-2 for the adaptive LASSO solutions, where in step 2 the penalty function is modified to $\gamma'_T \mathbf{v}_j^T |\boldsymbol{\eta}_j|$, with the components in $\mathbf{v}_j$ having the form $1/|\tilde{\boldsymbol{\xi}}_j|$.

We choose the penalization parameters γ_T and γ'_T above by two BIC criteria as discussed next. Let $\tilde{\boldsymbol{\xi}}_{\gamma_T}$ be the LASSO solution from (2.5), and $\hat{\boldsymbol{\xi}}_{\gamma'_T}$ be the one-step Adaptive LASSO solution from (2.7). Let $\tilde{S}_{\gamma_T}$ and $\hat{S}_{\gamma'_T}$ be the number of nonzero elements in $\tilde{\boldsymbol{\xi}}_{\gamma_T}$ and $\hat{\boldsymbol{\xi}}_{\gamma'_T}$ respectively. Then we define

$$\text{BIC}_1(\gamma_T) = \log \left(N^{-2} \left\| \mathbf{B}^T \mathbf{y} - \mathbf{B}^T \mathbf{M}_{\boldsymbol{\beta}(\tilde{\boldsymbol{\xi}}_{\gamma_T})} \tilde{\boldsymbol{\xi}}_{\gamma_T} \right\|^2 \right) + |\tilde{S}_{\gamma_T}| \frac{\log(N)}{N} \log(\log(2N - 2)), \quad (4.6)$$

$$\text{BIC}_2(\gamma'_T) = \log \left(N^{-2} \left\| \mathbf{B}^T \mathbf{y} - \mathbf{B}^T \mathbf{M}_{\boldsymbol{\beta}(\tilde{\boldsymbol{\xi}}_{\gamma_T})} \hat{\boldsymbol{\xi}}_{\gamma'_T} \right\|^2 \right) + |\hat{S}_{\gamma'_T}| \frac{\log(N)}{N} \log(\log(2N - 2)). \quad (4.7)$$

The parameters γ_T and γ'_T are chosen to minimize the criterion (4.6) and (4.7) above respectively, representing the penalization parameter for the LASSO problem (2.5) and the adaptive LASSO problem (2.7) respectively. From our experience, for exactly sparse spatial weight matrices, the two criteria produce very similar penalization parameter, which is consistent with our discussion in Section 3.2. These two criteria allow for different penalization parameters for the LASSO and the adaptive LASSO problems, which is essential when $\lambda_T = o(\|\boldsymbol{\xi}_{J_2}^*\|_1)$ as in Assumption R5, when potentially γ'_T for the adaptive LASSO problem is different from γ_T for the LASSO problem.

The above BIC criteria are inspired by the one proposed in Wang et al. (2009). In fact, if we define $\mathbf{u}_j = T^{-1/2} N^{-a/2} \sum_{t=1}^T (\mathbf{B}_t - \bar{\mathbf{B}}) \gamma_{tj}$, $j = 1, \dots, N$ which is a portion of the vector $\mathbf{B}^T \mathbf{y}$, and $(\mathbf{B}^T \mathbf{M}_{\beta(\tilde{\xi}_{\gamma_T})} \tilde{\xi}_{\gamma_T})_j$ to be the portion of the vector $\mathbf{B}^T \mathbf{M}_{\beta(\tilde{\xi}_{\gamma_T})} \tilde{\xi}_{\gamma_T}$ corresponding to $\mathbf{u}_j$, then

$$\text{bic}(\gamma_T) = \log \left(N^{-1} \left\| \mathbf{u}_j - \left(\mathbf{B}^T \mathbf{M}_{\beta(\tilde{\xi}_{\gamma_T})} \tilde{\xi}_{\gamma_T} \right)_j \right\|^2 \right) + |\tilde{S}_{\gamma_T}| \frac{\log(N)}{N} \log(\log(2N - 2))$$

is an individual BIC criterion parallel to that defined in Wang et al. (2009) for the j -th portion of the vector $\mathbf{B}^T \mathbf{y} - \mathbf{B}^T \mathbf{M}_{\beta(\tilde{\xi}_{\gamma_T})} \tilde{\xi}_{\gamma_T}$, which has N elements. The BIC criteria in (4.6) and (4.7) consider the whole vector (with N^2 elements) instead of just a portion of it. Although the value of a is unknown in the definition of $\mathbf{B}$, the locations of the minimum for BIC₁ or BIC₂ above are in fact independent of the value of a , due to the logarithmic operation in the first term of (4.6) or (4.7). In practice, we set $a = 1$.

5 Simulations

We next give detailed simulation results for our one-step Adaptive LASSO estimator. In the next section, we apply the estimator to provide concrete evidence of cross-sectional dissemination of a price shock among disaggregated components of the Consumer Price Index (CPI) in the United States.

First, we generate data from a simple benchmark model for studying the performance of the estimator in which we assume: (i) dependence between the regressors and the disturbance term; (ii) absence of spatial dependence and (iii) absence of time dependence. To this end, we generate spatial matrices $\mathbf{W}_1^*$ and $\mathbf{W}_2^*$ by randomly selecting its elements (except diagonal elements) to be either 0.3 or 0, with an overall predetermined sparsity level. If the sum of a row is larger than one, we normalise by 1.1 times the L_1 norm of that row. We set $K = 2$ and $\beta^* = [1; 0.5]^T$. The disturbance ϵ_t is normal and independently distributed with mean 0 and variance 1. We define the covariates as $\text{vec}(\mathbf{X}_t) = 0.2 \cdot \iota_K \otimes \epsilon_t + \epsilon_t^X$, where ϵ_t^X is an exogenous shock of size $NK \times 1$ sampled from a normal distribution with mean zero and covariance matrix

$$\Sigma_X = \begin{bmatrix} \Sigma_X^{block} & 0.5 \cdot I_N \\ 0.5 \cdot I_N & \Sigma_X^{block} \end{bmatrix} \quad (5.8)$$

where, for now, $\Sigma_X^{block} = 2 \cdot I_N$. Therefore, the covariates both correlate with the disturbance term and among themselves. Finally, the instruments are generated as $\text{vec}(\mathbf{B}_t) = 0.7 \cdot \epsilon_t^X + \epsilon_t^B$, where ϵ_t^B is drawn from the same distribution as ϵ_t^X . This setting ensures correlation with the covariates but not with the disturbance term.

We study the performance of the estimator for $N = 25, 50$ and 75 and $T = 50, 100$ and 200 and sparsity levels $\kappa = 0.95$ and 0.99 . These choices allow us to explore the effects of dimension on the

performance of the estimator when it can be larger than the sample size T . For each combination of N and T , we generate a pair $\mathbf{W}_1^*$ and $\mathbf{W}_2^*$ and simulate the data eight times and analyse the results. We then iterate over pairs of the spatial weight matrices fifty times (and hence a total of 200 sets of data for given N and T). This is done to ensure that results are not sensitive to a particular choice of $\mathbf{W}_1^*$ and $\mathbf{W}_2^*$, but rather robust across many specifications. All simulations contain fixed effects sampled at the individual level from a standard normal distribution.

Table 1 shows the results for the baseline simulation. For most combinations of N and T , the specificity, the proportion of true zeros estimated as zeros, is above 95%. The sensitivity, or the proportion of non-zeros estimated as non-zeros, improves hugely as the T increases. It is intuitive since the non-zero elements are relatively small, and hence when T is too small they cannot be picked up easily. Biases are mostly negative, meaning that we usually underestimate the non-zero values of the weight matrices. Also it is clear that the performance of the adaptive LASSO is much better than LASSO in general. It is of interest to note that while the L_1 error norm can be large, the L_2 error norm is usually much smaller. Moreover, the sparsity of the estimated $\widehat{\mathbf{W}}_1^*$ and $\widehat{\mathbf{W}}_2^*$ is usually close to the true sparsity, indicating the good performance of the BIC criteria described above.

In the next simulation scenario, we introduce correlation between the regressors and the disturbance term, along with spatial and time dependence. For this purpose, we implement several changes in the data generating process. First, we allow the disturbance term to follow the autoregressive process $\epsilon_t = 0.2 \cdot \epsilon_{t-1} + \epsilon_t^e$, where ϵ_t^e is a $N \times 1$ random vector with mean zero and variance Σ_e . The off-diagonal elements of Σ_e are randomly chosen to be either 0 or 0.25, such that the sparsity of this matrix is set at 0.95; the main diagonal is one. Second, the covariates are also allowed to follow the autoregressive process $\text{vec}(\mathbf{X}_t) = 0.2 \cdot \text{vec}(\mathbf{X}_{t-1}) + 0.2 \cdot \iota_K \otimes \epsilon_t^e + \epsilon_t^X$, but now with elements Σ_X^{block} also randomly chosen using the same procedure as Σ_e . Similar modifications of the data generating process also applies to the instruments $\mathbf{B}_t$. Table 2 shows that results are largely unchanged, demonstrating a estimator's robustness both with respect to time and spatial dependences.

In our last simulation scenario, we generate data with approximately sparse spatial weight matrices. As in the previous cases, we randomly select elements of $\mathbf{W}_1^*$ and $\mathbf{W}_2^*$ such that they attain a given sparsity level, in which case the elements are set at 0.30. We now take a further step: an additional 25% of their elements is randomly selected to be small but nonzero. We set them at 0.01. In this way, we allow the spatial weight matrices to be composed of a few strong elements in addition to a large number of weak elements, which is arguably a good representation of real-life scenarios. We expect these small elements to be shrunk to zero. Therefore, the specificity refers to the proportion of nonzeros and small elements estimated as zeros (and analogously to sensitivity). Results, shown in Table 3, are largely unchanged. Overall, the estimator shows a good performance and robustness relative to a number of scenarios.

Table 1: Simulations - Baseline

	$\kappa = 0.95$						$\kappa = 0.99$					
	$T = 50$		$T = 100$		$T = 200$		$T = 50$		$T = 100$		$T = 200$	
	$\mathbf{W}_1^*$	$\mathbf{W}_2^*$	$\mathbf{W}_1^*$	$\mathbf{W}_2^*$	$\mathbf{W}_1^*$	$\mathbf{W}_2^*$	$\mathbf{W}_1^*$	$\mathbf{W}_2^*$	$\mathbf{W}_1^*$	$\mathbf{W}_2^*$	$\mathbf{W}_1^*$	$\mathbf{W}_2^*$
$N = 25$												
Specificity	91.84% (0.021)	91.59% (0.024)	95.65% (0.017)	95.53% (0.023)	97.33% (0.010)	97.68% (0.013)	94.01% (0.023)	93.74% (0.024)	97.23% (0.017)	96.84% (0.021)	98.91% (0.006)	98.84% (0.007)
Sensitivity	76.37% (0.072)	62.07% (0.080)	85.01% (0.076)	73.66% (0.060)	89.02% (0.042)	82.62% (0.063)	72.92% (0.191)	59.38% (0.161)	78.75% (0.142)	65.62% (0.196)	78.04% (0.172)	65.54% (0.160)
Bias	-0.0496 (0.020)	-0.0542 (0.032)	-0.0503 (0.019)	-0.0481 (0.024)	-0.0339 (0.011)	-0.0512 (0.015)	-0.0199 (0.051)	-0.0566 (0.043)	-0.0536 (0.036)	-0.0678 (0.044)	-0.0252 (0.035)	-0.0062 (0.052)
LASSO L_1	15.8772 (1.880)	18.8917 (2.293)	12.3977 (1.164)	13.7595 (1.531)	9.7663 (1.026)	11.0006 (1.460)	12.5593 (1.675)	13.4585 (1.960)	9.2044 (1.001)	10.5228 (1.473)	7.1008 (0.777)	7.9494 (0.659)
LASSO L_2	8.7356 (1.470)	10.2930 (1.814)	5.8083 (0.922)	5.8343 (0.995)	4.1840 (0.923)	4.3746 (0.820)	3.8598 (1.436)	4.2457 (1.358)	1.9423 (0.664)	2.2660 (0.880)	1.3088 (0.358)	1.3749 (0.430)
AdaLASSO L_1	2.0073 (0.316)	2.8264 (0.483)	1.3376 (0.199)	1.6009 (0.218)	0.9920 (0.208)	1.1212 (0.234)	1.3047 (0.307)	1.5313 (0.339)	0.7647 (0.118)	0.9430 (0.165)	0.5299 (0.123)	0.6132 (0.110)
AdaLASSO L_2	1.5869 (0.297)	2.1436 (0.410)	1.0313 (0.199)	1.1549 (0.189)	0.7767 (0.218)	0.8778 (0.208)	0.5847 (0.239)	0.7005 (0.216)	0.3211 (0.106)	0.3978 (0.148)	0.2729 (0.103)	0.3040 (0.124)
Sparsity	0.8853 (0.023)	0.8900 (0.023)	0.9121 (0.018)	0.9252 (0.018)	0.9302 (0.010)	0.9386 (0.010)	0.9334 (0.023)	0.9321 (0.023)	0.9653 (0.017)	0.9617 (0.017)	0.9814 (0.006)	0.9820 (0.006)
LASSO βL_1	0.0175 (0.0073)		0.0142 (0.0077)		0.0098 (0.0048)		0.0192 (0.0096)		0.0114 (0.0067)		0.0093 (0.0052)	
AdaLASSO βL_1	0.0229 (0.0125)		0.0180 (0.0081)		0.0093 (0.0053)		0.0197 (0.0103)		0.0119 (0.0055)		0.0080 (0.0050)	
AdaLASSO FE L_1	0.1228 (0.0193)		0.0867 (0.0112)		0.0567 (0.0051)		0.1268 (0.0304)		0.0819 (0.0108)		0.0570 (0.0081)	
$N = 50$												
Specificity	94.88% (0.014)	97.37% (0.012)	98.61% (0.003)	99.72% (0.001)	98.99% (0.002)	99.87% (0.001)	99.71% (0.001)	99.84% (0.001)	99.69% (0.001)	99.87% (0.001)	99.84% (0.001)	99.91% (0.001)
Sensitivity	70.84% (0.043)	44.77% (0.076)	75.14% (0.051)	47.67% (0.062)	89.97% (0.059)	78.53% (0.059)	47.52% (0.105)	32.41% (0.106)	73.37% (0.088)	56.61% (0.121)	90.60% (0.062)	80.32% (0.094)
Bias	-0.0357 (0.008)	-0.0839 (0.014)	-0.0687 (0.009)	-0.1214 (0.015)	-0.0592 (0.011)	-0.1064 (0.012)	-0.1153 (0.046)	-0.1402 (0.042)	-0.1247 (0.031)	-0.1285 (0.026)	-0.0835 (0.016)	-0.0990 (0.021)
LASSO L_1	50.7003 (3.185)	45.8584 (6.446)	33.6714 (1.537)	32.2013 (2.023)	26.6569 (1.185)	26.5566 (1.358)	20.2389 (1.493)	16.4905 (1.353)	17.7394 (1.748)	16.8458 (1.261)	15.7369 (1.357)	14.7641 (1.450)
LASSO L_2	29.6230 (1.826)	29.1554 (3.993)	20.3573 (1.186)	27.1444 (1.306)	13.3291 (1.685)	19.6906 (1.894)	5.3541 (0.660)	7.3272 (0.811)	4.0822 (0.638)	6.4041 (0.959)	3.3502 (0.747)	3.6864 (0.638)
AdaLASSO L_1	5.8574 (0.518)	7.6036 (0.975)	3.3228 (0.233)	5.1367 (0.360)	2.0960 (0.239)	2.9375 (0.345)	1.9054 (0.157)	2.1807 (0.215)	1.3258 (0.146)	1.6786 (0.193)	0.9106 (0.085)	0.9953 (0.147)
AdaLASSO L_2	5.3001 (0.552)	7.1801 (0.796)	3.7785 (0.367)	7.1481 (0.527)	2.0553 (0.414)	4.2803 (0.675)	1.2904 (0.189)	2.0218 (0.257)	0.8521 (0.193)	1.6055 (0.371)	0.5734 (0.172)	0.7592 (0.215)
Sparsity	0.9128 (0.012)	0.9556 (0.012)	0.9494 (0.003)	0.9743 (0.003)	0.9487 (0.003)	0.9586 (0.003)	0.9930 (0.002)	0.9949 (0.002)	0.9912 (0.002)	0.9919 (0.002)	0.9892 (0.001)	0.9914 (0.001)
LASSO βL_1	0.0068 (0.0033)		0.0110 (0.0053)		0.0238 (0.0047)		0.0066 (0.0028)		0.0051 (0.0026)		0.0083 (0.0039)	
AdaLASSO βL_1	0.0215 (0.0073)		0.0273 (0.0058)		0.0257 (0.0055)		0.0135 (0.0068)		0.0092 (0.0036)		0.0079 (0.0037)	
AdaLASSO FE L_1	0.1315 (0.0162)		0.0883 (0.0135)		0.0618 (0.0086)		0.1211 (0.0118)		0.0858 (0.0101)		0.0605 (0.0057)	
$N = 75$												
Specificity	95.53% (0.010)	98.95% (0.006)	98.47% (0.002)	99.97% (0.000)	98.99% (0.002)	99.98% (0.000)	99.82% (0.002)	99.92% (0.001)	99.91% (0.000)	99.98% (0.000)	99.88% (0.000)	99.97% (0.000)
Sensitivity	58.80% (0.045)	25.26% (0.077)	58.69% (0.035)	14.83% (0.038)	76.63% (0.037)	45.48% (0.088)	36.04% (0.163)	15.91% (0.119)	51.79% (0.113)	24.66% (0.050)	83.31% (0.053)	63.94% (0.051)
Bias	-0.0187 (0.009)	-0.0946 (0.025)	-0.0269 (0.010)	-0.1484 (0.014)	-0.0409 (0.011)	-0.1636 (0.009)	-0.1474 (0.026)	-0.1618 (0.047)	-0.1550 (0.011)	-0.1695 (0.018)	-0.1372 (0.010)	-0.1592 (0.018)
LASSO L_1	101.3190 (5.295)	90.2505 (7.207)	70.2353 (1.706)	69.2148 (2.089)	51.6931 (1.583)	49.5096 (2.458)	35.3622 (6.639)	25.7506 (5.282)	25.7147 (2.554)	19.9387 (2.229)	21.6990 (2.085)	17.5505 (1.843)
LASSO L_2	66.6440 (2.388)	71.8595 (3.470)	50.7667 (2.197)	75.3258 (1.676)	35.3242 (2.839)	64.6499 (3.734)	14.4121 (1.394)	15.1707 (1.509)	12.7050 (0.734)	15.0379 (0.363)	9.9994 (0.684)	11.5433 (0.987)
AdaLASSO L_1	11.7041 (0.554)	18.8388 (0.835)	7.5203 (0.372)	15.1930 (0.797)	4.1818 (0.446)	7.8471 (0.913)	3.5269 (0.527)	4.1580 (0.484)	2.0991 (0.224)	2.7489 (0.302)	1.3683 (0.160)	1.5881 (0.170)
AdaLASSO L_2	12.1758 (0.967)	19.8006 (0.746)	9.9700 (0.750)	21.9446 (0.671)	5.6462 (0.800)	17.3861 (1.554)	3.6799 (0.597)	4.3649 (0.550)	3.0916 (0.284)	4.2768 (0.143)	1.9882 (0.245)	2.8267 (0.319)
Sparsity	0.9271 (0.010)	0.9783 (0.010)	0.9565 (0.003)	0.9925 (0.003)	0.9531 (0.002)	0.9777 (0.002)	0.9945 (0.004)	0.9977 (0.004)	0.9938 (0.001)	0.9974 (0.001)	0.9902 (0.001)	0.9935 (0.001)
LASSO βL_1	0.0056 (0.0035)		0.0122 (0.0024)		0.0214 (0.0029)		0.0045 (0.0024)		0.0053 (0.0027)		0.0077 (0.0025)	
AdaLASSO βL_1	0.0195 (0.0073)		0.0255 (0.0078)		0.0343 (0.0052)		0.0107 (0.0055)		0.0086 (0.0043)		0.0083 (0.0026)	
AdaLASSO FE L_1	0.1324 (0.0151)		0.0980 (0.0133)		0.0655 (0.0065)		0.1230 (0.0116)		0.0830 (0.0082)		0.0573 (0.0055)	

Table 2: Simulations - Spatial and Time Dependence

	$\kappa = 0.95$						$\kappa = 0.99$					
	$T = 50$		$T = 100$		$T = 200$		$T = 50$		$T = 100$		$T = 200$	
	$\mathbf{W}_1^*$	$\mathbf{W}_2^*$	$\mathbf{W}_1^*$	$\mathbf{W}_2^*$	$\mathbf{W}_1^*$	$\mathbf{W}_2^*$	$\mathbf{W}_1^*$	$\mathbf{W}_2^*$	$\mathbf{W}_1^*$	$\mathbf{W}_2^*$	$\mathbf{W}_1^*$	$\mathbf{W}_2^*$
$N = 25$												
Specificity	92.74% (0.020)	93.18% (0.026)	95.26% (0.016)	95.31% (0.019)	97.20% (0.010)	97.10% (0.011)	94.85% (0.020)	94.61% (0.025)	97.57% (0.010)	97.56% (0.011)	98.35% (0.007)	98.42% (0.008)
Sensitivity	76.26% (0.085)	56.32% (0.088)	85.07% (0.063)	73.61% (0.078)	87.80% (0.072)	85.19% (0.077)	73.12% (0.203)	57.38% (0.197)	77.32% (0.200)	68.08% (0.151)	75.71% (0.229)	71.20% (0.151)
Bias	-0.0459 (0.024)	-0.0661 (0.028)	-0.0412 (0.018)	-0.0568 (0.019)	-0.0303 (0.013)	-0.0450 (0.016)	-0.0580 (0.043)	-0.0553 (0.065)	-0.0416 (0.045)	-0.0564 (0.040)	-0.0384 (0.030)	-0.0343 (0.028)
LASSO L_1	15.4898 (1.383)	17.7233 (2.244)	12.3738 (1.099)	14.6067 (1.475)	9.5705 (0.847)	11.1200 (0.930)	11.6885 (1.360)	12.7548 (2.128)	9.0438 (0.827)	10.3241 (1.052)	7.5403 (0.581)	8.4417 (0.830)
LASSO L_2	8.3648 (1.205)	9.6417 (1.556)	5.7390 (0.977)	6.3804 (1.132)	3.9241 (0.796)	4.0879 (0.822)	3.3667 (1.068)	3.8400 (1.335)	1.6454 (0.517)	2.0996 (0.488)	1.3672 (0.434)	1.5487 (0.501)
AdaLASSO L_1	2.0091 (0.272)	2.6392 (0.404)	1.3720 (0.199)	1.7347 (0.272)	0.9076 (0.183)	1.0747 (0.193)	1.1873 (0.197)	1.4486 (0.319)	0.7440 (0.121)	0.9338 (0.128)	0.5568 (0.093)	0.6411 (0.122)
AdaLASSO L_2	1.5806 (0.278)	2.1013 (0.336)	1.0454 (0.229)	1.2920 (0.277)	0.6828 (0.205)	0.7723 (0.220)	0.5136 (0.175)	0.6443 (0.210)	0.2799 (0.119)	0.3909 (0.115)	0.2793 (0.109)	0.3219 (0.129)
Sparsity	0.8939 (0.020)	0.9071 (0.020)	0.9134 (0.017)	0.9195 (0.017)	0.9279 (0.011)	0.9340 (0.011)	0.9419 (0.020)	0.9409 (0.020)	0.9702 (0.010)	0.9674 (0.010)	0.9777 (0.006)	0.9759 (0.006)
LASSO βL_1	0.0238 (0.0107)		0.0151 (0.0078)		0.0104 (0.0061)		0.0194 (0.0105)		0.0131 (0.0073)		0.0097 (0.0057)	
AdaLASSO βL_1	0.0257 (0.0111)		0.0155 (0.0074)		0.0102 (0.0056)		0.0187 (0.0110)		0.0117 (0.0060)		0.0097 (0.0054)	
AdaLASSO FE L_1	0.1666 (0.0281)		0.1275 (0.0209)		0.0912 (0.0124)		0.1430 (0.0230)		0.0994 (0.0158)		0.0778 (0.0149)	
$N = 50$												
Specificity	95.71% (0.006)	98.32% (0.005)	98.41% (0.003)	99.71% (0.001)	99.03% (0.003)	99.81% (0.001)	99.50% (0.003)	99.74% (0.002)	99.72% (0.001)	99.85% (0.001)	99.77% (0.001)	99.89% (0.001)
Sensitivity	72.01% (0.047)	38.66% (0.069)	76.41% (0.044)	49.32% (0.058)	92.17% (0.035)	83.34% (0.043)	62.36% (0.104)	32.28% (0.104)	79.36% (0.091)	61.33% (0.098)	90.96% (0.059)	78.71% (0.075)
Bias	-0.0422 (0.011)	-0.0849 (0.019)	-0.0535 (0.011)	-0.1203 (0.015)	-0.0449 (0.008)	-0.0964 (0.009)	-0.1077 (0.027)	-0.1152 (0.042)	-0.1017 (0.024)	-0.1140 (0.025)	-0.0762 (0.013)	-0.0875 (0.015)
LASSO L_1	47.6215 (2.332)	46.3907 (2.542)	34.1784 (1.632)	33.4828 (1.830)	26.1206 (1.594)	27.2913 (1.584)	22.7535 (2.013)	17.8247 (1.712)	19.6147 (1.082)	16.6800 (1.414)	17.1737 (1.019)	16.6719 (1.089)
LASSO L_2	27.9199 (1.857)	32.6318 (2.173)	20.0024 (1.588)	26.5266 (1.468)	12.4228 (1.656)	16.5056 (1.663)	5.7482 (0.889)	6.5105 (0.683)	4.5973 (0.827)	5.0205 (1.140)	3.3391 (0.639)	3.7092 (0.633)
AdaLASSO L_1	5.7157 (0.410)	8.3506 (0.634)	3.4198 (0.314)	5.1637 (0.449)	1.9505 (0.254)	2.6861 (0.337)	2.1073 (0.278)	2.2561 (0.251)	1.4588 (0.163)	1.5380 (0.272)	1.0578 (0.144)	1.1722 (0.159)
AdaLASSO L_2	5.3565 (0.512)	8.5588 (0.709)	3.6949 (0.417)	6.9440 (0.513)	1.7613 (0.368)	3.3426 (0.478)	1.2697 (0.230)	1.7462 (0.211)	0.9261 (0.222)	1.2277 (0.330)	0.5808 (0.169)	0.7764 (0.178)
Sparsity	0.9243 (0.007)	0.9649 (0.007)	0.9469 (0.004)	0.9735 (0.004)	0.9446 (0.004)	0.9584 (0.004)	0.9886 (0.003)	0.9943 (0.003)	0.9890 (0.002)	0.9927 (0.002)	0.9888 (0.002)	0.9910 (0.002)
LASSO βL_1	0.0220 (0.0094)		0.0130 (0.0067)		0.0089 (0.0050)		0.0132 (0.0056)		0.0080 (0.0045)		0.0066 (0.0038)	
AdaLASSO βL_1	0.0309 (0.0107)		0.0279 (0.0080)		0.0290 (0.0056)		0.0156 (0.0068)		0.0107 (0.0058)		0.0078 (0.0047)	
AdaLASSO FE L_1	0.1869 (0.0225)		0.1478 (0.0173)		0.1011 (0.0102)		0.1488 (0.0163)		0.1019 (0.0128)		0.0771 (0.0081)	
$N = 75$												
Specificity	96.64% (0.002)	99.35% (0.001)	98.31% (0.002)	99.95% (0.000)	98.89% (0.001)	99.98% (0.000)	99.63% (0.003)	99.83% (0.002)	99.89% (0.000)	99.97% (0.000)	99.85% (0.000)	99.96% (0.000)
Sensitivity	55.89% (0.037)	20.16% (0.026)	62.36% (0.035)	19.23% (0.046)	79.32% (0.042)	52.62% (0.044)	50.36% (0.186)	24.78% (0.137)	59.38% (0.069)	32.23% (0.064)	86.77% (0.052)	67.89% (0.059)
Bias	-0.0155 (0.011)	-0.1019 (0.017)	-0.0178 (0.013)	-0.1513 (0.014)	-0.0322 (0.008)	-0.1577 (0.007)	-0.1129 (0.030)	-0.1379 (0.038)	-0.1461 (0.014)	-0.1727 (0.018)	-0.1180 (0.013)	-0.1508 (0.015)
LASSO L_1	92.8275 (2.486)	92.4929 (3.890)	69.3742 (2.532)	69.2975 (2.952)	51.5381 (2.338)	51.6653 (2.603)	40.9962 (7.601)	29.8741 (5.929)	28.2289 (1.327)	21.9629 (1.401)	24.1912 (1.167)	19.9526 (1.176)
LASSO L_2	61.7967 (3.772)	78.1250 (4.121)	49.8898 (2.629)	73.0270 (3.052)	34.7662 (2.727)	61.2724 (3.232)	12.5982 (1.414)	15.8618 (1.135)	11.5532 (1.420)	14.8798 (1.663)	8.7574 (1.069)	11.3774 (1.438)
AdaLASSO L_1	11.1504 (0.555)	20.2390 (1.156)	7.1826 (0.443)	14.4001 (0.814)	4.1518 (0.375)	7.6221 (0.652)	3.8671 (0.501)	4.4855 (0.409)	2.2226 (0.176)	2.8199 (0.323)	1.5020 (0.163)	1.7648 (0.226)
AdaLASSO L_2	12.1522 (0.955)	22.1401 (1.245)	9.5576 (0.713)	21.0697 (0.991)	5.5205 (0.604)	15.9890 (0.977)	2.9568 (0.619)	4.3990 (0.487)	2.7268 (0.386)	4.1405 (0.483)	1.6173 (0.274)	2.7199 (0.398)
Sparsity	0.9413 (0.003)	0.9838 (0.003)	0.9529 (0.003)	0.9903 (0.003)	0.9505 (0.002)	0.9742 (0.002)	0.9916 (0.005)	0.9957 (0.005)	0.9932 (0.001)	0.9964 (0.001)	0.9900 (0.001)	0.9928 (0.001)
LASSO βL_1	0.0283 (0.0089)		0.0187 (0.0063)		0.0123 (0.0051)		0.0115 (0.0055)		0.0078 (0.0036)		0.0058 (0.0027)	
AdaLASSO βL_1	0.0303 (0.0100)		0.0306 (0.0094)		0.0429 (0.0077)		0.0148 (0.0066)		0.0098 (0.0045)		0.0107 (0.0033)	
AdaLASSO FE L_1	0.2214 (0.0214)		0.1751 (0.0205)		0.1305 (0.0125)		0.1538 (0.0162)		0.1194 (0.0118)		0.0857 (0.0103)	

Table 3: Simulations - Approximately sparse $\mathbf{W}^*$

	$\kappa = 0.95$						$\kappa = 0.99$					
	$T = 50$		$T = 100$		$T = 200$		$T = 50$		$T = 100$		$T = 200$	
	$\mathbf{W}_1^*$	$\mathbf{W}_2^*$	$\mathbf{W}_1^*$	$\mathbf{W}_2^*$	$\mathbf{W}_1^*$	$\mathbf{W}_2^*$	$\mathbf{W}_1^*$	$\mathbf{W}_2^*$	$\mathbf{W}_1^*$	$\mathbf{W}_2^*$	$\mathbf{W}_1^*$	$\mathbf{W}_2^*$
$N = 25$												
Specificity	91.57% (0.022)	92.06% (0.022)	95.16% (0.014)	95.18% (0.018)	97.18% (0.008)	97.27% (0.009)	95.50% (0.024)	95.08% (0.027)	97.56% (0.013)	97.29% (0.013)	98.72% (0.005)	98.50% (0.007)
Sensitivity	73.88% (0.089)	58.37% (0.093)	84.62% (0.100)	73.95% (0.068)	88.63% (0.068)	84.27% (0.075)	71.26% (0.212)	59.94% (0.224)	80.89% (0.136)	70.00% (0.215)	78.34% (0.166)	71.05% (0.225)
Bias	-0.0347 (0.024)	-0.0639 (0.031)	-0.0412 (0.020)	-0.0564 (0.022)	-0.0336 (0.012)	-0.0383 (0.017)	-0.0631 (0.043)	-0.0656 (0.062)	-0.0506 (0.040)	-0.0442 (0.064)	-0.0352 (0.018)	-0.0457 (0.059)
LASSO L_1	16.3726 (1.709)	18.5910 (1.989)	12.3370 (0.788)	14.6180 (1.293)	9.6115 (0.713)	11.4436 (0.976)	11.1202 (1.722)	12.2280 (2.148)	9.1601 (0.878)	10.3994 (1.231)	7.3514 (0.590)	8.3721 (0.759)
LASSO L_2	9.1103 (1.413)	10.3456 (1.573)	5.6898 (0.918)	6.3619 (1.007)	3.9045 (0.651)	4.2321 (0.746)	2.9964 (1.096)	3.4395 (1.343)	1.7468 (0.573)	1.9543 (0.737)	1.3062 (0.358)	1.2293 (0.420)
AdaLASSO L_1	2.1423 (0.307)	2.7727 (0.403)	1.3588 (0.192)	1.7104 (0.246)	0.9493 (0.140)	1.1287 (0.176)	1.0853 (0.211)	1.3509 (0.302)	0.7370 (0.112)	0.9102 (0.173)	0.5309 (0.107)	0.6050 (0.125)
AdaLASSO L_2	1.7230 (0.290)	2.1949 (0.370)	1.0050 (0.249)	1.2551 (0.255)	0.7103 (0.165)	0.8235 (0.191)	0.4609 (0.158)	0.5706 (0.210)	0.2703 (0.104)	0.3432 (0.146)	0.2512 (0.110)	0.2483 (0.104)
Sparsity	0.8848 (0.022)	0.8955 (0.022)	0.9135 (0.014)	0.9173 (0.014)	0.9320 (0.009)	0.9319 (0.009)	0.9481 (0.024)	0.9457 (0.024)	0.9672 (0.013)	0.9667 (0.013)	0.9770 (0.006)	0.9801 (0.006)
LASSO βL_1	0.0200 (0.0088)		0.0176 (0.0072)		0.0104 (0.0053)		0.0202 (0.0118)		0.0123 (0.0071)		0.0085 (0.0054)	
AdaLASSO βL_1	0.0232 (0.0098)		0.0188 (0.0089)		0.0122 (0.0051)		0.0195 (0.0093)		0.0113 (0.0070)		0.0081 (0.0047)	
AdaLASSO FE L_1	0.1613 (0.0343)		0.1154 (0.0223)		0.0817 (0.0149)		0.1274 (0.0219)		0.0900 (0.0151)		0.0692 (0.0137)	
$N = 50$												
Specificity	95.58% (0.006)	98.35% (0.004)	98.07% (0.003)	99.74% (0.001)	98.69% (0.002)	99.83% (0.001)	99.64% (0.002)	99.75% (0.001)	99.72% (0.001)	99.86% (0.001)	99.80% (0.001)	99.88% (0.001)
Sensitivity	70.96% (0.045)	36.94% (0.069)	76.91% (0.064)	46.12% (0.062)	90.35% (0.040)	77.85% (0.041)	56.78% (0.100)	32.89% (0.105)	80.48% (0.102)	60.67% (0.111)	90.25% (0.059)	80.89% (0.086)
Bias	-0.0333 (0.012)	-0.0932 (0.020)	-0.0416 (0.016)	-0.1280 (0.014)	-0.0329 (0.010)	-0.1059 (0.010)	-0.1104 (0.029)	-0.1180 (0.037)	-0.1063 (0.021)	-0.1254 (0.026)	-0.0779 (0.014)	-0.0919 (0.018)
LASSO L_1	47.4160 (2.043)	46.7864 (2.656)	34.8124 (1.649)	34.4674 (2.040)	27.3692 (1.252)	28.5235 (1.428)	22.0065 (1.530)	18.1341 (1.637)	19.3076 (1.129)	17.4868 (1.359)	17.3026 (0.881)	16.8456 (1.179)
LASSO L_2	28.0977 (1.898)	31.8434 (1.947)	20.0399 (1.752)	27.8665 (1.905)	13.4883 (1.493)	18.4677 (1.384)	5.7025 (0.826)	6.4853 (0.889)	4.3369 (0.747)	5.0966 (0.833)	3.3312 (0.637)	3.6195 (0.717)
AdaLASSO L_1	5.5796 (0.342)	8.3495 (0.511)	3.4505 (0.274)	5.4219 (0.486)	2.1406 (0.215)	3.0267 (0.311)	2.0286 (0.232)	2.2609 (0.285)	1.4152 (0.150)	1.6103 (0.232)	1.0429 (0.131)	1.1550 (0.186)
AdaLASSO L_2	5.3234 (0.521)	8.3706 (0.621)	3.6244 (0.448)	7.3861 (0.634)	2.0232 (0.349)	4.0148 (0.421)	1.3263 (0.247)	1.7456 (0.258)	0.8646 (0.215)	1.2434 (0.257)	0.5720 (0.158)	0.7381 (0.216)
Sparsity	0.9227 (0.006)	0.9665 (0.006)	0.9441 (0.004)	0.9749 (0.004)	0.9422 (0.003)	0.9612 (0.003)	0.9907 (0.002)	0.9943 (0.002)	0.9893 (0.002)	0.9926 (0.002)	0.9889 (0.001)	0.9910 (0.001)
LASSO βL_1	0.0230 (0.0092)		0.0139 (0.0056)		0.0091 (0.0044)		0.0130 (0.0066)		0.0085 (0.0046)		0.0077 (0.0038)	
AdaLASSO βL_1	0.0289 (0.0102)		0.0305 (0.0082)		0.0315 (0.0068)		0.0147 (0.0082)		0.0106 (0.0047)		0.0100 (0.0044)	
AdaLASSO FE L_1	0.1819 (0.0254)		0.1406 (0.0206)		0.0952 (0.0108)		0.1361 (0.0170)		0.0984 (0.0114)		0.0735 (0.0087)	
$N = 75$												
Specificity	96.08% (0.002)	99.45% (0.001)	97.86% (0.001)	99.98% (0.000)	98.46% (0.001)	99.99% (0.000)	99.78% (0.002)	99.93% (0.001)	99.88% (0.000)	99.98% (0.000)	99.85% (0.001)	99.97% (0.000)
Sensitivity	54.68% (0.031)	18.35% (0.027)	60.28% (0.027)	14.90% (0.030)	78.27% (0.024)	45.94% (0.040)	43.41% (0.148)	16.95% (0.099)	56.58% (0.091)	31.48% (0.065)	88.20% (0.045)	67.25% (0.058)
Bias	0.0117 (0.010)	-0.1171 (0.017)	0.0253 (0.013)	-0.1657 (0.016)	0.0047 (0.008)	-0.1663 (0.008)	-0.1249 (0.026)	-0.1258 (0.046)	-0.1413 (0.017)	-0.1709 (0.019)	-0.1212 (0.011)	-0.1547 (0.015)
LASSO L_1	92.8770 (2.493)	89.9153 (3.313)	69.7351 (2.428)	69.7805 (3.715)	52.7383 (1.568)	51.9687 (1.608)	36.7055 (5.339)	26.7234 (4.567)	27.8182 (1.167)	21.9811 (1.181)	24.8133 (0.971)	20.1337 (0.757)
LASSO L_2	68.7268 (4.132)	74.2535 (3.921)	55.9263 (3.619)	74.1796 (3.652)	38.1962 (2.341)	63.4999 (2.540)	13.1770 (1.129)	15.6883 (1.418)	11.3665 (1.022)	14.9145 (1.259)	8.7583 (0.958)	11.4794 (0.918)
AdaLASSO L_1	10.9390 (0.493)	19.3428 (0.911)	7.1855 (0.425)	14.8527 (1.015)	4.1802 (0.235)	7.7629 (0.384)	3.5669 (0.459)	4.2456 (0.510)	2.1246 (0.179)	2.7447 (0.218)	1.4897 (0.121)	1.7537 (0.157)
AdaLASSO L_2	13.5695 (1.157)	21.1167 (1.110)	11.2269 (1.019)	21.6238 (1.092)	6.2378 (0.515)	17.0020 (0.876)	3.2520 (0.458)	4.5083 (0.486)	2.6269 (0.329)	4.1693 (0.389)	1.5833 (0.243)	2.7568 (0.274)
Sparsity	0.9353 (0.003)	0.9861 (0.003)	0.9494 (0.003)	0.9927 (0.003)	0.9462 (0.002)	0.9780 (0.002)	0.9935 (0.003)	0.9975 (0.003)	0.9933 (0.001)	0.9966 (0.001)	0.9898 (0.001)	0.9930 (0.001)
LASSO βL_1	0.0346 (0.0092)		0.0198 (0.0069)		0.0125 (0.0043)		0.0115 (0.0058)		0.0083 (0.0053)		0.0060 (0.0030)	
AdaLASSO βL_1	0.0265 (0.0104)		0.0206 (0.0090)		0.0365 (0.0065)		0.0154 (0.0065)		0.0109 (0.0056)		0.0114 (0.0033)	
AdaLASSO FE L_1	0.2151 (0.0247)		0.1862 (0.0218)		0.1294 (0.0199)		0.1519 (0.0193)		0.1103 (0.0093)		0.0856 (0.0099)	

Note: Specificity refers to the proportion of nonzero and small elements estimated as zeros. Sensitivity refers to the proportion of large elements estimated as nonzeros.

6 Application

In this application, we are interested in understanding the heterogeneity of the price response in different sectors of the economy to unexpected macroeconomic shocks. We use a disaggregated database on personal consumption expenditure price indexes (PCE) and explain by its own lags, macroeconomic indicators and oil prices in international markets. This allows us to build impulse-response functions to sector-specific shocks, taking into account complex sectoral inter-linkages that are often unobserved by researchers.

Understanding to what extent heterogenous effect takes place is an useful policy information for two reasons. First, sectoral dependence implies that a shock usually has effects beyond the sectors immediately affected. Second, since some sectors have a central role in the economy, they are able to disseminate inflation to a greater extent than others. Hence, identifying those sectors provides valuable information on the current and future state of aggregate price index.

Our approach has striking similarities with Vector Autoregressions (VARs) which are often used in the macroeconomics literature. One of the widely-used methods to identify these models in the structural form is the Cholesky decomposition, which requires a specific ordering of the variables such that any specific variable in a given period is affected by but do not affect the next variable in the order. In the current setting, this is equivalent to placing known zeros in $\mathbf{W}_1$. In our case, identification is also achieved through limiting the degree of simultaneity. However, instead of known zeros, our identification hypothesis is the sparsity of matrices $\mathbf{W}_1$ and $\mathbf{W}_2$. As shown in Corollary 6, this is an alternative way of achieving identification.

Our starting point is a VAR which contains the consumer price and producer price indexes, industrial production index, personal income, unemployment rate, stock price index, foreign exchange rate with the United Kingdom, interest rate and the exogenous oil price. However, we disaggregate the consumer price index in 66 subcategories, adding up to 74 time series in total. We use third-level disaggregation from the PCE, published by the Bureau of Economic Analysis. In this level of disaggregation, the database contains price index for public transportation, educational books, garments, and so on. The macroeconomic indicator data is obtained from Boivin et al. (2009). Similar to the paper, we analyze the period from 1976:1 to 2005:6, adding up to 353 observations over time. This period was chosen for two reasons. First, it maintains consistency between previous results in the literature. Second, it does not consider a later period in which the boom of shale gas production may have endogenized oil prices. The penalization parameter is chosen by BIC, as discussed in Section 4. In all specifications, we use six lags of the endogenous and exogenous variables.

We calculate and rank sectors according to the eigenvector centrality of the estimated $\mathbf{W}_1$. The eigenvector centrality is widely accepted as a measure of the influence of a node in a network (Jackson, 2010). The top five most influential series were, respectively, "food an nonalcoholic beverages purchased for off-premises consumption", "motor vehicle fuels, lubricants, and fluids",

"garments", "accommodations", and "household utilities", all of which were largely expected.

In Figure 1, we show the impulse-response function of a one-standard deviation oil price shock. Each line represents the response of one sector. For clarity, we highlight two series: "fuel oil and other fuels" (black line) and "food and nonalcoholic beverages purchased for off-premises consumption" (red line). As expected, the impulse-response function shows a strong and almost immediate reaction of fuel to changes in international oil prices. Yet, food prices show a year-lagged reaction to the same shock, as oil price variation sinks through the economy, possibly affecting the food production costs.

We proceed with two additional exercises: the impulse-response functions to one standard deviation in the monetary policy variable (real interest rates) and exchange rates (dollar-pounds). These results are also expected. Fuel prices respond negatively to monetary shock and is mostly unaffected by exchange rates. Food prices, meanwhile, appear much more sticky and resilient to monetary shock and slightly increase due to exchange rate shock.

This application aims to show how our procedure unveils complex interactions between sectors without the need to pre-specify the interaction matrix. The results obtained herein are largely expected and contribute to the understanding of the dissemination of price shocks through the economy.

7 Conclusion

In this paper, to overcome the arbitrariness of the specification of spatial weight matrices in a spatial autoregressive model, we introduce sparse estimation of the spatial weight matrices through penalization. With the help of instrumental variables, our major contribution is to develop a spatial weight matrix estimator which enjoys partial sign-consistency with high probability, when the true matrices are sparse or approximately sparse. Approximate sparseness is arguably closer to real life scenarios than exact sparseness when spatial interactions structure is concerned. Theoretical contribution include the use of a Nagaev-type inequality for finding a set on which the error upper bounds for the spatial weight matrices, the spatial regression parameters and the spatial fixed effects are valid with probability converging to one. Another contribution is to spell out the rate of convergence for our estimators. Practical contribution include the development of a modified block coordinate descent algorithm for numerical computations of the one-step estimator of the spatial weight matrices. Two BIC criteria are also developed for finding the best penalization parameters for both the LASSO and the adaptive LASSO problems.

While approximate sparseness is more relaxed than exact sparsity, there are situations when the spatial weight matrices are not sparse at all. In this case, it is important to develop a technique to incorporate expert knowledge on top of data estimation for finding the underlying spatial interactions structure. One way to do this can be penalization on the adjustment of an expert-specified spatial weight matrix. In certain applications, a low dimensional specification can be used

for regularizing a spatial weight matrix. These are two future research directions for estimation of spatial weight matrix.

8 Appendix

We present an important Lemma before proving the Theorems in Section 3.3.

Lemma 1 *For a zero mean time series process $\mathbf{x}_t = \mathbf{f}(\mathcal{F}_t)$ defined in (3.1) with dependence measure $\theta_{t,d,j}^x$ defined in (3.2), assume $\Theta_{m,a}^x \leq Cm^{-\alpha}$ as in Assumption T1. Then there exists constants C_1, C_2 and C_3 independent of v, T and the index j such that*

$$P\left(\left|\frac{1}{T} \sum_{t=1}^T x_{t,j}\right| > v\right) \leq \frac{C_1 T^{w(\frac{1}{2}-\tilde{\alpha})}}{(Tv)^w} + C_2 \exp(-C_3 T^{\tilde{\beta}} v^2),$$

where $\tilde{\alpha} = \alpha \wedge (1/2 - 1/w)$, and $\tilde{\beta} = (3 + 2\tilde{\alpha}w)/(1 + w)$.

Furthermore, assume another zero mean time series process $\{\mathbf{z}_t\}$ (can be the same process $\{\mathbf{x}_t\}$) with both $\Theta_{m,2w}^x, \Theta_{m,2w}^z \leq Cm^{-\alpha}$, as in Assumption T1. Then provided $\max_j \|x_{tj}\|_{2w}, \max_j \|z_{tj}\|_{2w} \leq c_0 < \infty$ where c_0 is a constant, the above Nagaev-type inequality holds for the product process $\{x_{tj}z_{t\ell} - E(x_{tj}z_{t\ell})\}$.

This is Lemma 1 of Lam and Souza (2014). This lemma concerns with the tail probability of the average of a general time series process as defined in (3.1), and is an important foundation for all the theorems in Section 3.3. Note that if $\alpha > 1/2 - 1/w$, then $w(1/2 - \tilde{\alpha}) = \tilde{\beta} = 1$, simplifying the form of the inequality. This is what we assumed in the theorems presented in Section 3.3. However, this can be relaxed at the expense of more complicated rates in those theorems, which we chose not to pursue for the sake of simplicity in presentation.

Proof of Theorem 1. Our aim is to apply Lemma 1 on the processes defined in the sets $\mathcal{A}_1$ to $\mathcal{A}_7$. To this end, we first show that with tail Assumption D3, a process $\{\mathbf{z}_t\}$ has $\max_j \|z_{tj}\|_{2w} < \infty$ for any $w > 0$. Indeed, by the Fubini's Theorem,

$$\begin{aligned} E|z_{tj}|^{2w} &= E \int_0^{|z_{tj}|^{2w}} ds = \int_0^\infty P(|z_{tj}| > s^{1/2w}) ds \\ &\leq \int_0^\infty D_1 \exp(-D_2 s^{q/2w}) ds \\ &= \frac{4wD_1}{q} \int_0^\infty x^{4w/q-1} e^{-D_2 x^2} dx \\ &= \frac{2wD_1}{qD_2^{2w/q}} \Gamma(2w/q) =: \mu_{2w}^{2w} < \infty, \end{aligned} \tag{8.1}$$

so that $\|z_{tj}\|_{2w} \leq \mu_{2w} < \infty$ for any $w > 0$. Together with Assumption T1, Lemma 1 can be applied for the processes $\{B_{t,ij}X_{t,k\ell} - E(B_{t,ij}X_{t,k\ell})\}$, $\{B_{t,ij}\epsilon_{t,k}\}$, $\{B_{t,ij} - \mu_{b,ij}\}$, $\{\epsilon_{t,j}\}$ and $\{X_{t,ij}\}$ in the sets

$\mathcal{A}_1, \mathcal{A}_2, \mathcal{A}_4, \mathcal{A}_5$ and $\mathcal{A}_6$ respectively. Since we assumed $\alpha > 1/2 - 1/w$, we have $w(1/2 - \tilde{\alpha}) = \tilde{\beta} = 1$ in Lemma 1. By the union sum inequality, we then have

$$\begin{aligned}
P(\mathcal{A}_1^c) &\leq \sum_{\substack{1 \leq i, k \leq N \\ 1 \leq j, \ell \leq K}} P\left(\left|T^{-1} \sum_{t=1}^T B_{t,ij} X_{t,k\ell} - E(B_{t,ij} X_{t,k\ell})\right| \geq \lambda_T\right) \\
&\leq N^2 K^2 \left(\frac{C_1 T}{(T \lambda_T)^w} + C_2 \exp(-C_3 T \lambda_T^2) \right) \\
&\leq C_1 K^2 \left(\frac{C_3}{3} \right)^{w/2} \frac{N^2}{T^{w/2-1} \log^{w/2}(T \vee N)} + \frac{C_2 K^2 N^2}{T^3 \vee N^3}.
\end{aligned} \tag{8.2}$$

Similarly, we have

$$\begin{aligned}
P(\mathcal{A}_2^c) &\leq C_1 K \left(\frac{C_3}{3} \right)^{w/2} \frac{N^2}{T^{w/2-1} \log^{w/2}(T \vee N)} + \frac{C_2 K N^2}{T^3 \vee N^3}, \\
P(\mathcal{A}_4^c) &\leq C_1 K \left(\frac{C_3}{3} \right)^{w/2} \frac{N}{T^{w/2-1} \log^{w/2}(T \vee N)} + \frac{C_2 K N}{T^3 \vee N^3}, \\
P(\mathcal{A}_5^c) &\leq C_1 \left(\frac{C_3}{3} \right)^{w/2} \frac{N}{T^{w/2-1} \log^{w/2}(T \vee N)} + \frac{C_2 N}{T^3 \vee N^3}, \\
P(\mathcal{A}_6^c) &\leq C_1 K \left(\frac{C_3}{3} \right)^{w/2} \frac{N}{T^{w/2-1} \log^{w/2}(T \vee N)} + \frac{C_2 K N}{T^3 \vee N^3}.
\end{aligned} \tag{8.3}$$

To find an upper bound for $P(\mathcal{A}_3^c)$, define $\mathbf{B}_{t,k}$ to be the k th column of $\mathbf{B}_t$. If we can show that

$$\max_{1 \leq k \leq K} \left\| N^{-\frac{1}{2} - \frac{1}{2w}} \boldsymbol{\epsilon}_t^T \mathbf{B}_{t,k} \right\|_{2w} < \infty, \tag{8.4}$$

$$\Theta_{m,2w} = \sum_{t=m}^{\infty} \max_{1 \leq k \leq K} \left\| N^{-\frac{1}{2} - \frac{1}{2w}} (\boldsymbol{\epsilon}_t^T \mathbf{B}_{t,k} - \boldsymbol{\epsilon}_t'^T \mathbf{B}_{t,k}') \right\|_{2w} \leq a m^{-\alpha}, \tag{8.5}$$

for some $a > 0$ and all $m \geq 1$, then we can apply Lemma 1 for $\mathcal{A}_3$ to obtain

$$\begin{aligned}
P(\mathcal{A}_3^c) &\leq \sum_{k=1}^K P\left(\left|T^{-1} \sum_{t=1}^T N^{-\frac{1}{2} - \frac{1}{2w}} \boldsymbol{\epsilon}_t^T \mathbf{B}_{t,k}\right| \geq \lambda_T\right) \\
&\leq C_1 \left(\frac{C_3}{3} \right)^{w/2} \frac{K}{T^{w/2-1} \log^{w/2}(T \vee N)} + \frac{C_2 K}{T^3 \vee N^3}.
\end{aligned} \tag{8.6}$$

To show (8.4), write $\sum_{j=1}^N B_{t,jk} \epsilon_{t,j} = \mathbf{B}_{t,k}^T \boldsymbol{\epsilon}_t = \mathbf{B}_{t,k}^T \boldsymbol{\Sigma}_{\epsilon}^{1/2} \boldsymbol{\epsilon}_t^* = \sum_{j=1}^N (\mathbf{B}_{t,k}^T \boldsymbol{\Sigma}_{\epsilon}^{1/2})_j \epsilon_{t,j}^*$, where $\{\boldsymbol{\epsilon}_t^*\}$ is as in Assumption D2. Then by the independence of $\{\mathbf{B}_t\}$ and $\{\boldsymbol{\epsilon}_t\}$ (thus $\{\boldsymbol{\epsilon}_t^*\}$) assumed in D1,

$$\begin{aligned}
&E((\mathbf{B}_{t,k}^T \boldsymbol{\Sigma}_{\epsilon}^{1/2})_j \epsilon_{t,j}^* | (\mathbf{B}_{t,k}^T \boldsymbol{\Sigma}_{\epsilon}^{1/2})_s, \epsilon_{t,s}^*, s \leq j-1) \\
&= E((\mathbf{B}_{t,k}^T \boldsymbol{\Sigma}_{\epsilon}^{1/2})_j | (\mathbf{B}_{t,k}^T \boldsymbol{\Sigma}_{\epsilon}^{1/2})_s, s \leq j-1) \cdot E(\epsilon_{t,j}^* | \epsilon_{t,s}^*, s \leq j-1) = 0,
\end{aligned}$$

since $\{\epsilon_{t,j}^*\}_{1 \leq j \leq N}$ is a martingale difference. Hence $\{(\mathbf{B}_{t,k}^\top \Sigma_\epsilon^{1/2})_j \epsilon_{t,j}^*\}_{1 \leq j \leq N}$ is a martingale difference. By Lemma 2.1 of Li (2003), Assumptions D1, D2 and (8.1), we then have,

$$\begin{aligned}
E \left| N^{-\frac{1}{2} - \frac{1}{2w}} \mathbf{B}_{t,k}^\top \epsilon_t \right|^{2w} &= E \left| N^{-\frac{1}{2} - \frac{1}{2w}} \sum_{j=1}^N (\mathbf{B}_{t,k}^\top \Sigma_\epsilon^{1/2})_j \epsilon_{t,j}^* \right|^{2w} \\
&\leq N^{-2} (36w)^{2w} (1 + (2w-1)^{-1})^w \sum_{j=1}^N E |(\mathbf{B}_{t,k}^\top \Sigma_\epsilon^{1/2})_j \epsilon_{t,j}^*|^{2w} \\
&= N^{-2} (36w)^{2w} (1 + (2w-1)^{-1})^w \sum_{j=1}^N E |(\mathbf{B}_{t,k}^\top \Sigma_\epsilon^{1/2})_j|^{2w} E |\epsilon_{t,j}^*|^{2w} \\
&\leq N^{-2} (36w \mu_{2w})^{2w} (1 + (2w-1)^{-1})^w \sum_{j=1}^N E \left| \max_{1 \leq j \leq N} |B_{t,jk}| \right|^{2w} \|\Sigma_\epsilon^{1/2}\|_\infty^{2w} \\
&\leq N^{-2} (36w \mu_{2w} S_\epsilon)^{2w} (1 + (2w-1)^{-1})^w \sum_{j=1}^N N \max_{1 \leq j \leq N} E |B_{t,jk}|^{2w} \\
&\leq (36w \mu_{2w}^2 S_\epsilon)^{2w} (1 + (2w-1)^{-1})^w < \infty,
\end{aligned}$$

so that $\max_{1 \leq k \leq K} \|N^{-\frac{1}{2} - \frac{1}{2w}} \mathbf{B}_{t,k}^\top \epsilon_t\|_{2w} < \infty$, which is (8.4).

To show (8.5), observe that

$$\begin{aligned}
\Theta_{m,2w} &\leq \sum_{t=m}^{\infty} \max_{1 \leq k \leq K} N^{-\frac{1}{2} - \frac{1}{2w}} \left[\|\mathbf{B}_{t,k}^\top \Sigma_\epsilon^{1/2} (\epsilon_t^* - \epsilon_t'^*)\|_{2w} \right. \\
&\quad \left. + \|(\mathbf{B}_{t,k}^\top \Sigma_\epsilon^{1/2} - \mathbf{B}_{t,k}'^\top \Sigma_\epsilon^{1/2}) \epsilon_t'^*\|_{2w} \right] \\
&\leq \sum_{t=m}^{\infty} \max_{1 \leq k \leq K} N^{-\frac{1}{2} - \frac{1}{2w}} \left[\left\| \sum_{j=1}^N (\mathbf{B}_{t,k}^\top \Sigma_\epsilon^{1/2})_j (\epsilon_{t,j}^* - \epsilon_{t,j}'^*) \right\|_{2w} \right. \\
&\quad \left. + \left\| \sum_{j=1}^N (\mathbf{B}_{t,k}^\top \Sigma_\epsilon^{1/2} - \mathbf{B}_{t,k}'^\top \Sigma_\epsilon^{1/2})_j \epsilon_{t,j}'^* \right\|_{2w} \right].
\end{aligned}$$

The terms $\{(\mathbf{B}_{t,k}^\top \Sigma_\epsilon^{1/2})_j (\epsilon_{t,j}^* - \epsilon_{t,j}'^*)\}_j$ and $\{(\mathbf{B}_{t,k}^\top \Sigma_\epsilon^{1/2} - \mathbf{B}_{t,k}'^\top \Sigma_\epsilon^{1/2})_j \epsilon_{t,j}'^*\}_j$ can be shown to be martingale differences with respect to the filtration

$$\mathcal{F}_j = \sigma(\epsilon_{t,s}^*, \epsilon_{t,s}'^*, (\mathbf{B}_{t,k}^\top \Sigma_\epsilon^{1/2})_s, (\mathbf{B}_{t,k}'^\top \Sigma_\epsilon^{1/2})_s, s \leq j),$$

using similar arguments as before. Hence we can use Lemma 2.1 of Li (2003), Assumptions D1, D2, T1 and (8.1) to show that

$$\begin{aligned}
& \left\| N^{-\frac{1}{2}-\frac{1}{2w}} \sum_{j=1}^N (\mathbf{B}_{t,k}^T \boldsymbol{\Sigma}_\epsilon^{1/2})_j (\epsilon_{t,j}^* - \epsilon_{t,j}'^*) \right\|_{2w} \\
& \leq 36w(1 + (2w-1)^{-1})^{1/2} \\
& \quad \cdot \left[N^{-2} \sum_{j=1}^N E \left| \max_{1 \leq j \leq N} |B_{t,jk}| \right|^{2w} \left\| \boldsymbol{\Sigma}_\epsilon^{1/2} \right\|_\infty^{2w} (\theta_{t,2w,j}^{\epsilon^*})^{2w} \right]^{1/2w} \\
& \leq 36w\mu_w S_\epsilon (1 + (2w-1)^{-1})^{1/2} \max_{1 \leq j \leq N} \theta_{t,2w,j}^{\epsilon^*}.
\end{aligned}$$

Similarly,

$$\begin{aligned}
& \left\| N^{-\frac{1}{2}-\frac{1}{2w}} \sum_{j=1}^N (\mathbf{B}_{t,k}^T \boldsymbol{\Sigma}_\epsilon^{1/2} - \mathbf{B}_{t,k}'^T \boldsymbol{\Sigma}_\epsilon'^{1/2})_j \epsilon_{t,j}'^* \right\|_{2w} \\
& \leq 36w\mu_w S_\epsilon (1 + (2w-1)^{-1})^{1/2} \max_{1 \leq j \leq NK} \theta_{t,2w,j}^b.
\end{aligned}$$

Hence combining and using Assumption T1, we have

$$\begin{aligned}
\Theta_{m,2w} & \leq 36w\mu_w S_\epsilon (1 + (2w-1)^{-1})^{1/2} (\Theta_{m,2w}^{\epsilon^*} + \Theta_{m,2w}^b) \\
& \leq 72Cw\mu_w S_\epsilon (1 + (2w-1)^{-1})^{1/2} m^{-\alpha},
\end{aligned}$$

which is (8.5).

Finally, to find an upper bound for $P(\mathcal{A}_7^c)$, write $\sum_{j=1}^N \bar{B}_{\cdot,jk} \bar{\epsilon}_{\cdot,j} = \sum_{j=1}^N (\bar{\mathbf{B}}_{\cdot,k}^T \boldsymbol{\Sigma}_\epsilon^{1/2})_j \bar{\epsilon}_{\cdot,j}^*$, where $\bar{\mathbf{B}}_{\cdot,k}$ is the sample mean of $\{\mathbf{B}_{t,k}\}$, and similarly for $\bar{\epsilon}_{\cdot,j}$. By the independence of $\{\mathbf{B}_t\}$ and $\{\epsilon_t\}$,

$$\begin{aligned}
& E((\bar{\mathbf{B}}_{\cdot,k}^T \boldsymbol{\Sigma}_\epsilon^{1/2})_j \bar{\epsilon}_{\cdot,j}^* | (\mathbf{B}_{t,k}^T \boldsymbol{\Sigma}_\epsilon^{1/2})_s, \epsilon_{t,s}^*, t = 1, \dots, T, s \leq j-1) \\
& = E((\bar{\mathbf{B}}_{\cdot,k}^T \boldsymbol{\Sigma}_\epsilon^{1/2})_j | (\mathbf{B}_{t,k}^T \boldsymbol{\Sigma}_\epsilon^{1/2})_s, t = 1, \dots, T, s \leq j-1) \cdot E(\bar{\epsilon}_{\cdot,j}^* | \epsilon_{t,s}^*, t = 1, \dots, T, s \leq j-1) = 0,
\end{aligned}$$

since $\{\epsilon_{t,j}^*\}_{1 \leq j \leq N}$ is a martingale difference. Hence $\{(\bar{\mathbf{B}}_{\cdot,k}^T \boldsymbol{\Sigma}_\epsilon^{1/2})_j \bar{\epsilon}_{\cdot,j}^*\}_{1 \leq j \leq N}$ is a martingale difference. Moreover, on $\mathcal{A}_4 \cap \mathcal{A}_5$, it is easy to show that $\max_j |(\bar{\mathbf{B}}_{\cdot,k}^T \boldsymbol{\Sigma}_\epsilon^{1/2})_j \bar{\epsilon}_{\cdot,j}^*| \leq \lambda_T(\mu_{b,\max} + \lambda_T)S_\epsilon$.

Hence, we can apply the Azuma's inequality to get

$$\begin{aligned}
P(\mathcal{A}_7^c) &\leq P(\mathcal{A}_7^c \cap \mathcal{A}_4 \cap \mathcal{A}_5) + P(\mathcal{A}_4^c \cup \mathcal{A}_5^c), \\
&\leq KP \left(\left| \sum_{j=1}^N (\bar{\mathbf{B}}_{:,k}^T \boldsymbol{\Sigma}_\epsilon^{1/2})_j \bar{\epsilon}_{:,j}^* \right| \geq 2^{1/2} \lambda_T N^{1/2} (\mu_{b,\max} + \lambda_T) S_\epsilon \log^{1/2}(T \vee N), \mathcal{A}_4 \cap \mathcal{A}_5 \right) \\
&\quad + P(\mathcal{A}_4^c) + P(\mathcal{A}_5^c) \\
&\leq 2K \exp \left(\frac{-2\lambda_T^2 N (\mu_{b,\max} + \lambda_T)^2 S_\epsilon^2 \log(T \vee N)}{2N\lambda_T^2 (\mu_{b,\max} + \lambda_T)^2 S_\epsilon^2} \right) + P(\mathcal{A}_4^c) + P(\mathcal{A}_5^c) \\
&= \frac{2K}{T \vee N} + P(\mathcal{A}_4^c) + P(\mathcal{A}_5^c). \tag{8.7}
\end{aligned}$$

Combining (8.2), (8.3), (8.6) and (8.7), and using $P(\mathcal{M}) \geq 1 - \sum_{j=1}^7 P(\mathcal{A}_j^c)$, we can arrive at the result as stated in the theorem. $\square$

Proof of Theorem 2. Define the matrices $\tilde{\mathbf{V}}$ and $\mathbf{U}$, where

$$\tilde{\mathbf{V}} = \begin{pmatrix} \mathbf{I}_K \otimes \tilde{\mathbf{w}}_{2,1} \\ \vdots \\ \mathbf{I}_K \otimes \tilde{\mathbf{w}}_{2,N} \end{pmatrix}, \quad \mathbf{U} = \mathbf{I}_N \otimes T^{-1} \sum_{t=1}^T (\mathbf{b}_t - \bar{\mathbf{b}}) \mathbf{x}_t^T,$$

with $\tilde{\mathbf{w}}_{2,j}^T$ the j th row of $\tilde{\mathbf{W}}_2$. Also, define

$$E\mathbf{U}_0 = \mathbf{I}_N \otimes E \left(T^{-1} \sum_{t=1}^T \mathbf{b}_t \mathbf{x}_t^T \right) = \mathbf{I}_N \otimes E(\mathbf{b}_1 \mathbf{x}_1^T).$$

Assumption D4 in Section 3.2 ensures that there is a constant $u > 0$ so that $\lambda_{\min}(E\mathbf{U}_0 E\mathbf{U}_0^T) \geq u > 0$ uniformly as $T, N \rightarrow \infty$, where $\lambda_{\min}(\mathbf{A})$ is the smallest eigenvalue for the matrix $\mathbf{A}$. Then we can decompose $\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta} = \sum_{i=1}^5 I_i$, where

$$\begin{aligned}
I_1 &= (\tilde{\mathbf{V}}^T E\mathbf{U}_0^T \tilde{\mathbf{V}} \tilde{\mathbf{V}}^T E\mathbf{U}_0 \tilde{\mathbf{V}})^{-1} \tilde{\mathbf{V}}^T [E\mathbf{U}_0^T \tilde{\mathbf{V}} \tilde{\mathbf{V}}^T E\mathbf{U}_0 - \mathbf{U}^T \tilde{\mathbf{V}} \tilde{\mathbf{V}}^T \mathbf{U}] \tilde{\mathbf{V}} (\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta}^*), \\
I_2 &= (\tilde{\mathbf{V}}^T E\mathbf{U}_0^T \tilde{\mathbf{V}} \tilde{\mathbf{V}}^T E\mathbf{U}_0 \tilde{\mathbf{V}})^{-1} T^{-2} \mathbf{X}^T \tilde{\mathbf{W}}_2^{\otimes T} \tilde{\mathbf{W}}_2^{\otimes} \mathbf{B}^v \mathbf{B}^{vT} \tilde{\mathbf{W}}_2^{\otimes T} (\mathbf{W}_1^{*\otimes} - \tilde{\mathbf{W}}_1^{\otimes}) \Pi_1^{*\otimes} \mathbf{W}_2^{*\otimes} \mathbf{X} \boldsymbol{\beta}^*, \\
I_3 &= (\tilde{\mathbf{V}}^T E\mathbf{U}_0^T \tilde{\mathbf{V}} \tilde{\mathbf{V}}^T E\mathbf{U}_0 \tilde{\mathbf{V}})^{-1} T^{-2} \mathbf{X}^T \tilde{\mathbf{W}}_2^{\otimes T} \tilde{\mathbf{W}}_2^{\otimes} \mathbf{B}^v \mathbf{B}^{vT} \tilde{\mathbf{W}}_2^{\otimes T} (\mathbf{W}_1^{*\otimes} - \tilde{\mathbf{W}}_1^{\otimes}) \Pi_1^{*\otimes} \boldsymbol{\epsilon}^v, \\
I_4 &= (\tilde{\mathbf{V}}^T E\mathbf{U}_0^T \tilde{\mathbf{V}} \tilde{\mathbf{V}}^T E\mathbf{U}_0 \tilde{\mathbf{V}})^{-1} T^{-2} \mathbf{X}^T \tilde{\mathbf{W}}_2^{\otimes T} \tilde{\mathbf{W}}_2^{\otimes} \mathbf{B}^v \mathbf{B}^{vT} \tilde{\mathbf{W}}_2^{\otimes T} (\mathbf{W}_2^{*\otimes} - \tilde{\mathbf{W}}_2^{\otimes}) \mathbf{X} \boldsymbol{\beta}^*, \\
I_5 &= (\tilde{\mathbf{V}}^T E\mathbf{U}_0^T \tilde{\mathbf{V}} \tilde{\mathbf{V}}^T E\mathbf{U}_0 \tilde{\mathbf{V}})^{-1} T^{-2} \mathbf{X}^T \tilde{\mathbf{W}}_2^{\otimes T} \tilde{\mathbf{W}}_2^{\otimes} \mathbf{B}^v \mathbf{B}^{vT} \tilde{\mathbf{W}}_2^{\otimes T} \boldsymbol{\epsilon}^v,
\end{aligned}$$

where $\boldsymbol{\epsilon}^v = (\boldsymbol{\epsilon}_1^T, \dots, \boldsymbol{\epsilon}_T^T)^T$. Hence we have $\|\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\|_1 \leq \sum_{i=1}^5 \|I_i\|_1$. To bound these terms, first note that

$$\lambda_{\min}(\tilde{\mathbf{V}}^T E\mathbf{U}_0^T \tilde{\mathbf{V}} \tilde{\mathbf{V}}^T E\mathbf{U}_0 \tilde{\mathbf{V}}) \geq \lambda_{\min}^2(\tilde{\mathbf{V}}^T \tilde{\mathbf{V}}) \lambda_{\min}(E\mathbf{U}_0^T E\mathbf{U}_0) \geq uN^2,$$

since $\tilde{\mathbf{V}}^T \tilde{\mathbf{V}}$ is diagonal with each diagonal element at least N . Hence

$$\|(\tilde{\mathbf{V}}^T E \mathbf{U}_0^T \tilde{\mathbf{V}} \tilde{\mathbf{V}}^T E \mathbf{U}_0 \tilde{\mathbf{V}})^{-1}\|_\infty \leq \frac{K^{1/2}}{\lambda_{\min}(\tilde{\mathbf{V}}^T E \mathbf{U}_0^T \tilde{\mathbf{V}} \tilde{\mathbf{V}}^T E \mathbf{U}_0 \tilde{\mathbf{V}})} \leq \frac{K^{1/2}}{uN^2}. \quad (8.8)$$

To bound $\|I_1\|$, on $\mathcal{A}_1 \cap \mathcal{A}_4 \cap \mathcal{A}_6$, we have $\|E \mathbf{U}_0 - \mathbf{U}\|_{\max} \leq \lambda_T(1 + \mu_{b,\max} + \lambda_T)$. Also by Assumption D4, $\|E \mathbf{U}_0\|_{\max} \leq \sigma_{bx} < \infty$ uniformly for some constant $\sigma_{bx} > 0$. Hence,

$$\begin{aligned} \|I_1\|_1 &\leq \|(\tilde{\mathbf{V}}^T E \mathbf{U}_0^T \tilde{\mathbf{V}} \tilde{\mathbf{V}}^T E \mathbf{U}_0 \tilde{\mathbf{V}})^{-1}\|_1 \left\{ \|\tilde{\mathbf{V}}^T (E \mathbf{U}_0 - \mathbf{U})^T \tilde{\mathbf{V}} \tilde{\mathbf{V}}^T E \mathbf{U}_0\|_1 \right. \\ &\quad \left. + \|\tilde{\mathbf{V}}^T \mathbf{U}^T \tilde{\mathbf{V}} \tilde{\mathbf{V}}^T (E \mathbf{U}_0 - \mathbf{U})\|_1 \right\} \|\tilde{\mathbf{V}}(\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta}^*)\|_1 \\ &\leq \frac{K^{1/2}}{uN^2} \left\{ \|\tilde{\mathbf{V}}^T (E \mathbf{U}_0 - \mathbf{U})^T\|_1 \cdot 2N \cdot \|\tilde{\mathbf{V}}^T E \mathbf{U}_0\|_1 + \|\tilde{\mathbf{V}}^T (E \mathbf{U}_0 - \mathbf{U})^T\|_1^2 \cdot 2N \right. \\ &\quad \left. + \|\tilde{\mathbf{V}}^T E \mathbf{U}_0^T\|_1 \cdot 2N \cdot \|\tilde{\mathbf{V}}^T (E \mathbf{U}_0 - \mathbf{U})\|_1 \right\} \cdot 2N \|\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\|_1 \\ &\leq \frac{K^{1/2}}{uN^2} \cdot 16N^2 K^2 \lambda_T(1 + \mu_{b,\max} + \lambda_T)(2\sigma_{bx} + \lambda_T(1 + \mu_{b,\max} + \lambda_T)) \|\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\|_1 \\ &= O(\lambda_T \|\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\|_1). \end{aligned}$$

Similarly, on $\mathcal{A}_1 \cap \mathcal{A}_4 \cap \mathcal{A}_6$,

$$\begin{aligned} \|I_2\|_1 &\leq \frac{K^{1/2}}{uN^2} \cdot \|T^{-1} \mathbf{X}^T \tilde{\mathbf{W}}_2^{\otimes T} \tilde{\mathbf{W}}_2^{\otimes} \mathbf{B}^v\|_1 \cdot \|T^{-1} \mathbf{B}^{vT} \tilde{\mathbf{W}}_2^{\otimes T} (\mathbf{W}_1^{*\otimes} - \tilde{\mathbf{W}}_1^{\otimes}) \boldsymbol{\Pi}_1^{*\otimes} \mathbf{W}_2^{*\otimes} \mathbf{X} \boldsymbol{\beta}^*\|_1 \\ &\leq \frac{K^{1/2}}{uN^2} \cdot 4KN[\sigma_{bx} + \lambda_T(1 + \mu_{b,\max} + \lambda_T)] \cdot \frac{4K\|\boldsymbol{\beta}^*\|_1(\sigma_{bx} + \lambda_T(1 + \mu_{b,\max} + \lambda_T))}{1 - \eta} \|\tilde{\boldsymbol{\xi}}_1 - \boldsymbol{\xi}_1^*\|_1 \\ &= O(\|\tilde{\boldsymbol{\xi}}_1 - \boldsymbol{\xi}_1^*\|_1 N^{-1}). \end{aligned}$$

On $\mathcal{A}_1 \cap \mathcal{A}_2 \cap \mathcal{A}_4 \cap \mathcal{A}_5 \cap \mathcal{A}_6$, we have

$$\begin{aligned} \|I_3\|_1 &\leq \frac{K^{1/2}}{uN^2} \cdot 4KN[\sigma_{bx} + \lambda_T(1 + \mu_{b,\max} + \lambda_T)] \cdot \frac{2K\lambda_T(1 + \mu_{b,\max} + \lambda_T)}{1 - \eta} \|\tilde{\boldsymbol{\xi}}_1 - \boldsymbol{\xi}_1^*\|_1 \\ &= O(\lambda_T \|\tilde{\boldsymbol{\xi}}_1 - \boldsymbol{\xi}_1^*\|_1 N^{-1}). \end{aligned}$$

On $\mathcal{A}_1 \cap \mathcal{A}_4 \cap \mathcal{A}_6$, we have

$$\begin{aligned} \|I_4\|_1 &\leq \frac{K^{1/2}}{uN^2} \cdot 4KN[\sigma_{bx} + \lambda_T(1 + \mu_{b,\max} + \lambda_T)] \cdot 2K\|\boldsymbol{\beta}^*\|_1(\sigma_{bx} + \lambda_T(1 + \mu_{b,\max} + \lambda_T)) \|\tilde{\boldsymbol{\xi}}_2 - \boldsymbol{\xi}_2^*\|_1 \\ &= O(\|\tilde{\boldsymbol{\xi}}_2 - \boldsymbol{\xi}_2^*\|_1 N^{-1}). \end{aligned}$$

Finally, on $\mathcal{M}$, we have

$$\begin{aligned} \|T^{-1}\mathbf{B}^{v\top}\widetilde{\mathbf{W}}_2^{\otimes\top}\boldsymbol{\epsilon}^v\|_1 &\leq K\left(\lambda_T(1+\mu_{b,\max}+\lambda_T)\|\tilde{\boldsymbol{\xi}}_2-\boldsymbol{\xi}_2^*\|_1+\max_{1\leq k\leq K}\left|T^{-1}\sum_{t=1}^T\sum_{s=1}^N(B_{t,sk}-\bar{B}_{\cdot,sk})\epsilon_{t,s}\right|\right. \\ &\quad \left.+\lambda_T s_2(1+\mu_{b,\max}+\lambda_T)\right) \\ &\leq K(\lambda_T(1+\mu_{b,\max}+\lambda_T)(s_2+\|\tilde{\boldsymbol{\xi}}_2-\boldsymbol{\xi}_2^*\|_1)+\lambda_T N^{\frac{1}{2}+\frac{1}{2w}} \\ &\quad +2^{1/2}S_\epsilon(\mu_{b,\max}+\lambda_T)\lambda_T N^{1/2}\log^{1/2}(T\vee N)), \end{aligned}$$

so that on $\mathcal{M}$,

$$\|I_5\|_1 = O\left(\frac{\lambda_T(s_2+N^{\frac{1}{2}+\frac{1}{2w}}+N^{1/2}\log^{1/2}(T\vee N))}{N}+\frac{\lambda_T\|\tilde{\boldsymbol{\xi}}_2-\boldsymbol{\xi}_2^*\|_1}{N}\right).$$

Hence, we can see that $\|I_2\|_1$, $\|I_4\|_1$ and $\|I_5\|_1$ all have potentially dominating terms. Note also that $\|\tilde{\boldsymbol{\xi}}_1-\boldsymbol{\xi}_1^*\|_1+\|\tilde{\boldsymbol{\xi}}_2-\boldsymbol{\xi}_2^*\|_1=\|\tilde{\boldsymbol{\xi}}-\boldsymbol{\xi}^*\|_1$, so that on $\mathcal{M}$,

$$\|\tilde{\boldsymbol{\beta}}-\boldsymbol{\beta}^*\|_1 = O\left(\frac{\|\tilde{\boldsymbol{\xi}}-\boldsymbol{\xi}^*\|_1}{N}+\frac{\lambda_T(s_2+N^{\frac{1}{2}+\frac{1}{2w}}+N^{1/2}\log^{1/2}(T\vee N))}{N}\right),$$

which is the result we want. $\square$

Proof of Theorem 3. For the LASSO solution $\tilde{\boldsymbol{\xi}}$ in (2.5), it satisfies

$$\frac{1}{2T}\|\mathbf{B}^\top\mathbf{y}-\mathbf{B}^\top\mathbf{M}_{\tilde{\boldsymbol{\beta}}}\tilde{\boldsymbol{\beta}}\|^2\leq\frac{1}{2T}\|\mathbf{B}^\top\mathbf{y}-\mathbf{B}^\top\mathbf{M}_{\boldsymbol{\beta}(\boldsymbol{\xi}^*)}\boldsymbol{\xi}^*\|^2+\gamma_T(\|\boldsymbol{\xi}^*\|_1-\|\tilde{\boldsymbol{\xi}}\|_1).$$

Expanding $\mathbf{y}$ using model (2.2), and noting that $\mathbf{B}^\top(\boldsymbol{\mu}^*\otimes\mathbf{1}_T)=\mathbf{0}$ and $(\mathbf{M}_{\boldsymbol{\beta}^*}-\mathbf{M}_{\boldsymbol{\beta}(\boldsymbol{\xi}^*)})\boldsymbol{\xi}=\mathbf{X}_{\boldsymbol{\beta}^*-\boldsymbol{\beta}(\boldsymbol{\xi}^*)}\boldsymbol{\xi}_2$, the above becomes

$$\begin{aligned} &\frac{1}{2T}\|(\mathbf{B}^\top\mathbf{X}_{\boldsymbol{\beta}^*-\boldsymbol{\beta}(\boldsymbol{\xi}^*)}\boldsymbol{\xi}_2^*+\mathbf{B}^\top\boldsymbol{\epsilon})+\mathbf{B}^\top(\mathbf{M}_{\boldsymbol{\beta}(\boldsymbol{\xi}^*)}\boldsymbol{\xi}^*-\mathbf{M}_{\boldsymbol{\beta}(\tilde{\boldsymbol{\xi}})}\tilde{\boldsymbol{\xi}})\|^2 \\ &\leq\frac{1}{2T}\|\mathbf{B}^\top\mathbf{X}_{\boldsymbol{\beta}^*-\boldsymbol{\beta}(\boldsymbol{\xi}^*)}\boldsymbol{\xi}_2^*+\mathbf{B}^\top\boldsymbol{\epsilon}\|^2+\gamma_T(\|\boldsymbol{\xi}^*\|_1-\|\tilde{\boldsymbol{\xi}}\|_1), \end{aligned}$$

so that expanding the square norm on the left hand side and simplifying, we have

$$\begin{aligned} \frac{1}{2T}\|\mathbf{B}^\top(\mathbf{M}_{\boldsymbol{\beta}(\boldsymbol{\xi}^*)}\boldsymbol{\xi}^*-\mathbf{M}_{\boldsymbol{\beta}(\tilde{\boldsymbol{\xi}})}\tilde{\boldsymbol{\xi}})\|^2 &\leq\sum_{j=1}^6D_j+\gamma_T(\|\boldsymbol{\xi}^*\|_1-\|\tilde{\boldsymbol{\xi}}\|_1), \text{ where} \\ D_1 &=\frac{1}{T}\boldsymbol{\epsilon}^\top\mathbf{B}\mathbf{B}^\top\mathbf{X}_{\boldsymbol{\beta}(\tilde{\boldsymbol{\xi}})-\boldsymbol{\beta}^*}\tilde{\boldsymbol{\xi}}_2, \quad D_2=\frac{1}{T}\boldsymbol{\epsilon}^\top\mathbf{B}\mathbf{B}^\top\mathbf{M}_{\boldsymbol{\beta}^*}(\tilde{\boldsymbol{\xi}}-\boldsymbol{\xi}^*), \quad D_3=\frac{1}{T}\boldsymbol{\epsilon}^\top\mathbf{B}\mathbf{B}^\top\mathbf{X}_{\boldsymbol{\beta}^*-\boldsymbol{\beta}(\boldsymbol{\xi}^*)}\boldsymbol{\xi}_2^*, \\ D_4 &=\frac{1}{T}\boldsymbol{\xi}_2^{*\top}\mathbf{X}_{\boldsymbol{\beta}^*-\boldsymbol{\beta}(\boldsymbol{\xi}^*)}^\top\mathbf{B}\mathbf{B}^\top\mathbf{X}_{\boldsymbol{\beta}(\tilde{\boldsymbol{\xi}})-\boldsymbol{\beta}^*}\tilde{\boldsymbol{\xi}}_2, \quad D_5=\frac{1}{T}\boldsymbol{\xi}_2^{*\top}\mathbf{X}_{\boldsymbol{\beta}^*-\boldsymbol{\beta}(\boldsymbol{\xi}^*)}^\top\mathbf{B}\mathbf{B}^\top\mathbf{M}_{\boldsymbol{\beta}^*}(\tilde{\boldsymbol{\xi}}-\boldsymbol{\xi}^*), \\ D_6 &=\frac{1}{T}\boldsymbol{\xi}_2^{*\top}\mathbf{X}_{\boldsymbol{\beta}^*-\boldsymbol{\beta}(\boldsymbol{\xi}^*)}^\top\mathbf{B}\mathbf{B}^\top\mathbf{X}_{\boldsymbol{\beta}^*-\boldsymbol{\beta}(\boldsymbol{\xi}^*)}\boldsymbol{\xi}_2^*. \end{aligned}$$

Hereafter, we use C to denote a generic constant, which can be different from line to line. On the set $\mathcal{M}$, by Assumption R2, we have

$$\begin{aligned}
|D_1| &\leq \|T^{-1}\boldsymbol{\epsilon}^T(\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)\|_{\max} \cdot N^{-a} \|T^{-1}(\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)^T \mathbf{X}_{\beta(\tilde{\boldsymbol{\xi}}) - \beta^*}\|_1 \cdot \|\tilde{\boldsymbol{\xi}}_2\|_1 \\
&\leq \lambda_T \|\gamma\|_1 (1 + \mu_{b,\max} + \lambda_T) \cdot N^{1-a} \|\gamma\|_1 \|\beta(\tilde{\boldsymbol{\xi}}) - \beta^*\|_1 (C_{bx} N^{a-1} + \lambda_T (1 + \mu_{b,\max} + \lambda_T)) \\
&\quad \cdot (s_2 + N + \|\tilde{\boldsymbol{\xi}}_2 - \boldsymbol{\xi}_2^*\|_1) \\
&\leq C \lambda_T \|\beta(\tilde{\boldsymbol{\xi}}) - \beta^*\|_1 (s_2 + N + \|\tilde{\boldsymbol{\xi}}_2 - \boldsymbol{\xi}_2^*\|_1).
\end{aligned}$$

For D_2 , it can be decomposed as $D_2 = I_1 + I_2$, where

$$I_1 = \frac{1}{T} \boldsymbol{\epsilon}^T \mathbf{B} \mathbf{B}^T \mathbf{Z} (\tilde{\boldsymbol{\xi}}_1 - \boldsymbol{\xi}_1^*), \quad I_2 = \frac{1}{T} \boldsymbol{\epsilon}^T \mathbf{B} \mathbf{B}^T \mathbf{X}_{\beta^*} (\tilde{\boldsymbol{\xi}}_2 - \boldsymbol{\xi}_2^*).$$

On $\mathcal{M}$, similar to the treatment for D_1 , we have

$$|I_2| \leq C \lambda_T \|\tilde{\boldsymbol{\xi}}_2 - \boldsymbol{\xi}_2^*\|_1.$$

For I_1 , we have

$$|I_1| \leq \|T^{-1}\boldsymbol{\epsilon}^T(\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)\|_{\max} \cdot N^{-a} \|T^{-1}(\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)^T \mathbf{Z}\|_1 \|\tilde{\boldsymbol{\xi}}_1 - \boldsymbol{\xi}_1^*\|_1.$$

With $\mathbf{b}_{ti}^T$ and $\boldsymbol{\pi}_{1j}^{*T}$ the i th row of $\mathbf{B}_t$ and $\boldsymbol{\Pi}_t^*$ respectively, an element in $T^{-1}(\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)^T \mathbf{Z}$ can be expressed as

$$T^{-1} \gamma^T \sum_{t=1}^T (\mathbf{b}_{ti} - \bar{\mathbf{b}}_{\cdot i}) \boldsymbol{\pi}_{1j}^{*T} \mathbf{W}_2^* \mathbf{X}_t \beta^* + T^{-1} \gamma^T \sum_{t=1}^T (\mathbf{b}_{ti} - \bar{\mathbf{b}}_{\cdot i}) \boldsymbol{\pi}_{1j}^{*T} \boldsymbol{\epsilon}_t,$$

so that on $\mathcal{M}$, we can bound

$$\begin{aligned}
&\|T^{-1}(\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)^T \mathbf{Z}\|_1 \\
&\leq \frac{2}{1-\eta} \sum_{i=1}^N \max_{1 \leq j \leq N} |\gamma^T T^{-1} \sum_{t=1}^T (\mathbf{b}_{ti} - \bar{\mathbf{b}}_{\cdot i}) \mathbf{x}_{tj}^T \beta^*| + \frac{\|\gamma\|_1}{1-\eta} \sum_{i=1}^N \|T^{-1} \sum_{t=1}^T (\mathbf{b}_{ti} - \bar{\mathbf{b}}_{\cdot i}) \boldsymbol{\epsilon}_t\|_{\max} \\
&\leq \frac{2\|\gamma\|_1 \|\beta^*\|_1}{1-\eta} (\lambda_T N (1 + \mu_{b,\max} + \lambda_T) + C_{bx} N^a) + \frac{\|\gamma\|_1}{1-\eta} \lambda_T N (1 + \mu_{b,\max} + \lambda_T) \\
&\leq C N^a, \text{ since } \lambda_T N^{1-a} = o(1).
\end{aligned}$$

Hence, on $\mathcal{M}$, $|I_1| \leq C \lambda_T \|\tilde{\boldsymbol{\xi}}_1 - \boldsymbol{\xi}_1^*\|_1$, so that $|D_2| \leq C \lambda_T \|\tilde{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\|_1$.

Similar argument shows that, on $\mathcal{M}$, we can show that

$$|D_3| \leq C \lambda_T \|\beta(\boldsymbol{\xi}^*) - \boldsymbol{\xi}^*\|_1 (s_2 + N).$$

For D_4 , on $\mathcal{M}$, we have

$$\begin{aligned}
|D_4| &\leq \|T^{-1}\tilde{\boldsymbol{\xi}}_2^T \mathbf{X}_{\beta(\tilde{\boldsymbol{\xi}})-\beta^*} \mathbf{B}\|_{\max} \cdot \|\mathbf{B}^T \mathbf{X}_{\beta^*-\beta(\boldsymbol{\xi}^*)} \boldsymbol{\xi}_2^*\|_1 \\
&\leq 2\|T^{-1}(\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)^T \mathbf{X}_{\beta(\tilde{\boldsymbol{\xi}})-\beta^*}\|_{\max} \cdot N^{-a} \|T^{-1}(\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)^T \mathbf{X}_{\beta^*-\beta(\boldsymbol{\xi}^*)} \boldsymbol{\xi}_2^*\|_1 \\
&\leq 2\|\gamma\|_1 \|\beta(\tilde{\boldsymbol{\xi}}) - \beta^*\|_1 (\sigma_{bx} + \lambda_T(1 + \mu_{b,\max} + \lambda_T)) \cdot N^{1-a} \|\gamma\|_1 \|\beta(\boldsymbol{\xi}^*) - \beta^*\|_1 \\
&\quad \cdot (C_{bx}N^{a-1} + \lambda_T(1 + \mu_{b,\max} + \lambda_T))(s_2 + N) \\
&\leq C\|\beta(\tilde{\boldsymbol{\xi}}) - \beta^*\|_1 \|\beta(\boldsymbol{\xi}^*) - \beta^*\|_1 (s_2 + N).
\end{aligned}$$

Similarly, we can prove that on $\mathcal{M}$,

$$\begin{aligned}
|D_5| &\leq C\|\beta(\boldsymbol{\xi}^*) - \beta^*\|_1 \|\tilde{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\|_1, \\
|D_6| &\leq C\|\beta(\boldsymbol{\xi}^*) - \beta^*\|_1^2 (s_2 + N).
\end{aligned}$$

To simplify the rates further, note that exactly parallel to the proof of Theorem 2, we can show that on $\mathcal{M}$,

$$\|\beta(\boldsymbol{\xi}^*) - \beta^*\|_1 \leq C\lambda_T N^{-1}(s_2 + N^{\frac{1}{2}+\frac{1}{2w}} + N^{1/2} \log^{1/2}(T \vee N)). \quad (8.9)$$

At the same time, Theorem 2 can be applied to the LASSO solution $\tilde{\boldsymbol{\beta}} = \beta(\tilde{\boldsymbol{\xi}})$, so that

$$\|\beta(\tilde{\boldsymbol{\xi}}) - \beta^*\|_1 = C\lambda_T N^{-1}(s_2 + N^{\frac{1}{2}+\frac{1}{2w}} + N^{1/2} \log^{1/2}(T \vee N)) + CN^{-1} \|\tilde{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\|_1. \quad (8.10)$$

With (8.9) and (8.10), the rates of $|D_1|$ to $|D_6|$ become

$$\begin{aligned}
|D_1| &\leq C\lambda_T^2(s_2 + N^{\frac{1}{2}+\frac{1}{2w}} + N^{1/2} \log^{1/2}(T \vee N)) + C\lambda_T \|\tilde{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\|_1, \\
|D_2| &\leq C\lambda_T \|\tilde{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\|_1, \\
|D_3| &\leq C\lambda_T^2(s_2 + N^{\frac{1}{2}+\frac{1}{2w}} + N^{1/2} \log^{1/2}(T \vee N)), \\
|D_4| &\leq C\lambda_T^2(s_2 + N^{\frac{1}{2}+\frac{1}{2w}} + N^{1/2} \log^{1/2}(T \vee N)) + C\lambda_T \|\tilde{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\|_1, \\
|D_5| &\leq C\lambda_T \|\tilde{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\|_1, \\
|D_6| &\leq C\lambda_T^2(s_2 + N^{\frac{1}{2}+\frac{1}{2w}} + N^{1/2} \log^{1/2}(T \vee N)).
\end{aligned}$$

The above mean that for some constant $D > 0$, we have

$$\begin{aligned}
\frac{1}{2T} \|\mathbf{B}^T(\mathbf{M}_{\beta^*} \boldsymbol{\xi}^* - \mathbf{M}_{\beta(\tilde{\boldsymbol{\xi}})} \tilde{\boldsymbol{\xi}})\|^2 &\leq D\lambda_T \|\tilde{\boldsymbol{\xi}} - \boldsymbol{\xi}^*\|_1 \\
&\quad + D\lambda_T^2(s_2 + N^{\frac{1}{2}+\frac{1}{2w}} + N^{1/2} \log^{1/2}(T \vee N)) + \gamma_T(\|\boldsymbol{\xi}^*\|_1 - \|\tilde{\boldsymbol{\xi}}\|_1).
\end{aligned}$$

Setting $\gamma_T = 2D\lambda_T$, and adding $D\lambda_T\|\tilde{\xi} - \xi^*\|$ on both sides of the above, we have

$$\begin{aligned}
& \frac{1}{2T} \|\mathbf{B}^T(\mathbf{M}_{\beta^*}\xi^* - \mathbf{M}_{\beta(\tilde{\xi})}\tilde{\xi})\|^2 + D\lambda_T\|\tilde{\xi} - \xi^*\|_1 \\
& \leq D\lambda_T^2(s_2 + N^{\frac{1}{2} + \frac{1}{2w}} + N^{1/2} \log^{1/2}(T \vee N)) + 2D\lambda_T(\|\tilde{\xi} - \xi^*\|_1 + \|\xi^*\|_1 - \|\tilde{\xi}\|_1) \\
& \leq D\lambda_T^2(s_2 + N^{\frac{1}{2} + \frac{1}{2w}} + N^{1/2} \log^{1/2}(T \vee N)) + 4D\lambda_T(\|\tilde{\xi}_{J_1} - \xi_{J_1}^*\|_1 + \|\xi_{J_2}^*\|_1) \\
& = 4D\lambda_T\|\tilde{\xi}_{J_1} - \xi_{J_1}^*\|_1 + D\lambda_T h_{N,T}.
\end{aligned}$$

The above means that

$$\frac{1}{2T} \|\mathbf{B}^T(\mathbf{M}_{\beta^*}\xi^* - \mathbf{M}_{\beta(\tilde{\xi})}\tilde{\xi})\|^2 \leq 4D\lambda_T\|\tilde{\xi}_{J_1} - \xi_{J_1}^*\|_1 + D\lambda_T h_{N,T}, \quad (8.11)$$

$$\|\tilde{\xi} - \xi^*\|_1 \leq 4\|\tilde{\xi}_{J_1} - \xi_{J_1}^*\|_1 + h_{N,T}. \quad (8.12)$$

The first result of Theorem 3 is immediate from (8.12), since $n = |J_1|$ and so $\|\tilde{\xi}_{J_1} - \xi_{J_1}^*\|_1 \leq n^{1/2}\|\tilde{\xi}_{J_1} - \xi_{J_1}^*\|$.

For the second result, define $\epsilon = \|T^{-1}\mathbf{M}^T\mathbf{B}\mathbf{B}^T\mathbf{M}^* - \mathbf{G}\|_{\max}$, with notations defined in Assumption R1. It is easy to see that on $\mathcal{M}$,

$$\begin{aligned}
\epsilon & \leq N^{1-a} \|T^{-1}(\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)^T \mathbf{M}^* - E(T^{-1}(\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)^T \mathbf{M}^*)\|_{\max}^2 \\
& \quad + 2\|T^{-1}(\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)^T \mathbf{M}^* - E(T^{-1}(\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)^T \mathbf{M}^*)\|_{\max} \cdot N^{-a} \|E(T^{-1}(\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)^T \mathbf{M}^*)\|_1 \\
& = O(\lambda_T^2 N^{1-a} + \lambda_T N^{-a}(N^a + NT^{-1})) = O(\lambda_T).
\end{aligned} \quad (8.13)$$

By (8.12), we have

$$\|\tilde{\xi}_{J_1^c} - \xi_{J_1^c}^*\|_1 \leq 3\|\tilde{\xi}_{J_1} - \xi_{J_1}^*\|_1 + h_{N,T}.$$

Hence, letting $\alpha = \tilde{\xi} - \xi^*$ and $R = J_1$, we have by Assumption R3 that

$$\begin{aligned}
\kappa_{N,T}(n)\|\tilde{\xi}_{J_1} - \xi_{J_1}^*\| & = \kappa_{N,T}(n)\|\alpha_R\| \leq \|\mathbf{G}^{1/2}\alpha\| \\
& \leq (\alpha^T \mathbf{G} \alpha - T^{-1} \alpha^T \mathbf{M}^T \mathbf{B} \mathbf{B}^T \mathbf{M}^* \alpha)^{1/2} + T^{-1/2} \|\mathbf{B}^T \mathbf{M}^* \alpha\| \\
& \leq \epsilon^{1/2} \|\alpha\|_1 + T^{-1/2} \|\mathbf{B}^T \mathbf{M}^* \xi^* - \mathbf{B}^T \mathbf{M}_{\beta(\tilde{\xi})} \tilde{\xi}\| + T^{-1/2} \|\mathbf{B}^T \mathbf{X}_{\beta(\tilde{\xi}) - \beta^*} \tilde{\xi}_2\|.
\end{aligned} \quad (8.14)$$

On $\mathcal{M}$, denoting $\tilde{w}_{2,\ell j}$ the (ℓ, j) element of $\tilde{\mathbf{W}}_2$,

$$\begin{aligned}
T^{-1/2} \|\mathbf{B}^T \mathbf{X}_{\beta(\tilde{\xi}) - \beta^*} \tilde{\xi}_2\| &\leq \left\{ \sum_{1 \leq \ell, i \leq N} N^{-a} \left| \sum_{j=1}^N \gamma^T \left(T^{-1} \sum_{t=1}^T (\mathbf{b}_{ti} - \bar{\mathbf{b}}_{\cdot i}) \mathbf{x}_{tj}^T \right) (\beta(\tilde{\xi}) - \beta^*) \tilde{w}_{2,\ell j} \right|^2 \right\}^{1/2} \\
&\leq N^{-a/2} N^{1/2} \|\gamma\|_1 \|\beta(\tilde{\xi}) - \beta^*\|_1 \cdot \left\{ \max_{1 \leq j \leq N} \sum_{i=1}^N \left\| T^{-1} \sum_{t=1}^T (\mathbf{b}_{ti} - \bar{\mathbf{b}}_{\cdot i}) \mathbf{x}_{tj}^T \right\|_{\max}^2 \right\}^{1/2} \\
&\leq N^{(1-a)/2} \|\beta(\tilde{\xi}) - \beta^*\|_1 \left\{ 3 \max_{1 \leq j \leq N} \sum_{i=1}^N \left\| T^{-1} \sum_{t=1}^T (\mathbf{b}_{ti} \mathbf{x}_{tj}^T - E(\mathbf{b}_{ti} \mathbf{x}_{tj}^T)) \right\|_{\max}^2 \right. \\
&\quad \left. + 3 \max_{1 \leq j \leq N} \sum_{i=1}^N \|E(\mathbf{b}_{ti} \mathbf{x}_{tj}^T)\|_{\max}^2 + 3 \max_{1 \leq j \leq N} \sum_{i=1}^N \left\| T^{-1} \sum_{t=1}^T \bar{\mathbf{b}}_{\cdot i} \bar{\mathbf{x}}_{\cdot j} \right\|_{\max}^2 \right\}^{1/2} \\
&\leq 3^{1/2} N^{(1-a)/2} \|\beta(\tilde{\xi}) - \beta^*\|_1 \left\{ \lambda_T^2 N + C_{bx} N^a + \lambda_T^2 N (\mu_{b,\max} + \lambda_T)^2 \right\}^{1/2} \\
&\leq \frac{C \lambda_T (s_2 + N^{\frac{1}{2} + \frac{1}{2w}} + N^{1/2} \log^{1/2}(T \vee N))}{N^{1/2}} + \frac{C \|\tilde{\xi} - \xi^*\|_1}{N^{1/2}} \\
&\leq C N^{-1/2} h_{N,T} + C n^{1/2} N^{-1/2} \|\tilde{\xi}_{J_1} - \xi_{J_1}^*\|,
\end{aligned}$$

where the last line used the first result of Theorem 3 which we have proved before. Using the above, (8.11) and (8.13), (8.14) becomes

$$\begin{aligned}
\kappa_{N,T}(n) \|\tilde{\xi}_{J_1} - \xi_{J_1}^*\| &\leq C \lambda_T^{1/2} (4n^{1/2} \|\tilde{\xi}_{J_1} - \xi_{J_1}^*\| + h_{N,T}) + C \lambda_T^{1/2} n^{1/4} \|\tilde{\xi}_{J_1} - \xi_{J_1}^*\|^{1/2} \\
&\quad + C \lambda_T^{1/2} h_{N,T}^{1/2} + C N^{-1/2} h_{N,T} + C n^{1/2} N^{-1/2} \|\tilde{\xi}_{J_1} - \xi_{J_1}^*\|, \text{ so that} \\
(\kappa_{N,T}(n) - C n^{1/2} (\lambda_T^{1/2} + N^{-1/2})) \|\tilde{\xi}_{J_1} - \xi_{J_1}^*\| &- C \lambda_T^{1/2} n^{1/4} \|\tilde{\xi}_{J_1} - \xi_{J_1}^*\|^{1/2} \\
&- C (\lambda_T^{1/2} + N^{-1/2}) h_{N,T} + C \lambda_T^{1/2} h_{N,T}^{1/2} \leq 0.
\end{aligned}$$

Solving the above quadratic inequality for $\|\tilde{\xi}_{J_1} - \xi_{J_1}^*\|^{1/2}$, we get

$$\|\tilde{\xi}_{J_1} - \xi_{J_1}^*\|^{1/2} \leq \frac{C \lambda_T^{1/2} n^{1/4} + \left\{ C^2 \lambda_T n^{1/2} + 4C \kappa_{N,T}(n) (\lambda_T^{1/2} h_{N,T}^{1/2} + (\lambda_T^{1/2} + N^{-1/2}) h_{N,T}) \right\}^{1/2}}{2\kappa_{N,T}(n) - 2C n^{1/2} (\lambda_T^{1/2} + N^{-1/2})}.$$

For large enough T, N , we then have

$$\|\tilde{\xi}_{J_1} - \xi_{J_1}^*\| \leq \frac{C \lambda_T n^{1/2}}{\kappa^2(n)} + \frac{C h_{N,T}^{1/2} (\lambda_T^{1/2} + \lambda_T^{1/2} h_{N,T}^{1/2} + N^{-1/2} h_{N,T}^{1/2})}{\kappa(n)},$$

for some constant $C > 0$, which is the second result of Theorem 3.

Combining the bound in Theorem 2 and the first two results for Theorem 3, we have on $\mathcal{M}$,

$$\begin{aligned}\|\beta(\tilde{\xi}) - \beta^*\|_1 &\leq \frac{h_{N,T}}{N} + \frac{4n^{1/2}\|\tilde{\xi}_{J_1} - \xi_{J_1}^*\| + h_{N,T}}{N} \\ &= O\left(\frac{\lambda_T n + h_{N,T} + (\lambda_T n h_{N,T})^{1/2} + (\lambda_T n)^{1/2} h_{N,T} + (n/N)^{1/2} h_{N,T}}{N}\right) \\ &= O\left(\frac{\lambda_T n + h_{N,T}}{N}\right),\end{aligned}$$

since $\lambda_T n, n/N = o(1)$ by Assumption R6. This completes the proof of the theorem. $\square$

Proof of Theorem 4. For notational convenience, we denote $\tilde{\xi}_{\text{OS}}$ by $\hat{\xi}$ in this proof. Also, define $D = J_0 \cup J_1 \cup J_2$.

The KKT condition implies that $\hat{\xi}$ is a solution to the adaptive LASSO problem in (2.7) if and only if there exists a subgradient

$$\mathbf{g} = \partial(\mathbf{v}^T |\hat{\xi}|) = \left\{ \mathbf{g} \in \mathbb{R}^{2N^2} : \begin{cases} g_i = 0, & i \in D^c; \\ g_i = v_i \text{sign}(\hat{\xi}_i), & \hat{\xi}_i \neq 0; \\ |g_i| \leq v_i, & \text{otherwise.} \end{cases} \right\},$$

such that differentiating with respect to ξ_D , we have

$$T^{-1} \tilde{\mathbf{M}}_D^T \mathbf{B} \mathbf{B}^T \tilde{\mathbf{M}}_D \hat{\xi} - T^{-1} \tilde{\mathbf{M}}_D^T \mathbf{B} \mathbf{B}^T \mathbf{y} + \gamma'_T \mathbf{g}_D + T^{-1} \tilde{\mathbf{M}}_D^T \mathbf{B} \mathbf{B}^T \mathbf{X}_{\beta(\hat{\xi})} \text{vec}(\mathbf{I}_N) = \mathbf{0},$$

where $\tilde{\mathbf{M}} = \mathbf{M}_{\beta(\hat{\xi})}$ and for a matrix A , A_S is the sub-matrix of A with only those columns indexed by S . Using $\mathbf{B}^T \mathbf{y} = \mathbf{B}^T \mathbf{M}^* \xi^* + \mathbf{B}^T \epsilon$, the above can be written as

$$T^{-1} \tilde{\mathbf{M}}_D^T \mathbf{B} \mathbf{B}^T (\tilde{\mathbf{M}}_D \hat{\xi}_D - \mathbf{M}_D^* \xi_D^*) - T^{-1} \tilde{\mathbf{M}}_D^T \mathbf{B} \mathbf{B}^T \epsilon + T^{-1} \tilde{\mathbf{M}}_D^T \mathbf{B} \mathbf{B}^T \mathbf{X}_{\beta(\hat{\xi}) - \beta^*} \text{vec}(\mathbf{I}_N) = -\gamma'_T \mathbf{g}_D.$$

Hence, couple with the above, there is a partial sign consistent solution $\hat{\xi}$ such that $\hat{\xi}_{J_0 \cup J_2} = \mathbf{0}$ and $\text{sign}(\hat{\xi}_{J_1}) = \text{sign}(\xi_{J_1}^*)$ if and only if

$$\begin{aligned}T^{-1} \tilde{\mathbf{M}}_D^T \mathbf{B} \mathbf{B}^T (\tilde{\mathbf{M}}_{J_1} \hat{\xi}_{J_1} - \mathbf{M}_{J_1}^* \xi_{J_1}^*) - T^{-1} \tilde{\mathbf{M}}_D^T \mathbf{B} \mathbf{B}^T \mathbf{M}_{J_2}^* \xi_{J_2}^* \\ + T^{-1} \tilde{\mathbf{M}}_D^T \mathbf{B} \mathbf{B}^T \epsilon + T^{-1} \tilde{\mathbf{M}}_D^T \mathbf{B} \mathbf{B}^T \mathbf{X}_{\beta(\hat{\xi}) - \beta^*} \text{vec}(\mathbf{I}_N) = -\gamma'_T \mathbf{g}_D.\end{aligned}$$

The above can be rewritten as two, one on J_1 and one on $J' = J_0 \cup J_2$:

$$\left\{ \begin{aligned} &T^{-1} \tilde{\mathbf{M}}_{J_1}^T \mathbf{B} \mathbf{B}^T \tilde{\mathbf{M}}_{J_1} \hat{\xi}_{J_1} - T^{-1} \tilde{\mathbf{M}}_{J_1}^T \mathbf{B} \mathbf{B}^T \mathbf{M}_{J_1}^* \xi_{J_1}^* - T^{-1} \tilde{\mathbf{M}}_{J_1}^T \mathbf{B} \mathbf{B}^T \mathbf{M}_{J_2}^* \xi_{J_2}^*, \\ &+ T^{-1} \tilde{\mathbf{M}}_{J_1}^T \mathbf{B} \mathbf{B}^T \mathbf{X}_{\beta(\hat{\xi}) - \beta^*} \text{vec}(\mathbf{I}_N) - T^{-1} \tilde{\mathbf{M}}_{J_1}^T \mathbf{B} \mathbf{B}^T \epsilon = -\gamma'_T \mathbf{g}_{J_1}, \\ &\left| T^{-1} \tilde{\mathbf{M}}_{J'}^T \mathbf{B} \mathbf{B}^T \tilde{\mathbf{M}}_{J_1} \hat{\xi}_{J_1} - T^{-1} \tilde{\mathbf{M}}_{J'}^T \mathbf{B} \mathbf{B}^T \mathbf{M}_{J_1}^* \xi_{J_1}^* - T^{-1} \tilde{\mathbf{M}}_{J'}^T \mathbf{B} \mathbf{B}^T \mathbf{M}_{J_2}^* \xi_{J_2}^* \right. \\ &\left. + T^{-1} \tilde{\mathbf{M}}_{J'}^T \mathbf{B} \mathbf{B}^T \mathbf{X}_{\beta(\hat{\xi}) - \beta^*} \text{vec}(\mathbf{I}_N) - T^{-1} \tilde{\mathbf{M}}_{J'}^T \mathbf{B} \mathbf{B}^T \epsilon \right| \leq \gamma'_T \mathbf{v}_{J'}. \end{aligned} \right. \quad (8.15)$$

By Assumption R1, the matrix $\mathbf{G}_{J_1 J_1} = N^{-a} E(T^{-1} \mathbf{M}_{J_1}^{*T} (\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)) E(T^{-1} (\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)^T \mathbf{M}_{J_1}^*)$ is invertible. The first equation of (8.15) can be expressed as $\hat{\boldsymbol{\xi}}_{J_1} = \boldsymbol{\xi}_{J_1}^* + \sum_{j=1}^8 I_j$, where

$$\begin{aligned} I_1 &= -\mathbf{G}_{J_1 J_1}^{-1} (T^{-1} \mathbf{M}_{J_1}^{*T} \mathbf{B} \mathbf{B}^T \mathbf{M}_{J_1}^* - \mathbf{G}_{J_1 J_1}) (\hat{\boldsymbol{\xi}}_{J_1} - \boldsymbol{\xi}_{J_1}^*), \\ I_2 &= -\mathbf{G}_{J_1 J_1}^{-1} T^{-1} (\widetilde{\mathbf{M}}_{J_1} - \mathbf{M}_{J_1}^*) \mathbf{B} \mathbf{B}^T (\widetilde{\mathbf{M}}_{J_1} - \mathbf{M}_{J_1}^*) \hat{\boldsymbol{\xi}}_{J_1}, \\ I_3 &= -\mathbf{G}_{J_1 J_1}^{-1} T^{-1} \mathbf{M}_{J_1}^{*T} \mathbf{B} \mathbf{B}^T (\widetilde{\mathbf{M}}_{J_1} - \mathbf{M}_{J_1}^*) \hat{\boldsymbol{\xi}}_{J_1}, \\ I_4 &= -\mathbf{G}_{J_1 J_1}^{-1} T^{-1} (\widetilde{\mathbf{M}}_{J_1} - \mathbf{M}_{J_1}^*)^T \mathbf{B} \mathbf{B}^T \mathbf{M}_{J_1}^* \hat{\boldsymbol{\xi}}_{J_1}, \\ I_5 &= \mathbf{G}_{J_1 J_1}^{-1} T^{-1} (\widetilde{\mathbf{M}}_{J_1} - \mathbf{M}_{J_1}^*)^T \mathbf{B} \mathbf{B}^T \mathbf{M}_{J_1}^* \boldsymbol{\xi}_{J_1}^*, \\ I_6 &= \mathbf{G}_{J_1 J_1}^{-1} T^{-1} \widetilde{\mathbf{M}}_{J_1}^T \mathbf{B} \mathbf{B}^T \mathbf{M}_{J_2}^* \boldsymbol{\xi}_{J_2}^*, \\ I_7 &= -\mathbf{G}_{J_1 J_1}^{-1} T^{-1} \widetilde{\mathbf{M}}_{J_1}^T \mathbf{B} \mathbf{B}^T \mathbf{X}_{\boldsymbol{\beta}(\tilde{\boldsymbol{\xi}}) - \boldsymbol{\beta}^*} \text{vec}(\mathbf{I}_N), \\ I_8 &= \mathbf{G}_{J_1 J_1}^{-1} (T^{-1} \widetilde{\mathbf{M}}_{J_1}^T \mathbf{B} \mathbf{B}^T \boldsymbol{\epsilon} - \gamma'_T \mathbf{g}_{J_1}). \end{aligned}$$

We first bound $\|\mathbf{G}_{J_1 J_1}^{-1}\|_\infty$. To this end, partition $\mathbf{G}_{J_1 J_1} = (\mathbf{A}_{ij})_{1 \leq i, j \leq 2}$, such that $\mathbf{A}_{12} = \mathbf{A}_{21}^T$, and

$$\begin{aligned} \mathbf{A}_{11} &= N^{-a} E(T^{-1} \mathbf{Z}_{J_1}^T (\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)) E(T^{-1} (\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)^T \mathbf{Z}_{J_1}), \\ \mathbf{A}_{12} &= N^{-a} E(T^{-1} \mathbf{Z}_{J_1}^T (\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)) E(T^{-1} (\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)^T \mathbf{X}_{J_1}^*), \\ \mathbf{A}_{22} &= N^{-a} E(T^{-1} \mathbf{X}_{J_1}^{*T} (\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)) E(T^{-1} (\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)^T \mathbf{X}_{J_1}^*). \end{aligned}$$

Then by Assumption R1, if $n_r < \infty$ is the maximum number of elements in J_1 in each row of $\mathbf{W}_1^*$ and $\mathbf{W}_2^*$, each block diagonal $\mathbf{A}_{ij}$ defined above has at most n_r non-zeros in each row. Hence on $\mathcal{M}$,

$$\begin{aligned} \|\mathbf{G}_{J_1 J_1}^{-1}\|_\infty &\leq \|(\mathbf{A}_{11} - \mathbf{A}_{12} \mathbf{A}_{22}^{-1} \mathbf{A}_{12}^T)^{-1}\|_\infty + \|\mathbf{A}_{11}^{-1} \mathbf{A}_{12} (\mathbf{A}_{22} - \mathbf{A}_{12}^T \mathbf{A}_{11}^{-1} \mathbf{A}_{12})^{-1}\|_\infty \\ &\leq n_r^{1/2} \lambda_{\max}\{(\mathbf{A}_{11} - \mathbf{A}_{12} \mathbf{A}_{22}^{-1} \mathbf{A}_{12}^T)^{-1}\} + n_r^{1/2} \lambda_{\max}(\mathbf{A}_{11}^{-1}) \|\mathbf{A}_{12}\|_\infty \|(\mathbf{A}_{22} - \mathbf{A}_{12}^T \mathbf{A}_{11}^{-1} \mathbf{A}_{12})^{-1}\|_\infty \\ &\leq n_r^{1/2} \lambda_{\max}(\mathbf{G}_{J_1 J_1}^{-1}) \{1 + \|\mathbf{A}_{12}\|_{\max} n_r^{3/2} \lambda_{\max}(\mathbf{G}_{J_1 J_1}^{-1})\} \\ &\leq n_r^{1/2} u^{-1} (1 + O(1) n_r^{3/2} u^{-1}) = O(1), \end{aligned}$$

where we used Assumption R1 such that $\lambda_{\max}(\mathbf{G}_{J_1 J_1}^{-1}) \leq u^{-1}$, and on $\mathcal{M}$,

$$\begin{aligned} \|\mathbf{A}_{12}\|_{\max} &\leq N^{-a} \|E(T^{-1} (\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)^T \mathbf{X}_{J_1}^*)\|_{\max} \cdot \|E(T^{-1} (\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)^T \mathbf{Z}_{J_1})\|_1 \\ &\leq N^{-a} (\sigma_{bx} \|\boldsymbol{\beta}^*\|_1 \|\boldsymbol{\gamma}\|_1 + O(T^{-1}) \|\boldsymbol{\beta}^*\|_1 \|\boldsymbol{\gamma}\|_1) \cdot \frac{2}{1-\eta} \|\boldsymbol{\gamma}\|_1 \|\boldsymbol{\beta}^*\|_1 (C_{bx} N^a + O(NT^{-1})) = O(1). \end{aligned}$$

Hence on $\mathcal{M}$,

$$\begin{aligned} \|I_1\|_{\max} &\leq \|\mathbf{G}_{J_1 J_1}^{-1}\|_\infty \|T^{-1} \mathbf{M}_{J_1}^{*T} \mathbf{B} \mathbf{B}^T \mathbf{M}_{J_1}^* - \mathbf{G}_{J_1 J_1}\|_\infty \|\tilde{\boldsymbol{\xi}}_{J_1} - \boldsymbol{\xi}_{J_1}^*\|_{\max} \\ &\leq O(1) \cdot n_r \epsilon \|\hat{\boldsymbol{\xi}}_{J_1} - \boldsymbol{\xi}_{J_1}^*\|_{\max} = O(\lambda_T \|\hat{\boldsymbol{\xi}}_{J_1} - \boldsymbol{\xi}_{J_1}^*\|_{\max}). \end{aligned}$$

For I_2 , on $\mathcal{M}$,

$$\begin{aligned}
\|I_2\|_{\max} &\leq \|\mathbf{G}_{J_1 J_1}^{-1}\|_{\max} \|T^{-1} \mathbf{X}_{\beta(\tilde{\xi}) - \beta^*, J_1}^T (\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma) \cdot N^{-a} T^{-1} (\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)^T \mathbf{X}_{\beta(\tilde{\xi}) - \beta^*, J_1}\|_{\infty} \|\hat{\xi}_{J_1}\|_{\max} \\
&\leq O(1) \cdot N^{-a} n_r \max_i \sum_{i=1}^N \left[(\beta(\tilde{\xi}) - \beta^*)^T T^{-1} \sum_{t=1}^T \mathbf{x}_{ti} (\mathbf{b}_{tj} - \bar{\mathbf{b}}_{\cdot j})^T \gamma \right]^2 \\
&\leq O(1) N^{-a} \|\beta(\tilde{\xi}) - \beta^*\|_1^2 \|\gamma\|_1^2 \cdot \max_i \sum_{j=1}^N \|T^{-1} \sum_{t=1}^T \mathbf{x}_{ti} (\mathbf{b}_{tj} - \bar{\mathbf{b}}_{\cdot j})^T\|_{\max}^2 \\
&= O(\|\beta(\tilde{\xi}) - \beta^*\|_1^2 N^{-a} (\lambda_T^2 N + C_{bx} N^a + \lambda_T^2 N (\mu_{b, \max} + \lambda_T)^2)) \\
&= O(\|\beta(\tilde{\xi}) - \beta^*\|_1^2).
\end{aligned}$$

Similar lines of proof show that on $\mathcal{M}$, we have

$$\|I_3\|_{\max}, \|I_4\|_{\max}, \|I_5\|_{\max}, \|I_7\|_{\max} = O(\|\beta(\tilde{\xi}) - \beta^*\|_1), \quad \|I_6\|_{\max} = O(\|\beta(\tilde{\xi}) - \beta^*\|_1 + \|\xi_{J_2}^*\|_{\max}).$$

For I_8 , on $\mathcal{M}$,

$$\begin{aligned}
\|I_8\|_{\max} &\leq \|\mathbf{G}_{J_1 J_1}^{-1}\|_{\infty} (\|T^{-1} \widetilde{\mathbf{M}}_{J_1}^T (\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma) N^{-a} T^{-1} (\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)^T \epsilon\|_{\max} + \gamma'_T \|\mathbf{g}_{J_1}\|_{\max}) \\
&\leq O(1) \cdot \left(N^{-a} \|T^{-1} \widetilde{\mathbf{M}}_{J_1}^T (\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)\|_{\infty} \cdot \max_j \|T^{-1} \sum_{t=1}^T (\mathbf{B}_t - \bar{\mathbf{B}}) \gamma \epsilon_{tj}\|_{\max} + \frac{\gamma'_T}{\min_{j \in J_1} |\tilde{\xi}_j|} \right) \\
&= O\left(\lambda_T + \frac{\gamma'_T}{\min_{j \in J_1} |\xi_j^*| - \|\tilde{\xi}_{J_1} - \xi_{J_1}^*\|} \right) = O(\lambda_T + \gamma'_T).
\end{aligned}$$

Combining the above and using the result of Theorem 2, it means that

$$\begin{aligned}
(1 - O(\lambda_T)) \|\hat{\xi}_{J_1} - \xi_{J_1}^*\|_{\max} &= O(\lambda_T + \gamma'_T + \|\xi_{J_2}^*\|_{\max} + h_{N,T} N^{-1}), \text{ so that} \\
\|\hat{\xi}_{J_1} - \xi_{J_1}^*\|_{\max} &= O(\lambda_T + \gamma'_T + \|\xi_{J_2}^*\|_{\max} + \|\xi_{J_2}^*\|_1 N^{-1}),
\end{aligned}$$

which is the first result of the theorem.

Also, by this result, we have $\|\hat{\xi}_{J_1} - \xi_{J_1}^*\|_{\max} = o(1)$ on $\mathcal{M}$. Since $\min_{j \in J_1} \xi_j^* \geq \tau > 0$ uniformly, it means that $\text{sign}(\hat{\xi}_{J_1}) = \text{sign}(\xi_{J_1}^*)$ when T, N are large enough. Hence it remains to show the remaining inequality of (8.15).

Since the lower bound on the right hand side of the inequality in (8.15) is

$$\frac{\gamma'_T}{\|\tilde{\xi}_{J'}\|_{\max}} \geq \frac{\gamma'_T}{\|\xi_{J_2}^*\|_{\max} + \|\tilde{\xi}_{J_1^c} - \xi_{J_1^c}^*\|},$$

if we can show that the max-norm of the left hand side of the inequality in (8.15) is, on $\mathcal{M}$, of order $o(\gamma'_T (\|\xi_{J_2}^*\|_{\max} + \|\tilde{\xi}_{J_1^c} - \xi_{J_1^c}^*\|)^{-1})$, then the proof of sign consistency is complete.

We first find the order of $\|\tilde{\xi}_{J_1^c} - \xi_{J_1^c}^*\|$. Using Assumption R3, with $\alpha = \tilde{\xi} - \xi$, we have

$$\begin{aligned}
\kappa_{N,T}(n)\|\tilde{\xi}_{J_1^c} - \xi_{J_1^c}^*\| &\leq \|\mathbf{G}^{1/2}\alpha\| \\
&\leq \epsilon^{1/2}\|\alpha\|_1 + T^{-1/2}\|\mathbf{B}^T\mathbf{M}^*\xi^* - \mathbf{B}^T\mathbf{M}_{\beta(\tilde{\xi})}\tilde{\xi}\| + T^{-1/2}\|\mathbf{B}^T\mathbf{X}_{\beta(\tilde{\xi})-\beta^*}\tilde{\xi}_2\| \\
&= O(\lambda_T^{1/2}(4n^{1/2}\|\tilde{\xi}_{J_1} - \xi_{J_1}^*\| + h_{N,T}) + \lambda_T^{1/2}n^{1/4}\|\tilde{\xi}_{J_1} - \xi_{J_1}^*\|^{1/2} \\
&\quad + N^{-1/2}h_{N,T} + n^{1/2}N^{-1/2}\|\tilde{\xi}_{J_1} - \xi_{J_1}^*\|) \\
&= O(n^{1/2}(\lambda_T^{1/2} + N^{-1/2})\|\tilde{\xi}_{J_1} - \xi_{J_1}^*\| + \lambda_T^{1/2}n^{1/4}\|\tilde{\xi}_{J_1} - \xi_{J_1}^*\|^{1/2} \\
&\quad + (\lambda_T^{1/2} + N^{-1/2})h_{N,T}) \\
&= O((\lambda_T n + h_{N,T})(\lambda_T^{1/2} + N^{-1/2}) \\
&\quad + \lambda_T^{1/2}n^{1/4}(\lambda_T^{1/2}n^{1/4} + \lambda_T^{1/4}h_{N,T}^{1/4} + \lambda_T^{1/4}h_{N,T}^{1/2} + N^{-1/4}h_{N,T}^{1/2})) \\
&= O(\lambda_T n^{1/2} + (\lambda_T^{1/2} + N^{-1/2})h_{N,T} + \lambda_T^{3/4}n^{1/4}h_{N,T}^{1/4}),
\end{aligned}$$

where we have used the same lines of proof after (8.14) and the result of Theorem 3 for the above. Hence we have

$$\|\tilde{\xi}_{J_1^c} - \xi_{J_1^c}^*\| \leq \|\mathbf{G}^{1/2}\alpha\| = O(\lambda_T n^{1/2} + (\lambda_T^{1/2} + N^{-1/2})h_{N,T} + \lambda_T^{3/4}n^{1/4}h_{N,T}^{1/4}) = o(1). \quad (8.16)$$

Similar to finding the order of $\|I_j\|_{\max}$, for the second inequality of (8.15), it is easy to show that

$$\begin{aligned}
\|T^{-1}\tilde{\mathbf{M}}_{J'}^T\mathbf{B}\mathbf{B}^T\mathbf{X}_{\beta(\tilde{\xi})-\beta^*,J_1}\hat{\xi}_{2,J_1}\|_{\max} &\leq n_r N^{-a}\|T^{-1}\tilde{\mathbf{M}}_{J'}^T(\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)\|_{\max} \\
&\quad \cdot \|T^{-1}(\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)^T\mathbf{X}_{\beta(\tilde{\xi})-\beta^*}\|_1 \\
&= O(\|\beta(\tilde{\xi}) - \beta^*\|_1), \\
\|T^{-1}\tilde{\mathbf{M}}_{J'}^T\mathbf{B}\mathbf{B}^T\mathbf{M}_{J_1}^*(\hat{\xi}_{J_1} - \xi_{J_1}^*)\|_{\max} &\leq n_r\|T^{-1}\tilde{\mathbf{M}}_{J'}^T\mathbf{B}\mathbf{B}^T\mathbf{M}_{J_1}^*\|_{\max} \cdot \|\hat{\xi}_{J_1} - \xi_{J_1}^*\|_{\max} \\
&= O(\|\hat{\xi}_{J_1} - \xi_{J_1}^*\|_{\max}), \\
\|T^{-1}\tilde{\mathbf{M}}_{J'}^T\mathbf{B}\mathbf{B}^T\mathbf{M}_{J_2}^*\xi_{J_2}^*\|_{\max} &\leq \|T^{-1}\tilde{\mathbf{M}}_{J'}^T(\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)\|_{\max} \\
&\quad \cdot N^{-a}\|T^{-1}(\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)^T\mathbf{M}_{J_2}^*\|_1 \cdot \|\xi_{J_2}^*\|_1 \\
&= O(\|\xi_{J_2}^*\|_1), \\
\|T^{-1}\mathbf{M}_{J'}^T\mathbf{B}\mathbf{B}^T\mathbf{X}_{\beta(\tilde{\xi})-\beta^*}\text{vec}(\mathbf{I}_N)\|_{\max} &= O(\|\beta(\tilde{\xi}) - \beta^*\|_1), \\
\|T^{-1}\mathbf{M}_{J'}^T\mathbf{B}\mathbf{B}^T\epsilon\|_{\max} &= O(\lambda_T).
\end{aligned}$$

Hence for the inequality of (8.15) to satisfy for large enough N, T , a sufficient condition is

$$(\lambda_T + \|\xi_{J_2}^*\|_1 + \|\hat{\xi}_{J_1} - \xi_{J_1}^*\|_{\max})(\|\xi_{J_2}^*\|_{\max} + \|\tilde{\xi}_{J_1^c} - \xi_{J_1^c}^*\|) = o(\gamma'_T).$$

By the first result of the theorem, this reduces to

$$(\lambda_T + \|\boldsymbol{\xi}_{J_2}^*\|_1)(\|\boldsymbol{\xi}_{J_2}^*\|_{\max} + \|\tilde{\boldsymbol{\xi}}_{J_1^c} - \boldsymbol{\xi}_{J_1^c}^*\|) = o(\gamma'_T), \quad (8.17)$$

since $\gamma'_T(\|\boldsymbol{\xi}_{J_2}^*\|_{\max} + \|\tilde{\boldsymbol{\xi}}_{J_1^c} - \boldsymbol{\xi}_{J_1^c}^*\|) = o(\gamma'_T)$ as $\|\boldsymbol{\xi}_{J_2}^*\|_{\max} = o(1)$ by Assumption R6 and $\|\tilde{\boldsymbol{\xi}}_{J_1^c} - \boldsymbol{\xi}_{J_1^c}^*\| = o(1)$ from (8.16). If $\|\boldsymbol{\xi}_{J_2}^*\|_1 = O(\lambda_T)$, then setting $\gamma'_T = \gamma_T = C\lambda_T$ for some constant $C > 0$ is sufficient since then (8.17) will satisfy automatically. If $\lambda_T = o(\|\boldsymbol{\xi}_{J_2}^*\|_1)$, then (8.17) becomes

$$\|\boldsymbol{\xi}_{J_2}^*\|_1(\|\boldsymbol{\xi}_{J_2}^*\|_{\max} + \lambda_T n^{1/2} + (\lambda_T^{1/2} + N^{-1/2})h_{N,T} + \lambda_T^{3/4} n^{1/4} h_{N,T}^{1/4}) = o(\gamma'_T),$$

which is equivalent to

$$\|\boldsymbol{\xi}_{J_2}^*\|_1 = o\left(\frac{\gamma'_T}{\|\boldsymbol{\xi}_{J_2}^*\|_{\max}}, \frac{\gamma'_T}{\lambda_T n^{1/2}}, \frac{\gamma_T'^{1/2}}{\lambda_T^{1/4} + N^{-1/4}}, \frac{\gamma'_T}{(\lambda_T^{1/2} + N^{-1/2})q_{N,T}}, \frac{\gamma_T'^{4/5}}{\lambda_T^{3/5} n^{1/5}}, \frac{\gamma'_T}{\lambda_T^{3/4} n^{1/4} q_{N,T}^{1/4}}\right),$$

where $q_{N,T} = \lambda_T(s_2 + N^{\frac{1}{2} + \frac{1}{2w}} + N^{1/2} \log^{1/2}(T \vee N))$. This is satisfied by Assumption R5. Hence the partial sign consistency of $\hat{\boldsymbol{\xi}}_{\text{OS}}$ as a solution of (2.7) is established.

Finally, with partial sign consistency on $\mathcal{M}$, we have

$$\begin{aligned} \|\hat{\boldsymbol{\xi}}_{\text{OS}} - \boldsymbol{\xi}^*\| &\leq \|\hat{\boldsymbol{\xi}}_{\text{OS}, J_1} - \boldsymbol{\xi}_{J_1}^*\| + \|\hat{\boldsymbol{\xi}}_{\text{OS}, J_2} - \boldsymbol{\xi}_{J_2}^*\| \\ &\leq n^{1/2} \|\hat{\boldsymbol{\xi}}_{\text{OS}, J_1} - \boldsymbol{\xi}_{J_1}^*\|_{\max} + \|\boldsymbol{\xi}_{J_2}^*\|, \\ &= O(\lambda_T n^{1/2} + \gamma'_T n^{1/2} + \|\boldsymbol{\xi}_{J_2}^*\|_{\max} n^{1/2} + \|\boldsymbol{\xi}_{J_2}^*\|_1 n^{1/2} N^{-1} + \|\boldsymbol{\xi}_{J_2}^*\|). \end{aligned}$$

This completes the proof of the theorem. $\square$

Proof of Theorem 5. Using model (2.1), denoting $\boldsymbol{\Pi}^* = (\mathbf{I}_N - \mathbf{W}_1^*)^{-1}$, we have $\bar{\mathbf{y}} = \boldsymbol{\Pi}^* \boldsymbol{\mu}^* + \boldsymbol{\Pi}^* \mathbf{W}_2^* \bar{\mathbf{X}} \boldsymbol{\beta}^* + \boldsymbol{\Pi}^* \bar{\boldsymbol{\epsilon}}$, where $\bar{\mathbf{y}} = T^{-1} \sum_{t=1}^T \mathbf{y}_t$ and similar definitions apply on $\bar{\mathbf{X}}$ and $\bar{\boldsymbol{\epsilon}}$. Hence from (2.8), writing $\widehat{\mathbf{W}}_i$ as a spatial weight matrix reconstructed from $\hat{\boldsymbol{\xi}}_{\text{OS}}$,

$$\begin{aligned} \hat{\boldsymbol{\mu}} &= (\mathbf{I}_N - \widehat{\mathbf{W}}_1)(\boldsymbol{\Pi}^* \boldsymbol{\mu}^* + \boldsymbol{\Pi}^* \mathbf{W}_2^* \bar{\mathbf{X}} \boldsymbol{\beta}^* + \boldsymbol{\Pi}^* \bar{\boldsymbol{\epsilon}}) - \widehat{\mathbf{W}}_2 \bar{\mathbf{X}} \tilde{\boldsymbol{\beta}}, \quad \text{so that} \\ \hat{\boldsymbol{\mu}} - \boldsymbol{\mu}^* &= (\mathbf{W}_1^* - \widehat{\mathbf{W}}_1)(\boldsymbol{\Pi}^* \boldsymbol{\mu}^* + \boldsymbol{\Pi}^* \mathbf{W}_2^* \bar{\mathbf{X}} \boldsymbol{\beta}^* + \boldsymbol{\Pi}^* \bar{\boldsymbol{\epsilon}}) + (\mathbf{W}_2^* - \widehat{\mathbf{W}}_2) \bar{\mathbf{X}} \boldsymbol{\beta}^* + \widehat{\mathbf{W}}_2 \bar{\mathbf{X}} (\boldsymbol{\beta}^* - \tilde{\boldsymbol{\beta}}). \end{aligned}$$

Hence, on $\mathcal{M}$ with large enough T, N , using the partial sign consistency of $\widehat{\xi}_{\text{OS}}$,

$$\begin{aligned}
\|\widehat{\mu} - \mu^*\|_{\max} &\leq \|\mathbf{W}_1^* - \widehat{\mathbf{W}}_1\|_{\infty} \|\Pi^* \mu^* + \Pi^* \mathbf{W}_2^* \bar{\mathbf{X}} \beta^* + \Pi^* \bar{\epsilon}\|_{\max} \\
&\quad + \|\mathbf{W}_2^* - \widehat{\mathbf{W}}_2\|_{\infty} \|\bar{\mathbf{X}}\|_{\max} \|\beta^*\|_1 + \|\widehat{\mathbf{W}}_2\|_{\infty} \|\bar{\mathbf{X}}\|_{\max} \|\tilde{\beta} - \beta^*\|_1 \\
&\leq (n_r \|\widehat{\xi}_{\text{OS}, J_1} - \xi_{J_1}^*\|_{\max} + \max_{1 \leq j \leq N} \|\mathbf{w}_{1,j,J_2}^*\|_1) \left(\frac{2\|\beta^*\|_1 \lambda_T + \|\mu^*\|_{\max} + \lambda_T}{1 - \eta} \right) \\
&\quad + (n_r \|\widehat{\xi}_{\text{OS}, J_1} - \xi_{J_1}^*\|_{\max} + \max_{1 \leq j \leq N} \|\mathbf{w}_{2,j,J_2}^*\|_1) \lambda_T \|\beta^*\|_1 + 2\lambda_T O\left(\frac{\lambda_T n + h_{N,T}}{N}\right) \\
&= O(\lambda_T + \gamma'_T + \max_{1 \leq j \leq N} \|\mathbf{w}_{1,j,J_2}^*\|_1),
\end{aligned}$$

where $\mathbf{w}_{i,j,J_2}^{*\text{T}}$ is the j th row of $\mathbf{W}_i^*$, restricted to J_2 .

For the remaining proof, writing

$$\widehat{\Pi} = (\mathbf{I}_N - \mathbf{W}_1^* - (\widehat{\mathbf{W}}_1 - \mathbf{W}_1^*))^{-1} = \Pi^* (\mathbf{I}_N - (\widehat{\mathbf{W}}_1 - \mathbf{W}_1^*)) = \Pi^* (\mathbf{I}_N - (\widehat{\mathbf{W}}_1 - \mathbf{W}_1^*) \Pi^*)^{-1},$$

and note that

$$(\mathbf{I}_N - \mathbf{W}_1^* - (\widehat{\mathbf{W}}_1 - \mathbf{W}_1^*)) \Pi^* (\mathbf{I}_N - (\widehat{\mathbf{W}}_1 - \mathbf{W}_1^*))^{-1} = \mathbf{I}_N,$$

It is easy to show that

$$\widehat{\Pi} - \Pi^* = \Pi^* (\widehat{\mathbf{W}}_1 - \mathbf{W}_1^*) \Pi^* (\mathbf{I}_N - (\widehat{\mathbf{W}}_1 - \mathbf{W}_1^*) \Pi^*)^{-1}.$$

From this, on $\mathcal{M}$ with large T, N and using the partial sign consistency property of $\widehat{\xi}_{\text{OS}}$,

$$\begin{aligned}
\|\widehat{\Pi} - \Pi^*\|_{\infty} &\leq \frac{1}{(1 - \eta)^2} \|\widehat{\mathbf{W}}_1 - \mathbf{W}_1^*\|_{\infty} \sum_{k \geq 0} \left(\frac{\|\widehat{\mathbf{W}}_1 - \mathbf{W}_1^*\|_{\infty}}{1 - \eta} \right)^k \\
&= \frac{1}{(1 - \eta)^2} \|\widehat{\mathbf{W}}_1 - \mathbf{W}_1^*\|_{\infty} \left(1 - \frac{\|\widehat{\mathbf{W}}_1 - \mathbf{W}_1^*\|_{\infty}}{1 - \eta} \right)^{-1} \\
&= O(\lambda_T + \gamma'_T + \max_{1 \leq j \leq N} \|\mathbf{w}_{1,j,J_2}^*\|_1).
\end{aligned}$$

Hence on $\mathcal{M}$ with large enough T, N , we have

$$\begin{aligned}
\|\widehat{\Pi} \widehat{\mu} - \Pi^* \mu^*\|_{\max} &\leq \|\widehat{\Pi} - \Pi^*\|_{\infty} (\|\widehat{\mu} - \mu^*\|_{\max} + \|\mu^*\|_{\max}) + \|\Pi^*\|_{\infty} \|\widehat{\mu} - \mu^*\|_{\max} \\
&= O(\|\widehat{\Pi} - \Pi^*\|_{\infty} + \|\widehat{\mu} - \mu^*\|_{\max}) \\
&= O(\lambda_T + \gamma'_T + \max_{1 \leq j \leq N} \|\mathbf{w}_{1,j,J_2}^*\|_1).
\end{aligned}$$

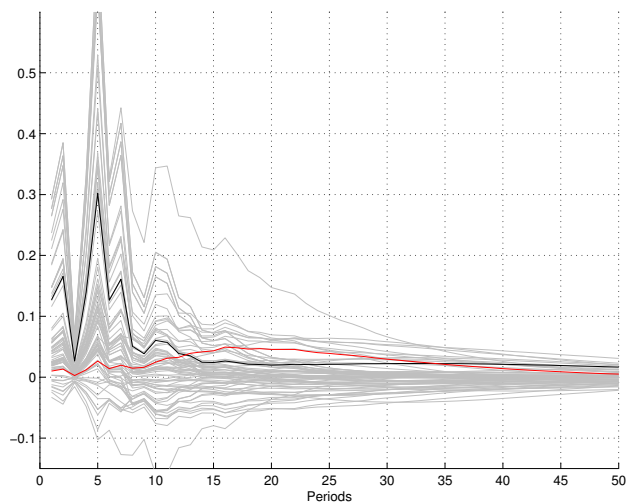
This completes the proof of the theorem. $\square$

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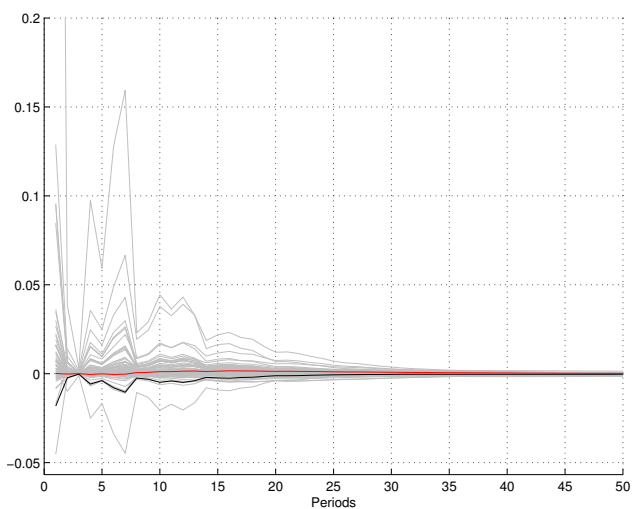
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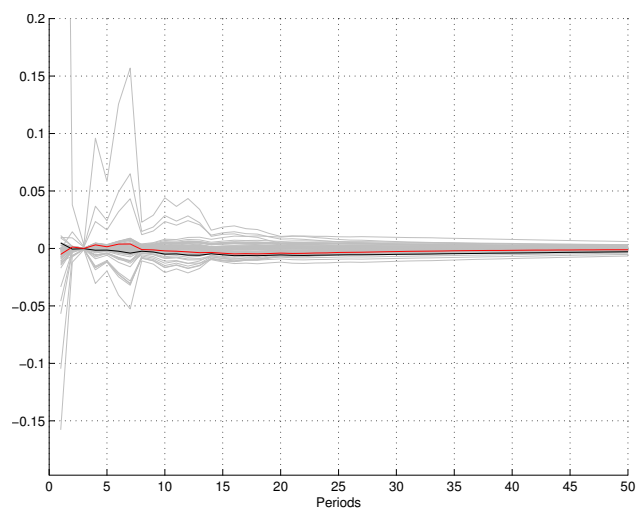
Note: "fuel oil and other fuels" in black line; "food and nonalcoholic beverages purchased for off-premises consumption" in red line.

Figure 1: Impulse-response function to one standard-deviation oil price shock



Note: "fuel oil and other fuels" in black line; "food and nonalcoholic beverages purchased for off-premises consumption" in red line.

Figure 2: Impulse-response function to one standard-deviation monetary policy shock



Note: "fuel oil and other fuels" in black line; "food and nonalcoholic beverages purchased for off-premises consumption" in red line.

Figure 3: Impulse-response function to one standard-deviation exchange rate shock